

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408384
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Nakazawa; Yasutaka et al.

Semiconductor device and display device including the semiconductor device

Abstract

A semiconductor device including an oxide semiconductor film that includes a transistor with excellent electrical characteristics is provided. It is a semiconductor device including a transistor. The transistor includes a gate electrode, a first insulating film, an oxide semiconductor film, a source electrode, a drain electrode, and a second insulating film. The source electrode and the drain electrode each include a first conductive film, a second conductive film over and in contact with the first conductive film, and a third conductive film over and in contact with the second conductive film. The second conductive film contains copper, the first conductive film and the third conductive film include a material that inhibits diffusion of copper, and an end portion of the second conductive film includes a region containing copper and silicon.

Inventors: Nakazawa; Yasutaka (Tochigi, JP), Koezuka; Junichi (Tochigi, JP), Hamochi; Takashi (Tochigi, JP)

Applicant: SEMICONDUCTOR ENERGY LABORATORY CO., LTD. (Atsugi, JP)

Family ID: 59397484

Assignee: Semiconductor Energy Laboratory Co., Ltd. (Atsugi, JP)

Appl. No.: 18/513803

Filed: November 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240088303 A1	Mar. 14, 2024

Foreign Application Priority Data

JP	2016-015730	Jan. 29, 2016
----	-------------	---------------

Related U.S. Application Data

continuation parent-doc US 17346359 20210614 US 11830950 child-doc US 18513803
continuation parent-doc US 16888892 20200601 US 11107930 20210831 child-doc US 17346359
continuation parent-doc US 16071770 US 10734529 20200804 WO PCT/IB2017/050230
20170117 child-doc US 16888892

Publication Classification

Int. Cl.: H10D30/67 (20250101); H01L21/768 (20060101); H10B12/00 (20230101); H10D64/62 (20250101); H10D84/08 (20250101); H10D84/80 (20250101); H10D84/83 (20250101); H10D86/40 (20250101); H10D86/60 (20250101); H10D87/00 (20250101); H10D88/00 (20250101); H10D99/00 (20250101); H10F39/12 (20250101); H10H29/10 (20250101)

U.S. Cl.:

CPC H10D30/6755 (20250101); H01L21/76843 (20130101); H01L21/76856 (20130101); H10B12/312 (20230201); H10D30/6713 (20250101); H10D30/6729 (20250101); H10D30/6734 (20250101); H10D30/6757 (20250101); H10D64/62 (20250101); H10D86/423 (20250101); H10D86/60 (20250101); H10D87/00 (20250101); H10D99/00 (20250101); H10F39/12 (20250101); H10H29/10 (20250101); H10D84/08 (20250101); H10D84/811 (20250101); H10D84/83 (20250101); H10D88/00 (20250101)

Field of Classification Search

CPC: H10D (30/6755); H10D (30/6713); H10D (30/6729); H10D (30/6734); H10D (30/6757); H01L (21/76843); H10H (29/10)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8054279	12/2010	Umezaki et al.	N/A	N/A
8314915	12/2011	Ota	N/A	N/A
8324626	12/2011	Kimura et al.	N/A	N/A
8400603	12/2012	Yoso et al.	N/A	N/A
8664036	12/2013	Yamazaki et al.	N/A	N/A
8743044	12/2013	Umezaki et al.	N/A	N/A
8759132	12/2013	Kimura et al.	N/A	N/A
8902145	12/2013	Umezaki et al.	N/A	N/A
8916865	12/2013	Yamazaki	257/E29.043	H01L 29/4908
9171803	12/2014	Koezuka et al.	N/A	N/A
9171867	12/2014	Kimura et al.	N/A	N/A
9246011	12/2015	Yamazaki et al.	N/A	N/A

9263539	12/2015	Yuan	N/A	N/A
9331207	12/2015	Yamazaki	N/A	H01L 29/201
9356098	12/2015	Yamazaki et al.	N/A	N/A
9391095	12/2015	Yamazaki et al.	N/A	N/A
9508592	12/2015	Koezuka et al.	N/A	N/A
9515093	12/2015	Xun et al.	N/A	N/A
9536903	12/2016	Umezaki et al.	N/A	N/A
9543449	12/2016	Kato	N/A	N/A
9583601	12/2016	Yamazaki et al.	N/A	N/A
9703409	12/2016	Ono	N/A	N/A
9748396	12/2016	Kato	N/A	N/A
9768315	12/2016	Koezuka et al.	N/A	N/A
9825057	12/2016	Yamazaki et al.	N/A	N/A
9831238	12/2016	Yamazaki et al.	N/A	N/A
9831347	12/2016	Yamazaki et al.	N/A	N/A
9842861	12/2016	Umezaki et al.	N/A	N/A
9954005	12/2017	Kimura et al.	N/A	N/A
9991392	12/2017	Koezuka et al.	N/A	N/A
10121903	12/2017	Yamazaki et al.	N/A	N/A
10128378	12/2017	Yamazaki et al.	N/A	N/A
10229906	12/2018	Yamazaki et al.	N/A	N/A
10297618	12/2018	Umezaki et al.	N/A	N/A
10734529	12/2019	Nakazawa et al.	N/A	N/A
10818795	12/2019	Yamazaki et al.	N/A	N/A
10930683	12/2020	Umezaki et al.	N/A	N/A
11967598	12/2023	Umezaki et al.	N/A	N/A
12107092	12/2023	Umezaki et al.	N/A	N/A
2007/0013078	12/2006	Lee et al.	N/A	N/A
2009/0134521	12/2008	Liu et al.	N/A	N/A
2010/0073268	12/2009	Matsunaga et al.	N/A	N/A
2011/0127524	12/2010	Yamazaki et al.	N/A	N/A
2013/0168671	12/2012	Koike et al.	N/A	N/A
2013/0207111	12/2012	Yamazaki	438/158	H01L 29/4908
2014/0001467	12/2013	Yamazaki et al.	N/A	N/A

2014/0021466	12/2013	Yamazaki et al.	N/A	N/A
2014/0291672	12/2013	Yamazaki	257/66	H01L 27/124
2015/0062451	12/2014	Ono	N/A	N/A
2015/0171116	12/2014	Okazaki et al.	N/A	N/A
2015/0200212	12/2014	Xun	257/43	H01L 29/78618
2015/0303309	12/2014	Koezuka et al.	N/A	N/A
2016/0005873	12/2015	Jintyou et al.	N/A	N/A
2017/0053940	12/2016	Xun et al.	N/A	N/A
2017/0338352	12/2016	Yamazaki	N/A	N/A
2019/0067486	12/2018	Yamazaki et al.	N/A	N/A
2020/0020810	12/2019	Yamazaki et al.	N/A	N/A
2021/0320128	12/2020	Umezaki et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101894760	12/2009	CN	N/A
103227208	12/2012	CN	N/A
103545318	12/2013	CN	N/A
104779254	12/2014	CN	N/A
106256017	12/2015	CN	N/A
1906414	12/2007	EP	N/A
3223283	12/2016	EP	N/A
2007-096055	12/2006	JP	N/A
2008-089915	12/2007	JP	N/A
2008-276211	12/2007	JP	N/A
2010-008758	12/2009	JP	N/A
2011-054951	12/2010	JP	N/A
2011-146697	12/2010	JP	N/A
2011-164369	12/2010	JP	N/A
2012-059860	12/2011	JP	N/A
2012-222171	12/2011	JP	N/A
2013-179290	12/2012	JP	N/A
2014-022492	12/2013	JP	N/A
2014-030014	12/2013	JP	N/A
2014-032999	12/2013	JP	N/A
2014-0300003	12/2013	JP	N/A
2014-197664	12/2013	JP	N/A
2015-049426	12/2014	JP	N/A
2015-130487	12/2014	JP	N/A
2015-130488	12/2014	JP	N/A
2015-133479	12/2014	JP	N/A
2015-144250	12/2014	JP	N/A
2015-213165	12/2014	JP	N/A
2016-006857	12/2015	JP	N/A
2013-0092463	12/2012	KR	N/A

2013-0131074	12/2012	KR	N/A
2014-0148305	12/2013	KR	N/A
2015-0083693	12/2014	KR	N/A
2016-0145031	12/2015	KR	N/A
201338174	12/2012	TW	N/A
201545359	12/2014	TW	N/A
WO-2008/126883	12/2007	WO	N/A
WO-2011/074409	12/2010	WO	N/A
WO-2012/002573	12/2011	WO	N/A
WO-2012/141089	12/2011	WO	N/A
WO-2014/003086	12/2013	WO	N/A
WO-2015/083034	12/2014	WO	N/A
WO-2015/083037	12/2014	WO	N/A
WO-2015/097597	12/2014	WO	N/A
WO-2015/159183	12/2014	WO	N/A

OTHER PUBLICATIONS

International Search Report (Application No. PCT/IB2017/050230) Dated May 9, 2017. cited by applicant

Written Opinion (Application No. PCT/IB2017/050230) Dated May 9, 2017. cited by applicant

Chinese Office Action (Application No. 201780006127.6) Dated Feb. 1, 2021. cited by applicant

Chinese Office Action (Application No. 202111335062.4) Dated May 30, 2022. cited by applicant

German Office Action (Application No. 112017000551.5) Dated Oct. 19, 2022. cited by applicant

Primary Examiner: Nguyen; Dao H

Attorney, Agent or Firm: Fish & Richardson P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/346,359, filed Jun. 14, 2021, now allowed, which is incorporated by reference and is a continuation of U.S. application Ser. No. 16/888,892, filed Jun. 1, 2020, now U.S. Pat. No. 11,107,930, which is incorporated by reference and is a continuation of U.S. application Ser. No. 16/071,770, filed Jul. 20, 2018, now U.S. Pat. No. 10,734,529, which is incorporated by reference and is a U.S. National Phase Application under U.S.C. § 3.71 of International Application No. PCT/IB2017/050230, filed Jan. 17, 2017, and claims the benefit of a foreign priority application filed in Japan as Application No. 2016-015730, on Jan. 29, 2016.

TECHNICAL FIELD

(1) One embodiment of the present invention relates to a semiconductor device including an oxide semiconductor film, and a display device including the semiconductor device.

(2) Note that one embodiment of the present invention is not limited to the foregoing technical field. The technical field of one embodiment of the invention disclosed in this specification and the like relates to an object, a method, or a manufacturing method. Another embodiment of the present invention relates to a process, a machine, manufacture, or a composition of matter (composition of matter). Thus, more specifically, examples of the technical field of one embodiment of the present invention disclosed in this specification include a semiconductor device, a display device, a liquid crystal display device, a light-emitting device, a lighting device, a power storage device, a memory

device, an imaging device, a driving method thereof, and a manufacturing method thereof.

(3) Note that in this specification and the like, a semiconductor device refers to every device that can function by utilizing semiconductor characteristics. A semiconductor element such as a transistor, a semiconductor circuit, an arithmetic device, and a memory device are embodiments of semiconductor devices. An imaging device, a display device, a liquid crystal display device, a light-emitting device, an electro-optical device, a power generation device (including a thin film solar cell, an organic thin film solar cell, and the like), and an electronic device may include a semiconductor device.

BACKGROUND ART

(4) Attention has been focused on a technique of forming a transistor (also referred to as a thin film transistor (TFT) or a field-effect transistor (FET)) using a semiconductor film formed over a substrate. The transistor is applied to a wide range of electronic devices such as an integrated circuit (IC) and an image device (display device). As semiconductor films applicable to the transistors, silicon-based semiconductor materials are widely known, but oxide semiconductors have been attracting attention as alternative materials.

(5) For example, a technique of fabricating a transistor using an In—Ga—Zn-based oxide as an oxide semiconductor is disclosed (see Patent Document 1, for example).

(6) Conventionally, aluminum has been widely used as a material used for a wiring, a signal line, and the like, but, development using copper (Cu) is extensively conducted to further reduce the resistance. However, copper (Cu) is disadvantageous in that adhesion thereof to a film used as a base is poor and that the characteristics of a transistor easily deteriorate due to diffusion of Cu into a semiconductor film of the transistor.

(7) Furthermore, a Cu—Mn alloy is disclosed as a material for an ohmic electrode formed over an oxide semiconductor film containing indium (see Patent Document 2, for example).

PRIOR ART DOCUMENT

Patent Document

(8) [Patent Document 1] Japanese Published Patent Application No. 2007-96055 [Patent Document 2] PCT International Publication No. 2012/002573

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(9) According to the structure disclosed in Patent Document 2, a Cu—Mn alloy film is deposited on an oxide semiconductor film, and then, heat treatment is performed on the Cu—Mn alloy film to form a Mn oxide at the joint interface between the oxide semiconductor film and the Cu—Mn alloy film. The Mn oxide is formed in such a manner that Mn in the Cu—Mn alloy film diffuses toward the oxide semiconductor film and is preferentially bonded to oxygen contained in the oxide semiconductor film. A region of the oxide semiconductor film which is reduced by Mn becomes oxygen vacancy, so that the region has a high carrier density and thus has high conductivity. Furthermore, Mn diffuses toward the oxide semiconductor film and thus the Cu—Mn alloy becomes pure Cu, whereby an ohmic electrode with low electric resistance is obtained.

(10) However, in the foregoing structure, an influence of Cu that diffuses from the ohmic electrode after the ohmic electrode is formed is not considered. For example, after an electrode including a Cu—Mn alloy film is formed over an oxide semiconductor film, heat treatment is performed, whereby a Mn oxide is formed at the joint interface between the oxide semiconductor film and the Cu—Mn alloy film. Because of formation of the Mn oxide, even if the amount of Cu which can diffuse into the oxide semiconductor film from the Cu—Mn alloy film in contact with the oxide semiconductor film can be reduced, Cu is reattached from a side surface of the Cu—Mn alloy film and a side surface or a surface of a pure Cu film obtained by release of Mn from the Cu—Mn alloy film to the surface of the oxide semiconductor film.

(11) In the case where a bottom-gate structure is used for a transistor using an oxide semiconductor film, part of a surface of the oxide semiconductor film serves as what is called a back-channel side,

and there have been the following problems when Cu is reattached to the back-channel side: the electrical characteristics (e.g., on-state current, field-effect mobility, and frequency characteristics) of the transistor deteriorate, and the transistor characteristics deteriorate in a gate BT stress test, which is a reliability test of a transistor.

(12) In view of the foregoing problems, an object of one embodiment of the present invention is to provide a novel semiconductor device using a conductive film containing copper for a transistor using an oxide semiconductor film. Another object of one embodiment of the present invention is to provide a semiconductor device including a transistor having excellent electrical characteristics (e.g., on-state current, field-effect mobility, and frequency characteristics) with the use of a conductive film containing copper for a transistor using an oxide semiconductor film. Another object of one embodiment of the present invention is to provide a semiconductor device including a transistor in which a change in the electrical characteristics is suppressed, with the use of a conductive film containing copper for a transistor using an oxide semiconductor film. Another object of one embodiment of the present invention is to provide a semiconductor device including a reliable transistor, with the use of a conductive film containing copper for a transistor using an oxide semiconductor film. Another object of one embodiment of the present invention is to provide a semiconductor device whose manufacturing cost is reduced, with the use of a conductive film containing copper for a transistor using an oxide semiconductor film. Another object of one embodiment of the present invention is to provide a semiconductor device with high productivity with the use of a conductive film containing copper for a transistor using an oxide semiconductor film. Another object of one embodiment of the present invention is to provide a novel semiconductor device. Another object of one embodiment of the present invention is to provide a method for manufacturing the novel semiconductor device.

(13) Note that the description of the aforementioned objects does not disturb the existence of other objects. Note that one embodiment of the present invention does not need to achieve all the objects. Other objects are apparent from and can be derived from the description of the specification and the like.

Means for Solving the Problems

(14) One embodiment of the present invention is a semiconductor device which includes a transistor and in which the transistor includes a gate electrode, a first insulating film over the gate electrode, an oxide semiconductor film including a region which overlaps with the gate electrode with the first insulating film therebetween, a source electrode electrically connected to the oxide semiconductor film, a drain electrode electrically connected to the oxide semiconductor film, and a second insulating film over the oxide semiconductor film, the source electrode, and the drain electrode; the source electrode and the drain electrode each contain copper; and an end portion of the source electrode and an end portion of the drain electrode each include a region containing copper and silicon.

(15) Another embodiment of the present invention is a semiconductor device which includes a transistor and in which the transistor includes a gate electrode, a first insulating film over the gate electrode, an oxide semiconductor film including a region which overlaps with the gate electrode with the first insulating film therebetween, a source electrode electrically connected to the oxide semiconductor film, a drain electrode electrically connected to the oxide semiconductor film, a second insulating film over the oxide semiconductor film, the source electrode, and the drain electrode; the source electrode and the drain electrode each contain copper; and an end portion of the source electrode and an end portion of the drain electrode each include a region including a compound containing copper and silicon.

(16) In each of the above structures, the end portion of the source electrode and the end portion of the drain electrode each preferably include a region in contact with the second insulating film.

(17) Another embodiment of the present invention is a semiconductor device which includes a transistor and in which the transistor includes a gate electrode, a first insulating film over the gate

electrode, an oxide semiconductor film including a region which overlaps with the gate electrode with the first insulating film therebetween, a source electrode electrically connected to the oxide semiconductor film, a drain electrode electrically connected to the oxide semiconductor film, a second insulating film over the oxide semiconductor film, the source electrode, and the drain electrode; the source electrode and the drain electrode each include a first conductive film, a second conductive film over and in contact with the first conductive film, and a third conductive film over and in contact with the second conductive film; the second conductive film contains copper; the first conductive film and the third conductive film include a material that inhibits diffusion of copper; and an end portion of the second conductive film includes a region containing copper and silicon.

(18) Another embodiment of the present invention is a semiconductor device which includes a transistor and in which the transistor includes a gate electrode, a first insulating film over the gate electrode, an oxide semiconductor film including a region which overlaps with the gate electrode with the first insulating film therebetween, a source electrode electrically connected to the oxide semiconductor film, a drain electrode electrically connected to the oxide semiconductor film, a second insulating film over the oxide semiconductor film, the source electrode, and the drain electrode; the source electrode and the drain electrode each include a first conductive film, a second conductive film over and in contact with the first conductive film, and a third conductive film over and in contact with the second conductive film; the second conductive film contains copper; the first conductive film and the third conductive film include a material that inhibits diffusion of copper; and an end portion of the second conductive film includes a region including a compound containing copper and silicon.

(19) In each of the above structures, the end portion of the second conductive film preferably includes a region in contact with the second insulating film.

(20) In each of the above structures, the first conductive film and the third conductive film preferably contain at least one of titanium, tungsten, tantalum, and molybdenum. Furthermore, it is preferable that the first conductive film and the third conductive film include an oxide and the oxide contain at least one of In and Zn.

(21) In each of the above structures, the oxide semiconductor film preferably contains In, Zn, and M (M represents Al, Ga, Y, or Sn). Furthermore, it is preferable that the oxide semiconductor film include a crystal part and the crystal part have c-axis alignment.

(22) Another embodiment of the present invention is a display device including the semiconductor device of any of the above embodiments and a display element. Another embodiment of the present invention is a display module including the display device of the above embodiment and a touch sensor. Another embodiment of the present invention is an electronic device including the semiconductor device of any of the above embodiments, the display device of the above embodiment or the display module of the above embodiment, and at least one of an operation key and a battery.

Effect of the Invention

(23) According to one embodiment of the present invention, a novel semiconductor device in which a conductive film containing copper is used for a transistor using an oxide semiconductor film can be provided. According to another embodiment of the present invention, a semiconductor device including a transistor having excellent electrical characteristics (e.g., on-state current, field-effect mobility, and frequency characteristics) can be provided using a conductive film containing copper for a transistor using an oxide semiconductor film. According to another embodiment of the present invention, a semiconductor device including a transistor in which a change in the electrical characteristics is suppressed can be provided using a conductive film containing copper for a transistor using an oxide semiconductor film. According to another embodiment of the present invention, a semiconductor device including a reliable transistor can be provided using a conductive film containing copper for a transistor using an oxide semiconductor film. According to

another embodiment of the present invention, a semiconductor device whose manufacturing cost is reduced can be provided using a conductive film containing copper for a transistor using an oxide semiconductor film. According to another embodiment of the present invention, a semiconductor device with high productivity can be provided using a conductive film containing copper for a transistor using an oxide semiconductor film. According to another embodiment of the present invention, a novel semiconductor device can be provided. According to another embodiment of the present invention, a method for manufacturing the novel semiconductor device can be provided. (24) Note that the description of these effects does not disturb the existence of other effects. Note that one embodiment of the present invention does not need to have all the effects. Other effects are apparent from and can be derived from the description of the specification, the drawings, the claims, and the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a top view and FIGS. 1B and 1C are cross-sectional views illustrating one embodiment of a semiconductor device.
- (2) FIG. 2A is a top view and FIGS. 2B and 2C are cross-sectional views illustrating one embodiment of a semiconductor device.
- (3) FIGS. 3A-3C are cross-sectional views illustrating one embodiment of a semiconductor device.
- (4) FIG. 4A is a top view and FIGS. 4B and 4C are cross-sectional views illustrating one embodiment of a semiconductor device.
- (5) FIGS. 5A and 5B are cross-sectional views illustrating one embodiment of a semiconductor device.
- (6) FIG. 6A is a top view and FIGS. 6B and 6C are cross-sectional views illustrating one embodiment of a semiconductor device.
- (7) FIGS. 7A and 7B are cross-sectional views illustrating one embodiment of a semiconductor device.
- (8) FIGS. 8A and 8B are cross-sectional views illustrating one embodiment of a semiconductor device.
- (9) FIGS. 9A-9C are cross-sectional views illustrating one embodiment of a semiconductor device.
- (10) FIGS. 10A-10C are cross-sectional views illustrating one embodiment of a semiconductor device.
- (11) FIGS. 11A and 11B are cross-sectional views illustrating one embodiment of a semiconductor device.
- (12) FIG. 12A is a top view and FIGS. 12B and 12C are cross-sectional views illustrating one embodiment of a semiconductor device.
- (13) FIG. 13A is a top view and FIGS. 13B and 13C are cross-sectional views illustrating one embodiment of a semiconductor device.
- (14) FIGS. 14A-14F are cross-sectional views illustrating an example of a manufacturing process of a semiconductor device.
- (15) FIGS. 15A-15F are cross-sectional views illustrating an example of a manufacturing process of a semiconductor device.
- (16) FIGS. 16A-16F are cross-sectional views illustrating an example of a manufacturing process of a semiconductor device.
- (17) FIGS. 17A-17F are cross-sectional views illustrating an example of a manufacturing process of a semiconductor device.
- (18) FIGS. 18A-18F are cross-sectional views illustrating an example of a manufacturing process of a semiconductor device.

- (19) FIGS. **19A-19D** are cross-sectional views illustrating an example of a manufacturing process of a semiconductor device.
- (20) FIGS. **20A** and **20B** are cross-sectional views illustrating an example of a manufacturing process of a semiconductor device.
- (21) FIGS. **21A-21C** are diagrams illustrating an atomic ratio range of an oxide semiconductor of one embodiment of the present invention.
- (22) FIG. **22** is a diagram illustrating a crystal of InMZnO_4 .
- (23) FIGS. **23A** and **23B** are band diagrams of a layered structure of an oxide semiconductor.
- (24) FIGS. **24A-24C** are diagrams illustrating structural analysis of a CAAC-OS and a single crystal oxide semiconductor by XRD and FIGS. **24D** and **24E** are diagrams showing a selected-area electron diffraction pattern of a CAAC-OS.
- (25) FIGS. **25A-25E** are cross-sectional TEM images and plan-view TEM images of a CAAC-OS and an image obtained through image analysis thereof.
- (26) FIGS. **26A-26D** are diagrams showing an electron diffraction pattern and cross-sectional TEM images of an nc-OS.
- (27) FIGS. **27A** and **27B** are cross-sectional TEM images of an a-like OS.
- (28) FIG. **28** is a diagram showing a change of a crystal part of an In—Ga—Zn oxide by electron irradiation.
- (29) FIG. **29** is a top view illustrating one embodiment of a display device.
- (30) FIG. **30** is a cross-sectional view illustrating one embodiment of a display device.
- (31) FIG. **31** is a cross-sectional view illustrating one embodiment of a display device.
- (32) FIG. **32** is a cross-sectional view illustrating one embodiment of a display device.
- (33) FIG. **33** is a cross-sectional view illustrating one embodiment of a display device.
- (34) FIG. **34** is a cross-sectional view illustrating one embodiment of a display device.
- (35) FIG. **35** is a cross-sectional view illustrating one embodiment of a display device.
- (36) FIGS. **36A-36C** are a block diagram and a circuit diagram illustrating a display device.
- (37) FIGS. **37A-37C** are a circuit diagram and a timing chart for explaining one embodiment of the present invention.
- (38) FIG. **38A** is a graph and FIGS. **38B** and **38C** are circuit diagrams for explaining one embodiment of the present invention.
- (39) FIGS. **39A** and **39B** are a circuit diagram and a timing chart for explaining one embodiment of the present invention.
- (40) FIGS. **40A** and **40B** are a circuit diagram and a timing chart for explaining one embodiment of the present invention.
- (41) FIG. **41A** is a block diagram, FIGS. **41B** and **41D** are circuit diagrams, and FIGS. **41C** and **41E** are waveform diagram for explaining one embodiment of the present invention.
- (42) FIGS. **42A** and **42B** are a circuit diagram and a timing chart for explaining one embodiment of the present invention.
- (43) FIGS. **43A** and **43B** are circuit diagrams for explaining one embodiment of the present invention.
- (44) FIGS. **44A-44C** are circuit diagrams for explaining one embodiment of the present invention.
- (45) FIG. **45** is a diagram illustrating a display module.
- (46) FIGS. **46A-46E** are diagrams illustrating an electronic device.
- (47) FIGS. **47A-47G** are diagrams illustrating an electronic device.
- (48) FIGS. **48A** and **48B** are perspective views illustrating a display device.
- (49) FIG. **49A** is a cross-sectional view and FIGS. **49B** and **49C** are circuit diagrams illustrating a structure of a semiconductor device relating to one embodiment of the present invention.
- (50) FIG. **50** is a block diagram illustrating a structure of a CPU relating to one embodiment of the present invention.
- (51) FIG. **51** is a circuit diagram illustrating a structure of a memory element of one embodiment of

the present invention.

(52) FIGS. 52A-52C are circuit diagrams illustrating an example of an imaging device.

(53) FIGS. 53A and 53B are diagrams illustrating a structure example of an imaging device.

(54) FIGS. 54A and 54B are diagrams illustrating a STEM photograph of a sample that relates to an example.

(55) FIG. 55 is a diagram illustrating XPS analysis results of a sample that relate to an example.

(56) FIGS. 56A and 56B are diagrams illustrating XPS analysis results of a sample that relate to an example.

(57) FIGS. 57A-57C are Id-Vg characteristics of a transistor that relate to an example.

(58) FIGS. 58A-58C are Id-Vg characteristics of a transistor that relate to an example.

(59) FIGS. 59A and 59B are diagrams illustrating a STEM photograph of a sample that relates to an example.

(60) FIG. 60 is a diagram illustrating EDX analysis results of a sample that relate to an example.

(61) Embodiments of the present invention are explained in detail below using the drawings. Note that the present invention is not limited to description below, and modes and details thereof can be variously modified without departing from the purpose and the scope of the present invention. Thus, the present invention should not be interpreted as being limited to the description of the embodiments described below.

(62) Note that the position, the size, the range, or the like of each structure illustrated in drawings and the like does not represent the actual position, size, range, or the like in some cases for easy understanding. Therefore, the disclosed invention is not necessarily limited to the position, the size, the range, or the like disclosed in the drawings and the like.

(63) The ordinal numbers such as first and second in this specification and the like are used for convenience and do not denote the order of steps or the stacking order of layers in some cases. Therefore, for example, description can be made even when first is replaced with second or third, as appropriate. In addition, the ordinal numbers in this specification and the like are not necessarily the same as those which specify one embodiment of the present invention.

(64) Note that in this specification, terms for describing arrangement, such as “over” and “under”, are used for convenience in describing a positional relation between components with reference to drawings. Furthermore, the positional relation between components is changed as appropriate in accordance with a direction in which each component is described. Thus, there is no limitation to terms described in this specification, and description can be made in other words appropriately depending on the situation.

(65) In describing structures of the invention with reference to the drawings in this specification and the like, common reference numerals are used for the same portions in different drawings.

(66) In this specification and the like, a “semiconductor” includes characteristics of an “insulator” in some cases when the conductivity is sufficiently low, for example. Further, a “semiconductor” and an “insulator” cannot be strictly distinguished from each other in some cases because a border between the “semiconductor” and the “insulator” is not clear. Accordingly, a “semiconductor” described in this specification and the like can be put into an “insulator” in some cases. Similarly, an “insulator” described in this specification and the like can be put into a “semiconductor” in some cases. Alternatively, an “insulator” described in this specification and the like can be put into a “semi-insulator” in some cases.

(67) In this specification and the like, a “semiconductor” includes characteristics of a “conductor” in some cases when the conductivity is sufficiently high, for example. Further, a “semiconductor” and a “conductor” cannot be strictly distinguished from each other in some cases because a border between the “semiconductor” and the “conductor” is not clear. Accordingly, a “semiconductor” described in this specification and the like can be put into a “conductor” in some cases. Similarly, a “conductor” described in this specification and the like can be put into a “semiconductor” in some cases.

(68) In this specification and the like, a transistor is an element having at least three terminals of a gate, a drain, and a source. In addition, the transistor has a channel region between a drain (a drain terminal, a drain region, or a drain electrode) and a source (a source terminal, a source region, or a source electrode), and current can flow through the drain, the channel region, and the source. Note that in this specification and the like, a channel region refers to a region through which current mainly flows.

(69) Furthermore, functions of a source and a drain might be switched when a transistor of different polarity is employed or a direction of current is changed in circuit operation, for example. Therefore, the terms source and drain can be switched in this specification and the like.

(70) Note that the channel length refers to, for example, a distance between a source (a source region or a source electrode) and a drain (a drain region or a drain electrode) in a region where a semiconductor (or a portion where a current flows in a semiconductor when a transistor is on) and a gate electrode overlap with each other or a region where a channel is formed in a top view of the transistor. Note that in one transistor, channel lengths in all regions are not necessarily the same. In other words, the channel length of one transistor is not limited to one value in some cases. Therefore, in this specification and the like, the channel length is any one of values, the maximum value, the minimum value, or the average value in a region where a channel is formed.

(71) A channel width refers to, for example, the length of a portion where a source and a drain face each other in a region where a semiconductor (or a portion where a current flows in a semiconductor when a transistor is on) and a gate electrode overlap with each other, or a region where a channel is formed. Note that in one transistor, channel widths in all regions do not necessarily have the same value. In other words, a channel width of one transistor is not fixed to one value in some cases. Thus, in this specification and the like, a channel width is any one of values, the maximum value, the minimum value, or the average value in a region where a channel is formed.

(72) Note that in this specification and the like, “electrically connected” includes the case of connection through an “object having any electric function”. Here, there is no particular limitation on an “object having any electric function” as long as electric signals can be transmitted and received between components that are connected. Examples of an “object having any electric function” include a switching element such as a transistor, a resistor, an inductor, a capacitor, and elements with a variety of functions as well as an electrode and a wiring.

(73) A voltage usually refers to a potential difference between a given potential and a reference potential (e.g., a ground potential (GND) or a source potential). Accordingly, a voltage can also be called a potential.

(74) Note that in this specification and the like, a silicon oxynitride film refers to a film in which the content of oxygen is higher than that of nitrogen, and preferably contains oxygen, nitrogen, silicon, and hydrogen in the ranges of 55 atomic % to 65 atomic %, 1 atomic % to 20 atomic %, 25 atomic % to 35 atomic %, and 0.1 atomic % to 10 atomic %, respectively. A silicon nitride oxide film refers to a film in which the content of nitrogen is higher than that of oxygen, and preferably contains nitrogen, oxygen, silicon, and hydrogen in the ranges of 55 atomic % to 65 atomic %, 1 atomic % to 20 atomic %, 25 atomic % to 35 atomic %, and 0.1 atomic % to 10 atomic %, respectively.

(75) In this specification and the like, the terms “film” and “layer” can be interchanged with each other. For example, the term “conductive layer” can be changed into the term “conductive film” in some cases. Alternatively, for example, the term “insulating film” can be changed into the term “insulating layer” in some cases.

(76) In this specification and the like, “parallel” refers to the state where two straight lines are arranged to form an angle of greater than or equal to -10° and less than or equal to 10° . Accordingly, “parallel” also includes the case where the angle is greater than or equal to -5° and less than or equal to 5° . In addition, “substantially parallel” refers to the state where two straight

lines are arranged to form an angle of greater than or equal to -30° and less than or equal to 30° . In addition, “perpendicular” refers to the state where two straight lines are arranged to form an angle of greater than or equal to 80° and less than or equal to 100° . Accordingly, “perpendicular” also includes the case where the angle is greater than or equal to 85° and less than or equal to 95° . In addition, “substantially perpendicular” refers to the state where two straight lines are arranged to form an angle of greater than or equal to 60° and less than or equal to 120° .

(77) In this specification and the like, a trigonal or rhombohedral crystal is indicated as a hexagonal crystal system.

Embodiment 1

(78) In this embodiment, a semiconductor device of one embodiment of the present invention and a method of manufacturing the semiconductor device are described below using FIG. 1 to FIG. 20.

(79) <1-1. Structure Example 1 of Semiconductor Device>

(80) FIG. 1(A) is a top view of a transistor **100** that is a semiconductor device of one embodiment of the present invention. FIG. 1(B) corresponds to a cross-sectional view taken along dashed-dotted line X1-X2 in FIG. 1(A), and FIG. 1(C) corresponds to a cross-sectional view taken along dashed-dotted line Y1-Y2 in FIG. 1(A). Note that some components (a substrate **102**, an insulating film, and the like) of the transistor **100** are not illustrated in FIG. 1(A) for clarification.

(81) In some cases, the direction of dashed-dotted line X1-X2 in FIG. 1(A) is referred to as the channel length direction of the transistor **100**, and the direction of dashed-dotted line Y1-Y2 in FIG. 1(A) is referred to as the channel width direction of the transistor **100**.

(82) The transistor **100** includes a conductive film **104** that functions as a gate electrode over a substrate **102**; an insulating film **106** over the substrate **102** and the conductive film **104**; an insulating film **107** over the insulating film **106**; an oxide semiconductor film **108** over the insulating film **107**; conductive films **112a** and **112b** which function as a pair of electrodes electrically connected to the oxide semiconductor film **108**; insulating films **114** and **116** over the oxide semiconductor film **108** and the conductive films **112a** and **112b**; and an insulating film **118** over the insulating film **116**.

(83) The oxide semiconductor film **108** preferably contains indium (In), zinc (Zn), and M (M represents aluminum (Al), gallium (Ga), yttrium (Y), or tin (Sn)).

(84) In the transistor **100**, the insulating films **106** and **107** have a function of a gate insulating film of the transistor **100**. In addition, in the transistor **100**, one of the conductive films **112a** and **112b** functioning as a pair of electrodes has a function of a source electrode, and the other has a function of a drain electrode.

(85) The conductive film **112a** includes a conductive film **112a_1**, a conductive film **112a_2** over and in contact with the conductive film **112a_1**, and a conductive film **112a_3** over and in contact with the conductive film **112a_2**, and the conductive film **112b** includes a conductive film **112b_1**, a conductive film **112b_2** over and in contact with the conductive film **112b_1**, and a conductive film **112b_3** over and in contact with the conductive film **112b_2**. Further, the conductive film **112a_2** includes a region **112a_2a** and a region **112a_2b**. Further, the conductive film **112b_2** includes a region **112b_2a** and a region **112b_2b**.

(86) The conductive film **112a_2**, and the conductive film **112b_2** each contain copper; the region **112a_2b** and the region **112b_2b** each contain copper and silicon; and the conductive film **112a_1**, the conductive film **112a_3**, the conductive film **112b_1**, and the conductive film **112b_3** each contain a material that inhibits diffusion of copper. The region **112a_2b** is located at the end portion of the conductive film **112a_2** and includes a region in contact with the insulating film **114**; and the region **112b_2b** is located at the end portion of the conductive film **112b_2** and includes a region in contact with the insulating film **114**. The end portion of the conductive film **112a_1** includes a region located outward from the end portion of the conductive film **112a_2**, and the end portion of the conductive film **112b_1** includes a region located outward from the end portion of the conductive film **112b_2**. The conductive film **112a_3** covers the top surface of the conductive film

112a_2, and the conductive film **112b_3** covers the top surface of the conductive film **112b_2**. Thus, the conductive film **112a_2a** is covered with the conductive film **112a_1**, the region **112a_2b**, and the conductive film **112a_3**, and the conductive film **112b_2a** is covered with the conductive film **112b_1**, the region **112b_2b**, and the conductive film **112b_3**.

(87) As the region **112a_2b** and the region **112b_2b**, copper silicide (copper silicide) is preferably formed. Copper silicide, which has a bond between copper and silicon, is more stable than copper and has a function of inhibiting copper from diffusing to the outside. In the case where the insulating film **114** contains silicon, when the regions **112a_2b** and **112b_2b** contain copper and silicon, an effect of increasing adhesion between the conductive films **112a_2** and **112b_2** and the insulating film **114** is achieved.

(88) The regions **112a_2b** and **112b_2b** may contain copper, silicon, and nitrogen, and copper silicide nitride (copper silicide nitride) may be formed. When the regions **112a_2b** and **112b_2b** contains copper silicide nitride, diffusion of copper to the outside can be inhibited. When the conductive film **112a** and the conductive film **112b** includes the conductive film **112a_2** and the conductive film **112b_2** each containing copper, respectively, the resistance of the conductive films **112a** and **112b** can be reduced. Furthermore, when the conductive films **112a** and **112b** have the foregoing structure, diffusion of a copper element to the outside of the conductive films **112a** and **112b**, in particular, to the oxide semiconductor film **108**, can be inhibited. Thus, a semiconductor device including a transistor having excellent electrical characteristics can be provided.

(89) <1-2. Structure Example 2 of Semiconductor Device>

(90) Next, structure examples different from that of the transistor **100** illustrated in FIGS. **1(A)**, **(B)**, and **(C)** are described using FIG. **2** to FIG. **11**. Note that in FIG. **2** to FIG. **11** below, a portion having a function similar to that of the transistor **100** is shown by the same hatching and not particularly denoted by a reference numeral in some cases.

(91) FIG. **2(A)** is a top view of a transistor **100A** that is a semiconductor device of one embodiment of the present invention, FIG. **2(B)** corresponds to a cross-sectional view taken along a dashed dotted line X1-X2 shown in FIG. **2(A)**, and FIG. **2(C)** corresponds to a cross-sectional view taken along a dashed dotted line Y1-Y2 in FIG. **2(A)**.

(92) The transistor **100A** includes the conductive film **104** functioning as a first gate electrode over the substrate **102**, the insulating film **106** over the substrate **102** and the conductive film **104**, the insulating film **107** over the insulating film **106**, the oxide semiconductor film **108** over the insulating film **107**, the conductive films **112a** and **112b** functioning as a pair of electrodes and electrically connected to the oxide semiconductor film **108**, the insulating film **114** and the insulating film **116** over the oxide semiconductor film **108** and the conductive films **112a** and **112b**, a conductive film **120a** provided over the insulating film **116** and electrically connected to one of the conductive films **112a** and **112b** (the conductive film **112b** in FIG. **2B**) through an opening **152c** provided in the insulating films **114** and **116**, a conductive film **120b** provided over the insulating film **116** and functioning as a second gate electrode, and an insulating film **118** over the insulating film **116** and the conductive films **120a** and **120b**.

(93) In the transistor **100A**, the insulating films **106** and **107** have a function of a first gate insulating film of the transistor **100A**, the insulating films **114** and **116** have a function of a second gate insulating film of the transistor **100A**, and the insulating film **118** has a function of a protective insulating film of the transistor **100A**. Note that in this specification and the like, in some cases, the insulating films **106** and **107** are referred to as a first gate insulating film, and the insulating films **114** and **116** are referred to as a second gate insulating film. In the transistor **100A**, one of the conductive films **112a** and **112b** functioning as a pair of electrodes has a function of a source electrode, and the other has a function of a drain electrode. The conductive film **120a** has a function of a pixel electrode used for a display device.

(94) <<S-Channel Structure>>

(95) The oxide semiconductor film **108** in the transistor **100A** illustrated in FIG. **2** is sandwiched

between the conductive films **104** and **120b** such that the first gate insulating film and the second gate insulating film are provided between the oxide semiconductor film **108** and the conductive films **104** and **120b**. The length in the channel length direction and the length in the channel width direction of the conductive film **104** are longer than the length in the channel length direction and the length in the channel width direction of the oxide semiconductor film **108**, respectively. In addition, the length in the channel length direction and the length in the channel width direction of the conductive film **120b** are longer than the length in the channel length direction and the length in the channel width direction of the oxide semiconductor film **108**, respectively. Therefore, the oxide semiconductor film **108** is entirely covered with the conductive films **104** and **120b** such that the first gate insulating film and the second gate insulating film are provided between the oxide semiconductor film **108** and the conductive films **104** and **120b**.

(96) In other words, in the channel width direction of the transistor **100A**, the conductive films **104** and **120b** surround the oxide semiconductor film **108** such that the first gate insulating film and the second gate insulating film are provided between the oxide semiconductor film **108** and the conductive films **104** and **120b**.

(97) With such a structure, the oxide semiconductor film **108** included in the transistor **100A** can be electrically surrounded by electric fields of the conductive films **104** and **120b**. A device structure of a transistor, like that of the transistor **100A**, in which electric fields of the conductive films **104** and **120b** electrically surround an oxide semiconductor film where a channel region is formed can be referred to as a surrounded channel (s-channel) structure.

(98) Since the transistor **100A** has an s-channel structure, an electric field for inducing a channel can be effectively applied to the oxide semiconductor film **108** from the conductive films **104** and **120b**. Accordingly, the current drive capability of the transistor **100A** is increased, so that high on-state current characteristics can be obtained. Since the on-state current can be high, the transistor **100A** can be miniaturized. Furthermore, since the transistor **100A** has a structure in which the oxide semiconductor film **108** is surrounded by the conductive films **104** and **120b**, the mechanical strength of the transistor **100A** can be increased.

(99) With the foregoing structure, carriers flow in a further wide range of the oxide semiconductor film **108**, that is, a region on the first gate insulating film side of the oxide semiconductor film **108** and a region on the second gate insulating film side of the oxide semiconductor film **108**.

Therefore, the amount of carriers that transfer in the transistor **100A** is increased. As a result, the on-state current of the transistor **100A** is increased, and the field-effect mobility of the transistor **100A** is also increased to, specifically, higher than or equal to 10 cm²/V·s. Note that here, the field-effect mobility is not an approximate value of the mobility as the physical property of the oxide semiconductor film but is an index of the current drive capability of the transistor in a saturation region and the apparent field-effect mobility.

(100) As in a transistor **100B** illustrated in FIGS. 3(A) and (B), the conductive films **120a** and **120b** may be provided over the insulating film **118**. In that case, the conductive film **120a** and one of the conductive films **112a** and **112b** are electrically connected through the opening **152c** provided in the insulating films **114**, **116**, and **118**. Note that the top view of the transistor **100B** is similar to that of the transistor **100A** illustrated in FIG. 2(A), FIG. 3(A) corresponds to a cross-sectional view taken along the dashed dotted line X1-X2 in FIG. 2(A), and FIG. 3(B) corresponds to a cross-sectional view taken along the dashed dotted line Y1-Y2 in FIG. 2(A). In addition, the other portions of the structure of the transistor **100B** are similar to those of the transistor **100A**; thus, the structure of the transistor **100A** can be referred to.

(101) As in the transistor **100C** illustrated in FIG. 3(C), the conductive films **112a** and **112b** may include a region where the end portion of the conductive film **112a_1** and the end portion of the conductive film **112a_2** are aligned with each other and a region where the end portion of the conductive film **112b_1** and the end portion of the conductive film **112b_2** are aligned with each other.

(102) As in a transistor **100D** illustrated in FIGS. 4(A), (B), and (C), the conductive film **120b** functioning as the second gate electrode may be connected to the conductive film **104** functioning as a first gate electrode through openings **152a** and **152b** provided in the first gate insulating film (the insulating films **106** and **107**) and the second gate insulating film (the insulating films **114**, and **116**). Note that FIG. 4(A) is a top view of the transistor **100D** that is a semiconductor device of one embodiment of the present invention. FIG. 4(B) corresponds to a cross-sectional view taken along dashed-dotted line X1-X2 in FIG. 4(A), and FIG. 4(C) corresponds to a cross-sectional view taken along dashed-dotted line Y1-Y2 in FIG. 4(A). In addition, the other portions of the structure of the transistor **100D** are similar to those of the transistor **100A**; thus, the structure of the transistor **100A** can be referred to.

(103) In the transistor **100D**, since the conductive film **104** and the conductive film **120b** are connected through the openings **152a** and **152b** provided in the first gate insulating film and the second gate insulating film, a side surface of the oxide semiconductor film **108** in the channel width direction faces the conductive film **120b** with the first gate insulating film and the second gate insulating film therebetween. In addition, the conductive films **104** and **120b** are supplied with the same potential. Accordingly, the oxide semiconductor film **108** included in the transistor **100D** can be electrically surrounded by electric fields of the conductive films **104** and **120b** effectively. Note that only one of the openings **152a** and **152b** may be provided.

(104) As in a transistor **100E** illustrated in FIGS. 5(A) and (B), the conductive films **120a** and **120b** may be provided over the insulating film **118**. In that case, the conductive film **120a** and one of the conductive films **112a** and **112b** are electrically connected through the opening **152c** provided in the insulating films **114**, **116**, and **118**, and the conductive film **120b** functioning as the second gate electrode and the conductive film **104** functioning as a first gate electrode are electrically connected through the openings **152a** and **152b** provided in the insulating films **106**, **107**, **114**, **116**, and **118**. Note that the top view of the transistor **100E** is similar to that of the transistor **100D** in FIG. 4(A), FIG. 5(A) corresponds to a cross-sectional view taken along the dashed dotted line X1-X2 in FIG. 4(A), and FIG. 5(B) corresponds to a cross-sectional view taken along the dashed dotted line Y1-Y2 in FIG. 4(A). In addition, the other portions of the structure of the transistor **100E** are similar to those of the transistor **100D**; thus, the structure of the transistor **100D** can be referred to.

(105) As in a transistor **100F** illustrated in FIGS. 6(A), (B), and (C), the conductive film **120b** functioning as the second gate electrode may be connected to the conductive film **104** functioning as a first gate electrode, through a conductive film **112c** functioning as a connecting electrode. Note that FIG. 6(A) is a top view of the transistor **100F** that is a semiconductor device of one embodiment of the present invention. FIG. 6(B) corresponds to a cross-sectional view taken along dashed-dotted line X1-X2 in FIG. 6(A), and FIG. 6(C) corresponds to a cross-sectional view taken along dashed-dotted line Y1-Y2 in FIG. 6(A).

(106) The transistor **100F** illustrated in FIGS. 6(A), (B), and (C) is different from the transistor **100A** in that the conductive film **112c** provided over the insulating film **107** and electrically connected to the conductive film **104** through an opening **151** provided in the first gate insulating film (the insulating films **106** and **107**) is included and the conductive film **120b** is provided over the insulating film **116** and electrically connected to the conductive film **112c** in the opening **152d** provided in the second gate insulating film (the insulating films **114** and **116**).

(107) In the transistor **100F**, the conductive film **104** and the conductive film **120b** are electrically connected through the conductive film **112c**, one of side surfaces in the channel width direction of the oxide semiconductor film **108** faces the conductive film **112c** with the first gate insulating film and the second gate insulating film therebetween. In addition, the conductive films **104** and **120b** are supplied with the same potential. Accordingly, the oxide semiconductor film **108** included in the transistor **100F** can be electrically surrounded by electric fields of the conductive films **104** and **120b** effectively.

(108) In contrast, for example, in the case where the conductive films **104** and **120b** are not

connected to each other as in the transistor **100A** and the transistor **100B**, it is possible to supply different potentials to the conductive films **104** and **120b**.

(109) In addition, one or both of the length in the channel length direction and the length in the channel width direction of the conductive film **120b** are not necessarily longer than the length in the channel length direction and/or the length in the channel width direction of the oxide semiconductor film **108**.

(110) The conductive film **112c** can be formed in the same process as that of the conductive films **112a** and **112b**. In that case, the conductive film **112c** includes a conductive film **112c_1**, a conductive film **112c_2** over and in contact with the conductive film **112c_1**, and a conductive film **112c_3** over and in contact with the conductive film **112c_2**. The conductive film **112c_2** includes a region **112c_2a** and a region **112c_2b**.

(111) The conductive film **112c_2** contains copper, the region **112c_2b** contains copper and silicon, and the conductive films **112c_1** and **112c_3** each include a material that inhibits diffusion of copper. The region **112c_2b** is located at the end portion of the conductive film **112c_2** and includes a region in contact with the insulating film **114**. The end portion of the conductive film **112c_1** includes a region located outward from the end portion of the conductive film **112c_2**. The conductive film **112c_3** covers the top surface of the conductive film **112c_2**. Accordingly, the conductive film **112c_2a** is covered with the conductive film **112c_1**, the region **112c_2b**, and the conductive film **112c_3**.

(112) The region **112c_2b** can be formed with the same materials and in the same process as those of the regions **112a_2b** and **112b_2b**.

(113) With the foregoing structure, the resistance of the conductive film **112c** can be reduced. Furthermore, diffusion of a copper element to the outside of the conductive film **112c**, in particular, to the oxide semiconductor film **108**, can be inhibited.

(114) Note that the other portions of the structure of the transistor **100F** are similar to those of the transistor **100A**; thus, the structure of the transistor **100A** can be referred to.

(115) <1-3. Structure Example 3 of Semiconductor Device>

(116) FIGS. 7(A) and (B) are cross-sectional views of a transistor **100G** that is a semiconductor device of one embodiment of the present invention, the top view of the transistor **100G** is similar to that of the transistor **100** in FIG. 1(A), FIG. 7(A) corresponds to a cross-sectional view taken along dashed-dotted line X1-X2 in FIG. 1(A), and FIG. 7(B) corresponds to a cross-sectional view taken along dashed-dotted line Y1-Y2 in FIG. 1(A).

(117) The transistor **100G** is different from the transistor **100** in that the oxide semiconductor film **108** includes an oxide semiconductor film **108a** on the conductive film **104** side, an oxide semiconductor film **108b** over the oxide semiconductor film **108a**, and an oxide semiconductor film **108c** over the oxide semiconductor film **108b**. That is, the oxide semiconductor film **108** has a three-layer structure. The other portions of the structure of the transistor **100G** are similar to those of the transistor **100** and have similar effects. The portions different from those in the transistor **100** are described below.

(118) The oxide semiconductor films **108a**, **108b**, and **108c** each contain In, Zn, and M (M is Al, Ga, Y, or Sn).

(119) For example, the oxide semiconductor film **108b** preferably includes a region where the atomic proportion of In is larger than the atomic proportion of M. Furthermore, the oxide semiconductor films **108a** and **108c** preferably include a region where the atomic proportion of In is smaller than that in the oxide semiconductor film **108b**.

(120) When the oxide semiconductor film **108b** includes the region where the atomic proportion of In is larger than the atomic proportion of M, the field-effect mobility of the transistor **100G** can be increased. Specifically, the field-effect mobility of the transistor **100G** can exceed $10 \text{ cm}^2/\text{Vs}$, more preferably, the field-effect mobility of the transistor **100G** can exceed $30 \text{ cm}^2/\text{Vs}$.

(121) For example, when the transistor with high field-effect mobility is used for a gate driver that

generates a gate signal (in particular, a demultiplexer connected to an output terminal of a shift register included in a gate driver), a semiconductor device or a display device whose frame is narrow (also referred to as a narrow frame) can be provided.

(122) On the other hand, when the oxide semiconductor film **108b** includes the region where the atomic proportion of In is larger than the atomic proportion of M, the electrical characteristics of the transistor **100G** are easily changed in light irradiation. However, in the semiconductor device of one embodiment of the present invention, the oxide semiconductor film **108c** is formed over the oxide semiconductor film **108b**. The oxide semiconductor film **108c** includes the region where the atomic proportion of In is smaller than that in the oxide semiconductor film **108b** and thus has larger E_g than the oxide semiconductor film **108b**. For this reason, the oxide semiconductor film **108** that is a layered structure of the oxide semiconductor film **108b** and the oxide semiconductor film **108c** can have increased resistance to a negative bias stress test with light.

(123) Impurities such as hydrogen or moisture entering the oxide semiconductor film **108**, in particular, a channel region of the oxide semiconductor film **108b**, affect the transistor characteristics and thus cause a problem. Therefore, it is preferable that the amount of impurities such as hydrogen or moisture in the channel region of the oxide semiconductor film **108b** be as small as possible. Furthermore, oxygen vacancies formed in the channel region of the oxide semiconductor film **108b** affect the transistor characteristics and thus cause a problem. For example, when oxygen vacancies are formed in the channel region of the oxide semiconductor film **108b**, the oxygen vacancies are bonded to hydrogen to serve as a carrier supply source. The carrier supply source generated in the channel region of the oxide semiconductor film **108b** causes a change in the electrical characteristics, typically, a shift in the threshold voltage, of the transistor **100G** including the oxide semiconductor film **108b**. Therefore, it is preferable that the amount of oxygen vacancies in the channel region of the oxide semiconductor film **108b** be as small as possible.

(124) Thus, in one embodiment of the present invention, insulating films in contact with the oxide semiconductor film **108**, specifically, the insulating films **114** and **116** formed above the oxide semiconductor film **108**, contain excess oxygen. Oxygen or excess oxygen is transferred from the insulating films **114** and **116** to the oxide semiconductor film **108**, whereby the oxygen vacancies in the oxide semiconductor film can be reduced.

(125) As in a transistor **100H** illustrated in FIGS. **8(A)** and **(B)**, the oxide semiconductor film **108** may have a two-layer structure with the oxide semiconductor film **108b** and the oxide semiconductor film **108c**. Note that a top view of the transistor **100H** is similar to that of the transistor **100** illustrated in FIG. **1(A)**, FIG. **8(A)** corresponds to a cross-sectional view taken along the dashed dotted line X1-X2 in FIG. **1(A)**, and FIG. **8(B)** corresponds to a cross-sectional view taken along the dashed dotted line Y1-Y2 in FIG. **1(A)**. The other portions of the structure of the transistor **100H** are similar to those of the transistor **100G**; thus, the structure of the transistor **100G** can be referred to.

(126) As in a transistor **100J** illustrated in FIGS. **9(A)** and **(B)**, the conductive film **120b** functioning as the second gate electrode may be included, and the oxide semiconductor film **108** may include the oxide semiconductor film **108b** and the oxide semiconductor film **108c**. Note that a top view of the transistor **100J** is similar to that of the transistor **100A** illustrated in FIG. **2(A)**, FIG. **9(A)** corresponds to a cross-sectional view taken along the dashed dotted line X1-X2 in FIG. **2(A)**, and FIG. **9(B)** corresponds to a cross-sectional view taken along the dashed dotted line Y1-Y2 in FIG. **2(A)**. The other portions of the structure of the transistor **100J** are similar to those of the transistor **100A**; thus, the structure of the transistor **100A** can be referred to.

(127) As in a transistor **100K** illustrated in FIG. **9(C)**, the conductive films **112a** and **112b** may include a region where the end portion of the conductive film **112a_1** and the end portion of the conductive film **112a_2** are aligned with each other and a region where the end portion of the conductive film **112b_1** and the end portion of the conductive film **112b_2** are aligned with each

other.

(128) As in a transistor **100L** illustrated in FIGS. **10(A)** and **(B)**, the conductive films **120a** and **120b** may be provided over the insulating film **118**. As in the transistor **100M** illustrated in FIG. **10(C)**, the conductive films **112a** and **112b** may include a region where the end portion of the conductive film **112a_1** and the end portion of the conductive film **112a_2** are aligned with each other and a region where the end portion of the conductive film **112b_1** and the end portion of the conductive film **112b_2** are aligned with each other.

(129) As in the transistors **100J**, **100K**, **100L**, and **100M**, with an s-channel structure, carriers flow in a further wide range of the oxide semiconductor film **108**, that is, a region on the first gate insulating film side of the oxide semiconductor film **108b** and a region on the second gate insulating film side of the oxide semiconductor film **108b**. Therefore, the amount of carriers that transfer in these transistors is increased. As a result, the on-state current and the field-effect mobility of the transistors are increased.

(130) The drawings illustrate an example in which the oxide semiconductor film **108** in the transistor **100** is reduced in thickness in a region which is not covered with the conductive films **112a** and **112b**, in other words, part of the oxide semiconductor film has a depressed portion. However, one embodiment of the present invention is not limited thereto, and the oxide semiconductor film in a region which is not covered with the conductive films **112a** and **112b** is not necessarily reduced in thickness and does not necessarily have a depressed region. FIGS. **11(A)** and **(B)** illustrate an example of this case. FIGS. **11(A)** and **(B)** are cross-sectional views illustrating an example of the semiconductor device. Note that FIGS. **11(A)** and **(B)** are cross-sectional views of the transistor **100N** having a structure in which the oxide semiconductor film **108** of the aforementioned transistor **100** does not have a depressed portion.

(131) <1-4. Structure Example 4 of Semiconductor Device>

(132) FIG. **12(A)** is a top view of a transistor **100P**, FIG. **12(B)** corresponds to a cross-sectional view taken along dashed-dotted line X1-X2 in FIG. **12(A)**, and FIG. **12(C)** corresponds to a cross-sectional view taken along dashed-dotted line Y1-Y2 in FIG. **12(A)**.

(133) The transistor **100P** illustrated in FIG. **12** includes the conductive film **104** over the substrate **102**, the insulating film **106** over the substrate **102** and the conductive film **104**, the insulating film **107** over the insulating film **106**, the oxide semiconductor film **108** over the insulating film **107**, the insulating film **114** over the oxide semiconductor film **108**, the insulating film **116** over the insulating film **114**, the conductive film **112a** electrically connected to the oxide semiconductor film **108** through an opening **151a** provided in the insulating films **114** and **116**, and the conductive film **112b** electrically connected to the oxide semiconductor film **108** through an opening **151b** provided in the insulating films **114** and **116**. In addition, the insulating film **118** is provided over the transistor **100P**, more specifically, over the conductive films **112a** and **112b** and the insulating film **116**.

(134) Note that in the transistor **100P**, the insulating films **106** and **107** have a function of a gate insulating film of the transistor **100P**, the insulating films **114** and **116** have a function of a protective insulating film of the oxide semiconductor film **108**, and the insulating film **118** has a function of a protective insulating film of the transistor **100P**. Furthermore, in the transistor **100P**, the conductive film **104** has a function of a gate electrode, the conductive film **112a** has a function of a source electrode, and the conductive film **112b** has a function of a drain electrode.

(135) Although the transistor **100** illustrated in FIG. **1** has a channel-etched structure, the transistor **100P** illustrated in FIGS. **12(A)**, **(B)**, and **(C)** has a channel-protective structure. A channel-protective transistor can also be suitably used as a semiconductor device of one embodiment of the present invention. The other portions of the structure of the transistor **100P** are similar to those of the transistor **100**; thus, the structure of the transistor **100** can be referred to.

(136) FIG. **13(A)** is a top view of a transistor **100Q**, FIG. **13(B)** corresponds to a cross-sectional view taken along dashed-dotted line X1-X2 in FIG. **13(A)**, and FIG. **13(C)** corresponds to a cross-

sectional view taken along dashed-dotted line Y1-Y2 in FIG. 13(A).

(137) The transistor **100Q** illustrated in FIG. 13 is different from the transistor **100P** illustrated in FIGS. 12(A), (B), and (C) in the shapes of the insulating films **114** and **116**. Specifically, the insulating films **114** and **116** of the transistor **100Q** are provided to have an island shape over a channel region of the oxide semiconductor film **108**. The other portions of the structure are similar to those of the transistor **100P**.

(138) The above structures of the transistors relating to this embodiment can be freely combined with each other.

(139) <1-5. Components of Semiconductor Device>

(140) Components included in the semiconductor device of this embodiment are described in detail below.

(141) <<Substrate>>

(142) There is no particular limitation on the property of a material and the like of the substrate **102** as long as it has heat resistance enough to withstand at least heat treatment to be performed later. For example, a glass substrate, a ceramic substrate, a quartz substrate, or a sapphire substrate may be used as the substrate **102**. Alternatively, a single crystal semiconductor substrate or a polycrystalline semiconductor substrate made of silicon or silicon carbide, a compound semiconductor substrate of silicon germanium or the like, an SOI substrate, or the like can be used, or these substrates over which a semiconductor element is provided may be used as the substrate **102**.

(143) Note that in the case where a glass substrate is used as the substrate **102**, a large-area substrate of the 6th generation (1500 mm×1850 mm), the 7th generation (1870 mm×2200 mm), the 8th generation (2200 mm×2400 mm), the 9th generation (2400 mm×2800 mm), the 10th generation (2950 mm×3400 mm), or the like is used, whereby a large-sized display device can be manufactured.

(144) Alternatively, a flexible substrate may be used as the substrate **102**, and the transistor **100** may be formed directly on the flexible substrate. Alternatively, a separation layer may be provided between the substrate **102** and the transistor **100**. The separation layer can be used when part or the whole of a semiconductor device formed thereover is separated from the substrate **102** and transferred onto another substrate. In such a case, the transistor **100** can be transferred to a substrate having low heat resistance or a flexible substrate as well.

(145) <<Conductive Film>>

(146) The conductive film **104** functioning as the first gate electrode, the conductive film **112a** functioning as the source electrode, the conductive film **112b** functioning as the drain electrode, the conductive film **112c** functioning as the connecting electrode, the conductive film **120b** functioning as the second gate electrode, and the conductive film **120a** functioning as the pixel electrode can each be formed using a metal element selected from chromium (Cr), copper (Cu), aluminum (Al), gold (Au), silver (Ag), zinc (Zn), molybdenum (Mo), tantalum (Ta), titanium (Ti), tungsten (W), manganese (Mn), nickel (Ni), iron (Fe), and cobalt (Co); an alloy containing the metal elements as its components; an alloy containing a combination of the metal elements; or the like.

(147) Furthermore, Cu is preferably used for the conductive films **104**, **112a**, **112b**, **112c**, **120a**, and **120b**. A Cu—X alloy film (X is Mn, Ni, Cr, Fe, Co, Mo, Ta, or Ti) may be used. The use of a Cu—X alloy film results in lower manufacturing cost because the film can be processed by a wet etching process.

(148) Note that Cu or the Cu—X alloy film can be suitably used for the conductive film **112a_2** of the conductive film **112a**, the conductive film **112b_2** of the conductive film **112b**, and the conductive film **112c_2** of the conductive film **112c**. A Cu—Mn alloy film is particularly preferable as the Cu—X alloy film. Note that one embodiment of the present invention is not limited thereto as long as the conductive films **112a_2**, **112b_2**, and **112c_2** contain at least copper.

(149) The regions **112a_2b**, **112b_2b**, and **112c_2b** preferably contain Cu and Si, and preferably

contain copper silicide. When the regions **112a_2b**, **112b_2b**, and **112c_2b** contain copper silicide nitride, diffusion of copper to the outside can be inhibited. Copper silicide is formed by depositing Cu or a Cu-containing alloy and then making it react with, for example, a silane gas. After the reaction with the silane gas, reaction with plasma containing, for example, nitrogen may be performed to form copper silicide nitride containing Cu, Si, and N. Note that in the case where a surface of copper or a copper-containing alloy is covered with an oxide film, reduction treatment for removing the oxide film is preferably performed using hydrogen, ammonia, or the like before the reaction.

(150) The conductive films **112a_1** and **112a_3** of the conductive film **112a**, the conductive films **112b_1** and **112b_3** of the conductive film **112b**, and the conductive films **112c_1** and **112c_3** of the conductive film **112c** preferably contain any one or more selected from, in particular, titanium, tungsten, tantalum, and molybdenum among the above metal elements. When the conductive films **112a_1**, **112a_3**, **112b_1**, **112b_3**, **112c_1**, and **112c_3** contain any one or more selected from titanium, tungsten, tantalum, and molybdenum, diffusion of copper contained in the conductive films **112a_2**, **112b_2**, and **112c_2**, to the outside can be inhibited. That is, the conductive films **112a_1**, **112a_3**, **112b_1**, **112b_3**, **112c_1**, and **112c_3** have a function of what is called a barrier metal.

(151) For the conductive films **112a_1**, **112a_3**, **112b_1**, **112b_3**, **112c_1**, and **112c_3**, what is called a tantalum nitride film, which contains nitrogen and tantalum, is suitably used. The tantalum nitride film has conductivity and a high barrier property against copper or hydrogen. Furthermore, the tantalum nitride film releases little hydrogen from itself; thus, it can be most suitably used as the conductive film in contact with the oxide semiconductor film **108**.

(152) Oxide conductors such as an oxide containing indium and tin, an oxide containing tungsten and indium, an oxide containing tungsten, indium, and zinc, an oxide containing titanium and indium, an oxide containing titanium, indium, and tin, an oxide containing indium and zinc, an oxide containing silicon, indium, and tin, and an oxide containing indium, gallium, and zinc may be used for the conductive films **104**, **112a_1**, **112a_3**, **112b_1**, **112b_3**, **112c_1**, **112c_3**, **120a**, and **120b**. When the conductive films **112a_1**, **112a_3**, **112b_1**, **112b_3**, **112c_1**, and **112c_3** include an oxide containing at least one of In and Zn, diffusion of copper contained in the conductive films **112a_2**, **112b_2**, and **112c_2**, to the outside can be inhibited.

(153) In particular, the oxide conductors can be suitably used for the conductive film **120a**. The conductive film **120a** and the oxide semiconductor film **108** (the oxide semiconductor films **108b** and **108c**) favorably include the same metal element. With the structure, the manufacturing cost can be reduced.

(154) Here, an oxide conductor is described. In this specification and the like, an oxide conductor may be referred to as OC (Oxide Conductor). As for an oxide conductor, for example, oxygen vacancies are formed in an oxide semiconductor and hydrogen is added to the oxygen vacancies to form a donor level in the vicinity of the conduction band. As a result, the oxide semiconductor has increased conductivity to be a conductor. The oxide semiconductor having become a conductor can be referred to as an oxide conductor. An oxide semiconductor generally has a visible light transmitting property because of its large energy gap. Meanwhile, an oxide conductor is an oxide semiconductor having a donor level in the vicinity of the conduction band. Therefore, the influence of absorption due to the donor level is small in an oxide conductor, and it has a visible light transmitting property comparable to that of an oxide semiconductor.

(155) <<Insulating Films Functioning as First Gate Insulating Film>>

(156) As the insulating films **106** and **107** functioning as the first gate insulating film of the transistor **100**, an insulating layer including at least one of the following films formed by a plasma enhanced chemical vapor deposition (PECVD: (Plasma Enhanced Chemical Vapor Deposition)) method, a sputtering method, or the like can be used: a silicon oxide film, a silicon oxynitride film, a silicon nitride oxide film, a silicon nitride film, an aluminum oxide film, a hafnium oxide film, an

yttrium oxide film, a zirconium oxide film, a gallium oxide film, a tantalum oxide film, a magnesium oxide film, a lanthanum oxide film, a cerium oxide film, and a neodymium oxide film. Note that instead of the layered structure of the insulating films **106** and **107**, an insulating film of a single layer selected from the materials or an insulating film of three or more layers may be used. (157) Furthermore, the insulating film **106** has a function of a blocking film that inhibits penetration of oxygen. For example, in the case where excess oxygen is supplied to the insulating films **107**, **114**, **116**, and/or the oxide semiconductor film **108**, the insulating film **106** can inhibit penetration of oxygen.

(158) Note that the insulating film **107** that is in contact with the oxide semiconductor film **108** functioning as a channel region of the transistor **100** is preferably an oxide insulating film and more preferably includes a region containing oxygen whose amount is in excess of that satisfying the stoichiometric composition (oxygen-excess region). In other words, the insulating film **107** is an insulating film capable of releasing oxygen. Note that in order to provide the oxygen-excess region in the insulating film **107**, the insulating film **107** is formed in an oxygen atmosphere, for example. Alternatively, the formed insulating film **107** is subjected to heat treatment in an oxygen atmosphere.

(159) Furthermore, in the case where hafnium oxide is used for the insulating film **107**, the following effect is attained. Hafnium oxide has a higher dielectric constant than silicon oxide and silicon oxynitride. Therefore, the insulating film **107** using hafnium oxide can have a larger thickness than the insulating film **107** using silicon oxide; thus, leakage current due to tunnel current can be low. That is, a transistor with low off-state current can be obtained. Moreover, hafnium oxide with a crystal structure has a higher dielectric constant than hafnium oxide with an amorphous structure. Therefore, it is preferable to use hafnium oxide with a crystal structure in order to obtain a transistor with low off-state current. Examples of the crystal structure include a monoclinic crystal structure and a cubic crystal structure. Note that one embodiment of the present invention is not limited thereto.

(160) In this embodiment, a silicon nitride film is formed as the insulating film **106**, and a silicon oxide film is formed as the insulating film **107**. The silicon nitride film has a higher dielectric constant than a silicon oxide film and needs a larger thickness for capacitance equivalent to that of the silicon oxide film. Thus, when the silicon nitride film is included in the gate insulating film of the transistor **100**, the thickness of the insulating film can be increased. This makes it possible to reduce a decrease in withstand voltage of the transistor **100** and furthermore to increase the withstand voltage, thereby reducing electrostatic discharge damage to the transistor **100**.

(161) <<Oxide Semiconductor Film>>

(162) As the oxide semiconductor film **108**, the materials described above can be used.

(163) In the case where the oxide semiconductor film **108b** is In-M-Zn oxide, it is preferable that the atomic ratio of metal elements of a sputtering target used for depositing the In-M-Zn oxide satisfy $\text{In} > \text{M}$. The atomic ratio of metal elements of such a sputtering target is, for example, $\text{In}:\text{M}:\text{Zn}=2:1:3$, $\text{In}:\text{M}:\text{Zn}=3:1:2$, or $\text{In}:\text{M}:\text{Zn}=4:2:4.1$.

(164) In the case where the oxide semiconductor films **108a** and **108c** are In-M-Zn oxide, it is preferable that the atomic ratio of metal elements of a sputtering target used for depositing the In-M-Zn oxide satisfy $\text{In} \geq \text{M}$. The atomic ratio of metal elements of such a sputtering target is, for example, $\text{In}:\text{M}:\text{Zn}=1:1:1$, $\text{In}:\text{M}:\text{Zn}=1:1:1.2$, $\text{In}:\text{M}:\text{Zn}=1:3:2$, $\text{In}:\text{M}:\text{Zn}=1:3:4$, or $\text{In}:\text{M}:\text{Zn}=1:3:6$.

(165) In the case where the oxide semiconductor film **108** (**108a**, **108b**, and **108c**) are each In-M-Zn oxide, it is preferable to use a target including polycrystalline In-M-Zn oxide as the sputtering target. The use of the target including polycrystalline In-M-Zn oxide facilitates formation of the oxide semiconductor films **108a**, **108b**, and **108c** having crystallinity. Note that the atomic ratio in each of the oxide semiconductor films **108a**, **108b**, and **108c** which are formed varies from the above atomic ratio of metal elements contained in the sputtering target within a range of $\pm 40\%$. For example, when a sputtering target of the oxide semiconductor film **108b** with an atomic ratio of

In:Ga:Zn=4:2:4.1 is used, the atomic ratio of the formed oxide semiconductor film **108b** may be In:Ga:Zn=4:2:3 or in the neighborhood thereof.

(166) The energy gap of the oxide semiconductor film **108** is 2 eV or more, preferably 2.5 eV or more, more preferably 3 eV or more. With the use of an oxide semiconductor having such a wide energy gap, the off-state current of the transistor **100** can be reduced. In particular, an oxide semiconductor film having an energy gap of 2 eV or more, preferably 2 eV or more and 3.0 eV or less, is suitably used as the oxide semiconductor film **108b**, and an oxide semiconductor film having an energy gap of 2.5 eV or more and 3.5 eV or less is suitably used as the oxide semiconductor films **108a** and **108c**. Furthermore, the oxide semiconductor films **108a** and **108c** preferably have a higher energy gap than the oxide semiconductor film **108b**.

(167) The thickness of each of the oxide semiconductor films **108a**, **108b**, and **108c** is more than or equal to 3 nm and less than or equal to 200 nm, preferably more than or equal to 3 nm and less than or equal to 100 nm, and more preferably more than or equal to 3 nm and less than or equal to 50 nm.

(168) An oxide semiconductor film with low carrier density is used as the oxide semiconductor films **108a** and **108c**. For example, the carrier density of the oxide semiconductor films **108a** and **108c** is lower than or equal to $1 \times 10^{17} \text{ cm}^{-3}$, preferably lower than or equal to $1 \times 10^{15} \text{ cm}^{-3}$, more preferably lower than or equal to $1 \times 10^{13} \text{ cm}^{-3}$, and still more preferably lower than or equal to $1 \times 10^{11} \text{ cm}^{-3}$.

(169) Note that without limitation to the above, a material with an appropriate composition can be used depending on required semiconductor characteristics and electrical characteristics (e.g., field-effect mobility and threshold voltage) of a transistor. Furthermore, in order to obtain required semiconductor characteristics of a transistor, it is preferable that the carrier density, the impurity concentration, the defect density, the atomic ratio between a metal element and oxygen, the interatomic distance, the density, and the like of the oxide semiconductor film **108** be set appropriately.

(170) Note that it is preferable to use, as the oxide semiconductor film **108**, an oxide semiconductor film in which the impurity concentration is low and the density of defect states is low, in which case the transistor having more excellent electrical characteristics can be manufactured. Here, the state in which impurity concentration is low and the density of defect states is low (the amount of oxygen vacancies is small) is referred to as highly purified intrinsic or substantially highly purified intrinsic. A highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor film has few carrier generation sources; thus, the carrier density can be low. Thus, a transistor in which a channel region is formed in the oxide semiconductor film rarely has electrical characteristics in which the threshold voltage is negative (also referred to as normally on). Furthermore, the highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor film has a low density of defect states and accordingly has a low density of trap states in some cases. Furthermore, the highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor film has an extremely low off-state current; an element with a channel width of $1 \times 10^6 \mu\text{m}$ and a channel length L of $10 \mu\text{m}$ can have a characteristic that the off-state current is less than or equal to the measurement limit of a semiconductor parameter analyzer, i.e., less than or equal to $1 \times 10^{-13} \text{ A}$, at voltage (drain voltage) between a source electrode and a drain electrode of from 1 V to 10 V.

(171) Accordingly, the transistor in which the channel region is formed in the highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor film can be a transistor in which a variation in electrical characteristics is small and reliability is high. Note that charge trapped by the trap states in the oxide semiconductor film takes a long time to disappear and may behave like fixed charge. Thus, the transistor whose channel region is formed in the oxide semiconductor film having a high density of trap states has unstable electrical characteristics in some cases. As the impurity, hydrogen, nitrogen, alkali metal, alkaline earth metal, or the like is

given.

(172) Hydrogen contained in the oxide semiconductor film reacts with oxygen bonded to a metal atom to be water, and an oxygen vacancy is formed in a lattice from which oxygen is released (or a portion from which oxygen is released). Entry of hydrogen into the oxygen vacancy generates an electron that is a carrier in some cases. Furthermore, in some cases, part of hydrogen is bonded to oxygen bonded to a metal atom to generate an electron that is a carrier. Thus, a transistor using an oxide semiconductor film that contains hydrogen is likely to have normally-on characteristics.

Accordingly, it is preferable that hydrogen be reduced as much as possible in the oxide semiconductor film **108**. Specifically, in the oxide semiconductor film **108**, the hydrogen concentration measured by SIMS analysis is lower than or equal to 2×10^{20} atoms/cm³, preferably lower than or equal to 5×10^{19} atoms/cm³, more preferably lower than or equal to 1×10^{19} atoms/cm³, more preferably lower than or equal to 5×10^{18} atoms/cm³, preferably lower than or equal to 1×10^{18} atoms/cm³, more preferably lower than or equal to 5×10^{17} atoms/cm³, and still more preferably lower than or equal to 1×10^{16} atoms/cm³.

(173) The oxide semiconductor film **108b** preferably includes a region where the hydrogen concentration is lower than that in the oxide semiconductor film **108c**. When the oxide semiconductor film **108b** includes a region where the hydrogen concentration is lower than that in the oxide semiconductor film **108c**, a semiconductor device can be reliable.

(174) When silicon or carbon that is one of Group 14 elements is contained in the oxide semiconductor film **108b**, oxygen vacancies are increased in the oxide semiconductor film **108b**, and it becomes n-type. Thus, the concentration of silicon or carbon in the oxide semiconductor film **108b** or the concentration of silicon or carbon (the concentration measured by SIMS analysis) in the vicinity of an interface with the oxide semiconductor film **108b** is set to be lower than or equal to 2×10^{18} atoms/cm³, preferably lower than or equal to 2×10^{17} atoms/cm³.

(175) In the oxide semiconductor film **108b**, the concentration of alkali metal or alkaline earth metal that is measured by SIMS analysis is lower than or equal to 1×10^{18} atoms/cm³, preferably lower than or equal to 2×10^{16} atoms/cm³. Alkali metal and alkaline earth metal might generate carriers when bonded to an oxide semiconductor, in which case the off-state current of the transistor might be increased. Therefore, it is preferable to reduce the concentration of alkali metal or alkaline earth metal in the oxide semiconductor film **108b**.

(176) When nitrogen is contained in the oxide semiconductor film **108b**, it easily becomes n-type because electrons that are carriers are generated to increase the carrier density. As a result, a transistor using an oxide semiconductor film that contains nitrogen is likely to have normally-on characteristics. Thus, in the oxide semiconductor film, the amount of nitrogen is preferably reduced as much as possible; for example, the nitrogen concentration measured by SIMS analysis is preferably lower than or equal to 5×10^{18} atoms/cm³.

(177) The oxide semiconductor film **108b** and the oxide semiconductor film **108c** may each have a non-single-crystal structure. Examples of the non-single-crystal structure include a CAAC-OS (C Axis Aligned Crystalline Oxide Semiconductor) described later, a polycrystalline structure, a microcrystalline structure, and an amorphous structure. Among the non-single crystal structures, the amorphous structure has the highest density of defect states, whereas the CAAC-OS has the lowest density of defect states.

(178) <<Insulating Films Functioning as Second Gate Insulating Film>>

(179) The insulating films **114** and **116** function as a second gate insulating film of the transistor **100**. Furthermore, the insulating films **114** and **116** have a function of supplying oxygen to the oxide semiconductor film **108**. That is, the insulating films **114** and **116** contain oxygen.

Furthermore, the insulating film **114** is an insulating film that is permeable to oxygen. Note that the insulating film **114** also functions as a film that relieves damage to the oxide semiconductor film **108** at the time of forming the insulating film **116** later.

(180) Silicon oxide, silicon oxynitride, or the like with a thickness greater than or equal to 5 nm and less than or equal to 150 nm, preferably greater than or equal to 5 nm and less than or equal to 50 nm, can be used for the insulating film **114**.

(181) It is preferable that the number of defects in the insulating film **114** be small and typically, the spin density of a signal that appears at $g=2.001$ due to a dangling bond of silicon be lower than or equal to 3×10^{17} spins/cm³ by ESR measurement. This is because when the density of defects in the insulating film **114** is high, oxygen is bonded to the defects and the amount of oxygen that penetrates the insulating film **114** is decreased.

(182) Note that in the insulating film **114**, not all oxygen entering the insulating film **114** from the outside move to the outside of the insulating film **114** and some oxygen remains in the insulating film **114**. Furthermore, movement of oxygen occurs in the insulating film **114** in some cases in such a manner that oxygen enters the insulating film **114** and oxygen contained in the insulating film **114** moves to the outside of the insulating film **114**. When an oxide insulating film that is permeable to oxygen is formed as the insulating film **114**, oxygen released from the insulating film **116** provided over the insulating film **114** can be moved to the oxide semiconductor film **108** through the insulating film **114**.

(183) The insulating film **114** can be formed using an oxide insulating film having a low density of states due to nitrogen oxide. Note that the density of states due to the nitrogen oxide can be formed between the energy of the valence band maximum (E_{v_os}) of the oxide semiconductor film and the energy of the conduction band minimum (E_{c_os}) of the oxide semiconductor film. A silicon oxynitride film that releases a small amount of nitrogen oxide, an aluminum oxynitride film that releases a small amount of nitrogen oxide, or the like can be used as the above oxide insulating film.

(184) Note that a silicon oxynitride film that releases a small amount of nitrogen oxide is a film of which the amount of released ammonia is larger than the amount of released nitrogen oxide in thermal desorption spectroscopy analysis; the amount of released ammonia is typically greater than or equal to 1×10^{18} molecules/cm³ and less than or equal to 5×10^{19} molecules/cm³. Note that the amount of released ammonia corresponds to the released amount by heat treatment with which the surface temperature of the film becomes higher than or equal to 50° C. and lower than or equal to 650° C., preferably higher than or equal to 50° C. and lower than or equal to 550° C.

(185) Nitrogen oxide ($\text{NO}_{\text{sub}.x}$; x is greater than 0 and less than or equal to 2, preferably greater than or equal to 1 and less than or equal to 2), typically $\text{NO}_{\text{sub}.2}$ or NO, forms levels in the insulating film **114**, for example. The level is positioned in the energy gap of the oxide semiconductor film **108**. Therefore, when nitrogen oxide diffuses to the interface between the insulating film **114** and the oxide semiconductor film **108**, an electron is in some cases trapped by the level on the insulating film **114** side. As a result, the trapped electron remains in the vicinity of the interface between the insulating film **114** and the oxide semiconductor film **108**; thus, the threshold voltage of the transistor is shifted in the positive direction.

(186) Furthermore, nitrogen oxide reacts with ammonia and oxygen in heat treatment. Since nitrogen oxide contained in the insulating film **114** reacts with ammonia contained in the insulating film **116** in heat treatment, nitrogen oxide contained in the insulating film **114** is reduced. Therefore, an electron is hardly trapped at the interface between the insulating film **114** and the oxide semiconductor film **108**.

(187) By using the above oxide insulating film as the insulating film **114**, the shift in the threshold voltage of the transistor can be reduced, which reduces a change in the electrical characteristics of the transistor.

(188) Note that in a spectrum of the insulating film **114** that is measured by ESR at 100 K or lower, by heat treatment of a manufacturing process of the transistor, typically heat treatment at higher than or equal to 300° C. and lower than 350° C., a first signal at a g -factor of greater than or equal

to 2.037 and less than or equal to 2.039, a second signal at a g-factor of greater than or equal to 2.001 and less than or equal to 2.003, and a third signal at a g-factor of greater than or equal to 1.964 and less than or equal to 1.966 are observed. Note that the split width of the first signal and the second signal and the split width of the second signal and the third signal that are obtained by ESR measurement with an X-band are approximately 5 mT. The sum of the spin densities of the first signal at a g-factor of greater than or equal to 2.037 and less than or equal to 2.039, the second signal at a g-factor of greater than or equal to 2.001 and less than or equal to 2.003, and the third signal at a g-factor of greater than or equal to 1.964 and less than or equal to 1.966 is lower than 1×10^{18} spins/cm³, typically higher than or equal to 1×10^{17} spins/cm³ and lower than 1×10^{18} spins/cm³.

(189) Note that in the ESR spectrum at 100 K or lower, the sum of the spin densities of the first signal at a g-factor of greater than or equal to 2.037 and less than or equal to 2.039, the second signal at a g-factor of greater than or equal to 2.001 and less than or equal to 2.003, and the third signal at a g-factor of greater than or equal to 1.964 and less than or equal to 1.966 corresponds to that of signals attributed to nitrogen oxide (NO_x; x is greater than 0 and less than or equal to 2, preferably greater than or equal to 1 and less than or equal to 2). Typical examples of the nitrogen oxide include nitrogen monoxide and nitrogen dioxide. That is, the lower the sum of the spin densities of the first signal at a g-factor of greater than or equal to 2.037 and less than or equal to 2.039, the second signal at a g-factor of greater than or equal to 2.001 and less than or equal to 2.003, and the third signal at a g-factor of greater than or equal to 1.964 and less than or equal to 1.966 is, the lower the content of nitrogen oxide in the oxide insulating film is.

(190) The nitrogen concentration of the above oxide insulating film measured by SIMS is lower than or equal to 6×10^{20} atoms/cm³. The above oxide insulating film is formed by a PECVD method at a substrate temperature higher than or equal to 220° C. and lower than or equal to 350° C. with the use of silane and dinitrogen monoxide, whereby a dense and very hard film can be formed.

(191) The insulating film **116** is formed using an oxide insulating film that contains oxygen whose amount is in excess of that satisfying the stoichiometric composition. Part of oxygen is released by heating from the oxide insulating film containing oxygen whose amount is in excess of that satisfying the stoichiometric composition. The oxide insulating film containing oxygen whose amount is in excess of that satisfying the stoichiometric composition is an oxide insulating film of which the amount of released oxygen converted into oxygen atoms is greater than or equal to 1.03×10^{19} atoms/cm³, preferably greater than or equal to 3.0×10^{20} atoms/cm³, in TDS analysis. Note that the film surface temperature in the TDS is preferably higher than or equal to 100° C. and lower than or equal to 700° C., or higher than or equal to 100° C. and lower than or equal to 500° C.

(192) Silicon oxide, silicon oxynitride, or the like with a thickness greater than or equal to 30 nm and less than or equal to 500 nm, or preferably greater than or equal to 50 nm and less than or equal to 400 nm, can be used for the insulating film **116**.

(193) Furthermore, it is preferable that the number of defects in the insulating film **116** be small, and typically, the spin density of a signal that appears at g=2.001 due to a dangling bond of silicon be lower than 1.5×10^{18} spins/cm³, more preferably lower than or equal to 1×10^{18} spins/cm³ by ESR measurement. Note that the insulating film **116** is provided more apart from the oxide semiconductor film **108** than the insulating film **114** is, and thus may have higher density of defects than the insulating film **114**.

(194) The insulating films **114** and **116** can be formed using insulating films formed of the same kinds of materials; thus, a boundary between the insulating film **114** and the insulating film **116** cannot be clearly observed in some cases. Thus, in this embodiment, the boundary between the insulating film **114** and the insulating film **116** is shown by a dashed line. Note that although a two-layer structure of the insulating film **114** and the insulating film **116** is described in this

embodiment, the present invention is not limited thereto, and for example, a single-layer structure of the insulating film **114** or a layered structure of three or more layers may be employed.

(195) <<Insulating Film Functioning as Protective Insulating Film>>

(196) The insulating film **118** functions as a protective insulating film of the transistor **100**.

(197) The insulating film **118** contains one or both of hydrogen and nitrogen. Alternatively, the insulating film **118** contains nitrogen and silicon. Furthermore, the insulating film **118** has a function of blocking oxygen, hydrogen, water, alkali metal, alkaline earth metal, or the like. It is possible to prevent outward diffusion of oxygen from the oxide semiconductor film **108**, outward diffusion of oxygen contained in the insulating films **114** and **116**, and entry of hydrogen, water, or the like into the oxide semiconductor film **108** from the outside by providing the insulating film **118**.

(198) A nitride insulating film can be used as the insulating film **118**, for example. For the nitride insulating film, silicon nitride, silicon nitride oxide, aluminum nitride, aluminum nitride oxide, or the like can be used.

(199) Note that although the variety of films such as the conductive films, the insulating films, and the oxide semiconductor film that are described above can be formed by a sputtering method or a PECVD method, such films may be formed by another method, e.g., a thermal CVD (Chemical Vapor Deposition) method. Examples of a thermal CVD method include an MOCVD (Metal Organic Chemical Vapor Deposition) method and an ALD (Atomic Layer Deposition) method.

(200) A thermal CVD method has an advantage that no defect due to plasma damage is generated since it does not utilize plasma to form a film.

(201) Deposition by a thermal CVD method may be performed in such a manner that a source gas and an oxidizer are supplied to a chamber at a time, the pressure in the chamber is set to an atmospheric pressure or a reduced pressure, and they are made to react with each other in the vicinity of the substrate or over the substrate to be deposited on the substrate.

(202) Deposition by an ALD method may be performed in such a manner that the pressure in a chamber is set to an atmospheric pressure or a reduced pressure, source gases for reaction are sequentially introduced into the chamber, and then the sequence of the gas introduction is repeated.

(203) The variety of films such as the conductive films, the insulating films, the oxide semiconductor film, and the metal oxide film in the above embodiment can be formed by a thermal CVD method such as an MOCVD method or an ALD method, and in the case where an In—Ga—ZnO film is formed, for example, trimethylindium, trimethylgallium, and dimethylzinc are used. Note that the chemical formula of trimethylindium is $\text{In}(\text{CH}_3)_3$. The chemical formula of trimethylgallium is $\text{Ga}(\text{CH}_3)_3$. The chemical formula of dimethylzinc is $\text{Zn}(\text{CH}_3)_2$. Furthermore, without limitation to the above combination, triethylgallium (chemical formula: $\text{Ga}(\text{C}_2\text{H}_5)_3$) can be used instead of trimethylgallium and diethylzinc (chemical formula: $\text{Zn}(\text{C}_2\text{H}_5)_2$) can be used instead of dimethylzinc.

(204) For example, in the case where a hafnium oxide film is formed by a deposition apparatus using ALD, two kinds of gases, ozone (O_3) as an oxidizer and a source gas which is obtained by vaporizing liquid containing a solvent and a hafnium precursor compound (hafnium alkoxide or hafnium amide such as tetrakis(dimethylamide)hafnium (TDMAH)), are used. Note that the chemical formula of tetrakis(dimethylamide)hafnium is $\text{Hf}[\text{N}(\text{CH}_3)_2]_4$. Furthermore, as other material liquids, tetrakis(ethylmethylethylamide)hafnium and the like are given.

(205) For example, in the case where an aluminum oxide film is formed by a deposition apparatus using ALD, two kinds of gases, H_2O as an oxidizer and a source gas which is obtained by vaporizing liquid containing a solvent and an aluminum precursor compound (e.g., trimethylaluminum (TMA)), are used. Note that the chemical formula of trimethylaluminum is $\text{Al}(\text{CH}_3)_3$. Furthermore, as other material liquids, tris(dimethylamide)aluminum, triisobutylaluminum, aluminum tris(2,2,6,6-tetramethyl-3,5-heptanedionate), and the like are given.

(206) For example, in the case where a silicon oxide film is formed by a deposition apparatus using

ALD, hexachlorodisilane is adsorbed on a deposition surface, chlorine contained in the adsorbate is removed, and radicals of an oxidizing gas (e.g., O₂ or dinitrogen monoxide) are supplied to react with the adsorbate.

(207) For example, in the case where a tungsten film is formed by a deposition apparatus using ALD, a WF₆ gas and a B₂H₆ gas are introduced sequentially and repeatedly to form an initial tungsten film, and then a tungsten film is formed using a WF₆ gas and an H₂ gas. Note that an SiH₄ gas may be used instead of a B₂H₆ gas.

(208) For example, in the case where an oxide semiconductor film, e.g., an In—Ga—ZnO film is formed by a deposition apparatus using ALD, an In(CH₃)₃ gas and an O₃ gas are introduced sequentially and repeatedly to form an In—O layer, then a Ga(CH₃)₃ gas and an O₃ gas are used to form a GaO layer, and then a Zn(CH₃)₂ gas and an O₃ gas are used to form a ZnO layer. Note that the order of these layers is not limited to this example. A mixed compound layer such as an In—Ga—O layer, an In—Zn—O layer, or a Ga—Zn—O layer may be formed by using these gases. Note that an H₂O gas which is obtained by bubbling with an inert gas such as Ar may be used instead of an O₃ gas, but it is preferable to use an O₃ gas, which does not contain H. Instead of an In(CH₃)₃ gas, an In(C₂H₅)₃ gas may be used. Instead of a Ga(CH₃)₃ gas, a Ga(C₂H₅)₃ gas may be used. A Zn(CH₃)₂ gas may be used.

(209) <1-6. Manufacturing Method 1 of Transistor>

(210) Next, a method for manufacturing the transistor **100J** that is a semiconductor device of one embodiment of the present invention is described with reference to FIG. **14** to FIG. **17**. Note that FIG. **14** to FIG. **17** are cross-sectional views illustrating a method for manufacturing the semiconductor device. FIGS. **14**(A), (C), and (E), FIGS. **15**(A), (C), and (E), FIGS. **16**(A), (C), and (E), and FIGS. **17**(A), (C), and (E) are cross-sectional views in the channel length direction shown by X1-X2, and FIGS. **14**(B), (D), and (F), FIGS. **15**(B), (D), and (F), FIGS. **16**(B), (D), and (F), and FIGS. **17**(B), (D), and (F) are cross-sectional views in the channel width direction shown by Y1-Y2.

(211) <<Step of Forming First Gate Electrode and First Gate Insulating Film>>

(212) First, a conductive film is formed over the substrate **102** and processed through a lithography process and an etching process, whereby the conductive film **104** that functions as the first gate electrode is formed. Then, the insulating films **106** and **107** serving as the first gate insulating film are formed over the conductive film **104** (see FIGS. **14**(A) and **14**(B)).

(213) In this embodiment, a glass substrate is used as the substrate **102**, and as the conductive film **104** serving as the first gate electrode, a 10-nm-thick titanium film, a 100-nm-thick copper film, and a 50-nm-thick tantalum nitride film are each formed by a sputtering method.

(214) As the insulating film **106**, a 400-nm-thick silicon nitride film is formed by a PECVD method, and as the insulating film **107**, a 15-nm-thick silicon oxynitride film is formed by a PECVD method. Note that the insulating film **106** can have a layered structure of silicon nitride films. Specifically, the insulating film **106** can have a three-layer structure of a first silicon nitride film, a second silicon nitride film, and a third silicon nitride film. An example of the three-layer structure can be formed as follows.

(215) For example, the first silicon nitride film can be formed to have a thickness of 50 nm under the conditions where silane at a flow rate of 200 sccm, nitrogen at a flow rate of 2000 sccm, and an ammonia gas at a flow rate of 100 sccm are supplied as a source gas to a reaction chamber of a PECVD apparatus, the pressure in the reaction chamber is controlled to 100 Pa, and the power of 2000 W is supplied using a 27.12 MHz high-frequency power source.

(216) The second silicon nitride film can be formed to have a thickness of 300 nm under the conditions where silane at a flow rate of 200 sccm, nitrogen at a flow rate of 2000 sccm, and an ammonia gas at a flow rate of 2000 sccm are supplied as a source gas to the reaction chamber of the PECVD apparatus, the pressure in the reaction chamber is controlled to 100 Pa, and the power

of 2000 W is supplied using a 27.12 MHz high-frequency power source.

(217) The third silicon nitride film can be formed to have a thickness of 50 nm under the conditions where silane at a flow rate of 200 sccm and nitrogen at a flow rate of 5000 sccm are supplied as a source gas to the reaction chamber of the PECVD apparatus, the pressure in the reaction chamber is controlled to 100 Pa, and the power of 2000 W is supplied using a 27.12 MHz high-frequency power source.

(218) Note that the first silicon nitride film, the second silicon nitride film, and the third silicon nitride film can each be formed at a substrate temperature of 350° C. or lower.

(219) For example, in the case where a conductive film containing copper (Cu) is used as the conductive film **104**, the use of the three-layer structure of silicon nitride films for the insulating film **106** provides the following effect.

(220) The first silicon nitride film can inhibit diffusion of a copper (Cu) element from the conductive film **104**. The second silicon nitride film has a function of releasing hydrogen and can improve withstand voltage of the insulating film that serves as a gate insulating film. The third silicon nitride film releases a small amount of hydrogen and can inhibit diffusion of hydrogen released from the second silicon nitride film.

(221) The insulating film **107** is preferably formed of an insulating film containing oxygen to improve the characteristics of an interface with the oxide semiconductor film **108** (more specifically the oxide semiconductor film **108b**) formed later. Oxygen may be added to the insulating film **107** after the insulating film **107** is formed. As the oxygen to be added to the insulating film **107**, an oxygen radical, an oxygen atom, an oxygen atomic ion, an oxygen molecular ion, and the like are given. Methods for adding oxygen include an ion doping method, an ion implantation method, a plasma treatment method, and the like.

(222) <<Step of Forming Oxide Semiconductor Film>>

(223) Next, the oxide semiconductor film **108b** and the oxide semiconductor film **108c** are formed over the insulating film **107** (see FIGS. **14(C)** and **(D)**).

(224) Note that FIGS. **14(C)** and **(D)** are schematic cross-sectional views of an inner portion of a deposition apparatus when an oxide semiconductor film to be the oxide semiconductor film **108** is formed over the insulating film **107**. In FIGS. **14(C)** and **(D)**, a sputtering apparatus is used as the deposition apparatus, and a target **191** placed inside the sputtering apparatus and plasma **192** formed under the target **191** are schematically illustrated.

(225) When the oxide semiconductor film is formed, plasma is discharged in an atmosphere containing a first oxygen gas. At this time, oxygen is added to the insulating film **107** over which the oxide semiconductor film is to be formed. When the oxide semiconductor film is formed, an inert gas (e.g., a helium gas, an argon gas, or a xenon gas) as well as the first oxygen gas may be mixed.

(226) The first oxygen gas is contained at least when the oxide semiconductor film is formed. The proportion of the first oxygen gas in a deposition gas for forming the oxide semiconductor film is higher than 0% and lower than or equal to 100%, preferably higher than or equal to 10% and lower than or equal to 100%, more preferably higher than or equal to 30% and lower than or equal to 100%. Note that in FIGS. **14(C)** and **(D)**, oxygen or excess oxygen to be added to the insulating film **107** is schematically shown by arrows of broken lines.

(227) Note that the oxide semiconductor films **108b** and **108c** may be formed at the same substrate temperature or different substrate temperatures. Note that the substrate temperatures for the oxide semiconductor films **108b** and **108c** are preferably the same, in which case the manufacturing cost can be reduced.

(228) The oxide semiconductor film **108** is formed at a substrate temperature higher than or equal to room temperature and lower than 340° C., preferably higher than or equal to room temperature and lower than or equal to 300° C., more preferably higher than or equal to 100° C. and lower than or equal to 250° C., still more preferably higher than or equal to 100° C. and lower than or equal to

200° C., for example. The oxide semiconductor film **108** is formed while being heated, so that the crystallinity of the oxide semiconductor film **108** can be increased. On the other hand, in the case where a large-sized glass substrate (e.g., the 6th generation to the 10th generation) is used as the substrate **102** and the oxide semiconductor film **108** is formed at a substrate temperature higher than or equal to 150° C. and lower than 340° C., the substrate **102** might be changed in shape (distorted or warped). In the case where a large-sized glass substrate is used, the change in the shape of the glass substrate can be suppressed by forming the oxide semiconductor film **108** at a substrate temperature higher than or equal to 100° C. and lower than 150° C.

(229) In addition, increasing the purity of a sputtering gas is necessary. For example, as an oxygen gas or an argon gas used as a sputtering gas, a gas which is highly purified to have a dew point of -40° C. or lower, preferably -80° C. or lower, more preferably -100° C. or lower, still more preferably -120° C. or lower is used, whereby entry of moisture or the like into the oxide semiconductor film can be prevented as much as possible.

(230) When the oxide semiconductor film is formed by a sputtering method, a chamber of a sputtering apparatus is preferably evacuated to a high vacuum (to the degree of approximately 5×10^{-7} Pa to 1×10^{-4} Pa) by an adsorption vacuum pump such as a cryopump so that water and the like acting as impurities for the oxide semiconductor film are removed as much as possible. Alternatively, a turbo molecular pump and a cold trap are preferably combined so as to prevent a backflow of a gas, especially a gas containing carbon or hydrogen from an exhaust system to the inside of the chamber.

(231) The oxide semiconductor film to be the oxide semiconductor film **108b** is formed, and subsequently, the oxide semiconductor film to be the oxide semiconductor film **108c** is formed. Note that when these oxide semiconductor films are formed, plasma is discharged in an atmosphere containing a second oxygen gas.

(232) Note that the proportion of the first oxygen gas for forming the oxide semiconductor film to be the oxide semiconductor film **108b** may be the same as or different from the proportion of the second oxygen gas for forming the oxide semiconductor film to be the oxide semiconductor film **108c**.

(233) In this embodiment, the oxide semiconductor film to be the oxide semiconductor film **108b** is formed by a sputtering method using an In—Ga—Zn metal oxide target (In:Ga:Zn=4:2:4.1 [atomic ratio]), and successively, the oxide semiconductor film to be the oxide semiconductor film **108c** is formed in a vacuum by a sputtering method using an In—Ga—Zn metal oxide target

(In:Ga:Zn=1:1:1.2 [atomic ratio]). The substrate temperature when the oxide semiconductor films are formed is set to 170° C. As the deposition gas for forming the oxide semiconductor film to be the oxide semiconductor film **108b**, an oxygen gas at a flow rate of 15 sccm and an argon gas at a flow rate of 35 sccm are used. As the deposition gas for forming the oxide semiconductor film to be the oxide semiconductor film **108c**, an oxygen gas at a flow rate of 25 sccm and an argon gas at a flow rate of 25 sccm are used.

(234) Next, the formed oxide semiconductor films are processed into desired shapes, so that the island-shaped oxide semiconductor films **108b** and **108c** are formed (see FIGS. **14(E)** and **(F)**). Note that in this embodiment, the oxide semiconductor film **108** is composed of the oxide semiconductor films **108b** and **108c** (see FIGS. **14(E)** and **(F)**).

(235) Furthermore, heat treatment (hereinafter referred to as first heat treatment) is favorably performed after the oxide semiconductor film **108** is formed. By the first heat treatment, hydrogen, water, and the like contained in the oxide semiconductor film **108** can be reduced. Note that the heat treatment for the purpose of reducing hydrogen, water, and the like may be performed before the oxide semiconductor film **108** is processed into an island shape. Note that the first heat treatment is one kind of treatment for increasing the purity of the oxide semiconductor film.

(236) The first heat treatment can be performed at a temperature of, for example, higher than or equal to 150° C. and lower than the strain point of the substrate, preferably higher than or equal to

200° C. and lower than or equal to 450° C., more preferably higher than or equal to 250° C. and lower than or equal to 350° C.

(237) Moreover, an electric furnace, an RTA (Rapid Thermal Anneal) apparatus, or the like can be used for the first heat treatment. With the use of an RTA apparatus, the heat treatment can be performed at a temperature higher than or equal to the strain point of the substrate if the heating time is short. Therefore, the heating time can be shortened. The first heat treatment may be performed under an atmosphere of nitrogen, oxygen, ultra-dry air (air with a water content of 20 ppm or less, preferably 1 ppm or less, preferably 10 ppb or less), or a rare gas (e.g., argon, helium). Note that the nitrogen, oxygen, ultra-dry air, or rare gas preferably does not contain hydrogen, water, and the like. Furthermore, after heat treatment performed under a nitrogen or rare gas atmosphere, heating may be performed in an oxygen or ultra-dry air atmosphere. As a result, hydrogen, water, and the like in the oxide semiconductor film can be released and oxygen can be supplied to the oxide semiconductor film at the same time. Consequently, oxygen vacancies in the oxide semiconductor film can be reduced.

(238) <<Step of Forming Source Electrode and Drain Electrode>>

(239) Next, a conductive film **112** to be a source electrode and a drain electrode is formed on the insulating film **107** and the oxide semiconductor film **108** by a sputtering method (see FIGS. **15(A)** and **(B)**).

(240) In this embodiment, conductive films **112_1**, **112_2**, and **112_3** are stacked to form the conductive film **112**. A laminated film in which a 50-nm-thick tungsten film as the conductive film **112_1**, a 200-nm-thick copper film as the conductive film **112_2**, and a 5-nm-thick tungsten film as the conductive film **112_3** are stacked in this order is formed as the conductive film **112_1** by a sputtering method. Note that in this embodiment, the conductive film **112_1** and the conductive film **112_3** are formed using the same material; however, there is no limitation thereto. For example, a laminated film in which a 50-nm-thick tungsten film as the conductive film **112_1**, a 200-nm-thick copper film as the conductive film **112_2**, and a 50-nm-thick titanium film as the conductive film **112_3** are formed in this order may be used. The conductive film **112** has a layered structure of three layers; however, there is no limitation thereto. For example, the conductive film **112** may have a layered structure of two layers or a layered structure of four or more layers.

(241) Then, masks **141a** and **141b** are formed in desired regions over the conductive film **112_3**. Subsequently, the conductive films **112_2** and **112_3** are processed using the masks **141a** and **141b**, whereby the island-shaped conductive films **112a_2**, **112b_2**, **112a_3**, and **112b_3** separated from each other are formed (see FIGS. **15(C)** and **(D)**).

(242) Note that in this embodiment, the conductive films **112_2** and **112_3** are processed with a wet etching apparatus. Note that the method for processing the conductive film **112** is not limited thereto, and a dry etching apparatus may be used, for example. Note that the manufacturing cost can be reduced when the conductive film **112** is processed with a wet etching apparatus compared with when the conductive film **112** is processed with a dry etching apparatus.

(243) Next, the end portions of the conductive films **112a_2** and **112b_2** are silicified, whereby the regions **112a_2a** and **112b_2a** containing copper and the regions **112a_2b** and **112b_2b** containing copper silicide are formed.

(244) FIGS. **15(E)** and **(F)** are schematic cross-sectional views of the inner portion of a plasma apparatus when the end portions of the conductive films **112a_2** and **112b_2** are silicified. In FIGS. **15(E)** and **(F)**, a PECVD apparatus is used as the plasma apparatus and plasma **195** generated inside the PECVD apparatus is schematically illustrated.

(245) Copper exposed at the end portions of the conductive films **112a_2** and **112b_2** easily forms an oxide film on the surface. For this reason, as pretreatment before silicification of copper, plasma is discharged in an atmosphere containing a reducing gas (e.g., a hydrogen gas or an ammonia gas) so that the oxide film that covers the surface of copper is removed. At that time, the oxide film that covers the copper surface is reduced and the copper is exposed at the end portions of the

conductive films **112a_2** and **112b_2**. Note that the reduction method for removing the oxide film is not limited to plasma treatment. For example, the reduction may be performed by exposing the surface of copper to an atmosphere containing a reducing gas (e.g., a hydrogen gas or an ammonia gas) and performing heat treatment. The substrate temperature when the plasma treatment and heat treatment are performed is preferably higher than or equal to 300° C., more preferably higher than or equal to 350° C. The substrate temperature when the oxide film covering the copper surface is removed is 350° C. in this embodiment.

(246) Subsequently, by performing exposure to an atmosphere containing a silane gas to make copper and the silane gas reach with each other, copper silicide containing copper and silicon is formed at the end portions of the conductive films **112a_2**, and **112b_2**, so that the regions **112a_2b** and **112b_2b** are formed. In the conductive films **112a_2**, and **112b_2**, regions where copper silicide is not formed are the regions **112a_2a** and **112b_2a**. The substrate temperature during the formation of copper silicide is preferably higher than or equal to 200° C. and lower than or equal to 400° C., more preferably higher than or equal to 220° C. and lower than or equal to 350° C. In this embodiment, the substrate temperature during the formation of copper silicide is 220° C., and a silane gas at a flow rate of 300 sccm and a nitrogen gas at a flow rate of 500 sccm are used.

(247) The substrate temperature during the removal of the oxide film on the copper surface and the substrate temperature during the formation of copper silicide are preferably the same because the removal of the oxide film and the formation of copper silicide can be performed in the same apparatus or the same chamber. In that case, the substrate temperature during the formation of copper silicide is preferably set to 350° C.

(248) Note that the gas for forming copper silicide contains at least silicon, and a gas containing silicon accounts for greater than 0% and less than or equal to 100%, preferably greater than or equal to 10% and less than or equal to 100%, more preferably greater than or equal to 30% and less than or equal to 100% of the whole gas for forming copper silicide.

(249) Note that in FIGS. 15(E) and (F), silicon or silane that is added to the conductive films **112a_2**, and **112b_2** is schematically shown by arrows of broken lines.

(250) Note that after copper silicide is formed, plasma may be discharged in an atmosphere of a gas containing nitrogen to form copper silicide nitride containing copper, silicon, and nitrogen in the conductive films **112a_2** and **112b_2**. The substrate may be exposed to an atmosphere of a gas containing nitrogen and heat treatment is performed to form copper silicide nitride containing copper, silicon, and nitrogen in the conductive films **112a_2** and **112b_2**.

(251) Then, masks **142a** and **142b** are formed over part of the conductive film **112_1** and desired regions of the conductive films **112a_2**, **112b_2**, **112a_3**, and **112b_3**. Subsequently, the conductive film **112_1** is processed using the masks **142a** and **142b**, whereby the island-shaped conductive films **112a_1** and **112b_1** separated from each other are formed. When the process is performed, the conductive film **112a** that includes the conductive film **112a_1**, the conductive film **112a_2** including the regions **112a_2a** and **112a_2b**, and the conductive film **112_3** and the conductive film **112b** that includes the conductive film **112b_1**, the conductive film **112b_2** including the regions **112b_2a** and **112b_2b**, and the conductive film **112b_3** are formed (see FIGS. 16(A) and (B)).

(252) Note that in this embodiment, the conductive film **112_1** is processed with a dry etching apparatus. Note that the method for processing the conductive film **112_1** is not limited thereto, and a wet etching apparatus may be used, for example. Note that a finer pattern can be formed when a dry etching apparatus is used for processing the conductive film **112_1** than when a wet etching apparatus is used for processing the conductive film **112_1**.

(253) After the conductive films **112a** and **112b** are formed, a surface (on the back channel side) of the oxide semiconductor film **108** (more specifically, the oxide semiconductor film **108b**) may be cleaned. An example of the cleaning method is cleaning using a chemical solution such as phosphoric acid. The cleaning using a chemical solution such as phosphoric acid can remove impurities (e.g., elements included in the conductive films **112a** and **112b**) attached to the surface

of the oxide semiconductor film **108b**. Note that the cleaning is not necessarily performed; in some cases, the cleaning does not need to be performed.

(254) In the step of forming the conductive films **112a** and **112b** and/or the cleaning step, the thickness of a region of the oxide semiconductor film **108** which is not covered with the conductive films **112a** and **112b** might be reduced.

(255) <<Step of Forming Second Gate Insulating Film>>

(256) Next, the insulating films **114** and **116** are formed over the oxide semiconductor film **108** and the conductive films **112a** and **112b** (see FIGS. **16(C)** and **(D)**).

(257) Note that after the insulating film **114** is formed, the insulating film **116** is preferably formed successively without exposure to the air. After the insulating film **114** is formed, the insulating film **116** is formed successively without exposure to the air with at least one of the flow rate of a source gas, the pressure, high-frequency power, and the substrate temperature adjusted, whereby the concentration of impurities attributed to the atmospheric component at the interface between the insulating films **114** and **116** can be reduced, and oxygen in the insulating films **114** and **116** can be transferred to the oxide semiconductor film **108**; accordingly, the amount of oxygen vacancies in the oxide semiconductor film **108** can be reduced.

(258) For example, as the insulating film **114**, a silicon oxynitride film can be formed by a PECVD method. In that case, a deposition gas containing silicon and an oxidizing gas are preferably used as a source gas. Typical examples of the deposition gas containing silicon include silane, disilane, trisilane, and silane fluoride. Examples of the oxidizing gas include dinitrogen monoxide and nitrogen dioxide. An insulating film containing nitrogen and having a small amount of defects can be formed as the insulating film **114** by a PECVD method under the conditions where the flow rate of the oxidizing gas is more than 20 times and less than 100 times, preferably more than or equal to 40 times and less than or equal to 80 times, that of the deposition gas, and the pressure in a treatment chamber is lower than 100 Pa, preferably lower than or equal to 50 Pa.

(259) In this embodiment, a silicon oxynitride film is formed as the insulating film **114** by a PECVD method under the conditions where the substrate **102** is held at a temperature of 220° C., silane at a flow rate of 50 sccm and dinitrogen monoxide at a flow rate of 2000 sccm are used as a source gas, the pressure in the treatment chamber is 20 Pa, and a high-frequency power of 100 W at 13.56 MHz (1.6×10^{-2} W/cm² as the power density) is supplied to parallel-plate electrodes.

(260) As the insulating film **116**, a silicon oxide film or a silicon oxynitride film is formed under the conditions where the substrate placed in the treatment chamber of the PECVD apparatus that is vacuum-evacuated is held at a temperature of higher than or equal to 180° C. and lower than or equal to 350° C., the pressure in the treatment chamber into which a source gas introduced is higher than or equal to 100 Pa and lower than or equal to 250 Pa, more preferably higher than or equal to 100 Pa and lower than or equal to 200 Pa, and a high-frequency power of greater than or equal to 0.17 W/cm² and less than or equal to 0.5 W/cm², more preferably greater than or equal to 0.25 W/cm² and less than or equal to 0.35 W/cm² is supplied to an electrode provided in the treatment chamber.

(261) As the formation conditions of the insulating film **116**, the high-frequency power having the above power density is supplied to the reaction chamber having the above pressure, whereby the degradation efficiency of the source gas in plasma is increased, oxygen radicals are increased, and oxidation of the source gas is promoted; thus, the oxygen content in the insulating film **116** becomes higher than that in the stoichiometric composition. In the film formed at the above substrate temperature, the cohesion between silicon and oxygen is weak, and accordingly, part of oxygen in the film is released by heat treatment in a later step. Thus, it is possible to form an oxide insulating film which contains more oxygen than that satisfying the stoichiometric composition and from which part of oxygen is released by heating.

(262) Note that the insulating film **114** functions as a protective film for the oxide semiconductor

film **108** in the step of forming the insulating film **116**. Therefore, the insulating film **116** can be formed using the high-frequency power having a high power density while damage to the oxide semiconductor film **108** is reduced.

(263) Note that in the deposition conditions of the insulating film **116**, when the flow rate of the deposition gas containing silicon with respect to the oxidizing gas is increased, the amount of defects in the insulating film **116** can be reduced. Typically, it is possible to form an oxide insulating film with less defects in which the spin density of a signal which appears at $g=2.001$ due to a dangling bond of silicon is lower than 6×10^{17} spins/cm³, preferably lower than or equal to 3×10^{17} spins/cm³, more preferably lower than or equal to 1.5×10^{17} spins/cm³ by ESR measurement. As a result, the reliability of the transistor **100** can be improved. Heat treatment (hereinafter referred to as second heat treatment) is preferably performed after the insulating films **114** and **116** are formed. The second heat treatment can reduce nitrogen oxide contained in the insulating films **114** and **116**. Alternatively, by the second heat treatment, part of oxygen contained in the insulating films **114** and **116** can be transferred to the oxide semiconductor film **108**, so that the amount of oxygen vacancies in the oxide semiconductor film **108** can be reduced.

(264) The temperature of the second heat treatment is typically lower than 400° C., preferably lower than 375° C., more preferably higher than or equal to 150° C. and lower than or equal to 350° C. The second heat treatment may be performed in an atmosphere of nitrogen, oxygen, ultra-dry air (air with a water content of less than or equal to 20 ppm, preferably less than or equal to 1 ppm, more preferably less than or equal to 10 ppb), or a rare gas (argon, helium, or the like). Note that an electric furnace, an RTA apparatus, or the like can be used for the heat treatment, in which it is preferable that hydrogen, water, and the like not be contained in the nitrogen, oxygen, ultra-dry air, or rare gas.

(265) Next, a mask is formed over the insulating film **116** through a lithography process, and the opening **152c** is formed in desired regions of the insulating films **114** and **116**. Note that the opening **152c** is formed to reach the conductive film **112b** (see FIGS. **16(E)** and **16(F)**).

(266) <<Step of Forming Second Gate Electrode>>

(267) Next, the conductive films **120a** and **120b** are formed over the insulating film **116** to cover the opening **152c** (see FIGS. **17(A)**, **(B)**, **(C)**, and **(D)**).

(268) FIGS. **17(A)** and **(B)** are schematic cross-sectional views of the inner portion of the deposition apparatus when the conductive films **120a** and **120b** are formed over the insulating film **116**. In FIGS. **17(A)** and **(B)**, a sputtering apparatus is used as the deposition apparatus, and a target **193** placed inside the sputtering apparatus and plasma **194** formed under the target **193** are schematically illustrated.

(269) First, when the conductive films **120a** and **120b** are formed, plasma is discharged in an atmosphere containing a third oxygen gas. At this time, oxygen is added to the insulating film **116** over which the conductive films **120a** and **120b** are formed. When the conductive films **120a** and **120b** are formed, an inert gas (e.g., a helium gas, an argon gas, or a xenon gas) as well as the third oxygen gas may be mixed. For example, it is preferable to use the argon gas and the third oxygen gas with a flow rate higher than that of the argon gas. When the flow rate of the third oxygen gas is set higher, oxygen can be favorably added to the insulating film **116**. As an example of the conditions for forming the conductive films **120a** and **120b**, the proportion of the fourth oxygen gas in a whole deposition gas is higher than or equal to 50% and lower than or equal to 100%, preferably higher than or equal to 80% and lower than or equal to 100%.

(270) Note that in FIGS. **17(A)** and **(B)**, oxygen or excess oxygen added to the insulating film **116** is schematically illustrated by arrows of broken lines.

(271) The substrate temperature during formation of the conductive films **120a** and **120b** is higher than or equal to room temperature and lower than 340° C., preferably higher than or equal to room temperature and lower than or equal to 300° C., more preferably higher than or equal to 100° C.

and lower than or equal to 250° C., more preferably higher than or equal to 100° C. and lower than or equal to 200° C. The conductive films **120a** and **120b** are formed while being heated, so that the crystallinity of the conductive films **120a** and **120b** can be increased. On the other hand, in the case where a large-sized glass substrate (e.g., the 6th generation to the 10th generation) is used as the substrate **102** and the substrate temperature during formation of the conductive films **120a** and **120b** is set to higher than or equal to 150° C. and lower than 340° C., the substrate **102** might be changed in shape (distorted or warped). Therefore, in the case where a large-sized glass substrate is used, the change in the shape of the glass substrate can be suppressed by setting the substrate temperature during formation of the conductive films **120a** and **120b** to higher than or equal to 100° C. and lower than 150° C.

(272) In this embodiment, the conductive films **120a** and **120b** are formed by a sputtering method using an In—Ga—Zn metal oxide target (In:Ga:Zn=4:2:4.1 [atomic ratio]). The substrate temperature during formation of the conductive films **120a** and **120b** is set to 170° C. As the deposition gas for forming the conductive films **120a** and **120b**, an oxygen gas at a flow rate of 100 sccm is used.

(273) Note that as the conductive films **120a** and **120b**, the above-described oxide semiconductor film (In:Ga:Zn=1:1:1 [atomic ratio], In:Ga:Zn=1:3:2 [atomic ratio], In:Ga:Zn=1:3:4 [atomic ratio], In:Ga:Zn=1:3:6 [atomic ratio], In:Ga:Zn=3:1:2 [atomic ratio], In:Ga:Zn=4:2:3 [atomic ratio], or In:Ga:Zn=5:1:6 [atomic ratio], for example) may be used.

(274) Note that although this embodiment describes, as an example, a method in which oxygen is added to the insulating film **116** when the conductive films **120a** and **120b** are formed, there is no limitation thereto. For example, oxygen may be further added to the insulating film **116** after the conductive films **120a** and **120b** are formed.

(275) As a method for adding oxygen to the insulating film **116**, for example, a 5-nm-thick ITSO film is formed using a target of an oxide containing indium, tin, and silicon (also referred to as ITSO) (In.sub.2O.sub.3: SnO.sub.2:SiO.sub.2=85:10:5 [weight %]) as an oxide conductive film.

(276) In that case, the thickness of the oxide conductive film is preferably greater than or equal to 1 nm and less than or equal to 20 nm or greater than or equal to 2 nm and less than or equal to nm, in which case oxygen is suitably transmitted and release of oxygen can be inhibited. Then, oxygen is added to the insulating film **116** through the oxide conductive film. As the method for adding oxygen, an ion doping method, an ion implantation method, a plasma treatment method, and the like are given. Furthermore, a bias voltage is applied to the substrate side when oxygen is added, whereby oxygen can be effectively added to the insulating film **116**. An ashing apparatus is used, for example, and the power density of the bias voltage applied to the substrate side of the ashing apparatus is set to greater than or equal to 1 W/cm.sup.2 and less than or equal to 5 W/cm.sup.2 as the bias voltage. Furthermore, the substrate temperature at which oxygen is added is higher than or equal to room temperature and lower than or equal to 300° C., preferably higher than or equal to 100° C. and lower than or equal to 250° C., whereby oxygen can be added efficiently to the insulating film **116**.

(277) Next, the formed conductive film is processed into a desired shape, so that the island-shaped conductive films **120a** and **120b** are formed (see FIGS. 17(C) and (D)).

(278) <<Step of Forming Protective Insulating Film>>

(279) Then, the insulating film **118** is formed over the insulating film **116** and the conductive films **120a** and **120b** (see FIGS. 17(E) and (F)).

(280) The insulating film **118** contains one or both of hydrogen and nitrogen. As the insulating film **118**, a silicon nitride film is suitably used, for example. Furthermore, the insulating film **118** can be formed by a sputtering method or a PECVD method, for example. In the case where the insulating film **118** is formed by a PECVD method, for example, the substrate temperature is lower than 400° C., preferably lower than 375° C., more preferably higher than or equal to 180° C. and lower than or equal to 350° C. The substrate temperature at which the insulating film **118** is formed is

preferably within the above range because a dense film can be formed. Furthermore, when the substrate temperature at which the insulating film **118** is formed is within the above range, oxygen or excess oxygen in the insulating films **114** and **116** can be moved to the oxide semiconductor film **108**.

(281) After the insulating film **118** is formed, heat treatment similar to the first heat treatment or the second heat treatment described above (hereinafter referred to as third heat treatment) may be performed. The heat treatment at a temperature of lower than 400° C., preferably lower than 375° C., more preferably higher than or equal to 180° C. and lower than or equal to 350° C. is thus performed after the addition of oxygen to the insulating film **116** when the insulating film **118** is formed, whereby oxygen or excess oxygen in the insulating film **116** can be transferred to the oxide semiconductor film **108** (particularly, the oxide semiconductor film **108b**) and compensate for oxygen vacancies in the oxide semiconductor film **108**.

(282) Furthermore, the insulating film **106** is provided under the insulating film **107**, and the insulating film **118** is provided over the insulating films **114** and **116**. When the insulating films **106** and **118** are formed using a material having low oxygen permeability, e.g., silicon nitride, oxygen contained in the insulating films **107**, **114**, and **116** can be confined to the oxide semiconductor film **108** side; thus, oxygen can be favorably transferred to the oxide semiconductor film **108**.

(283) The insulating film **118** contains one or both of hydrogen and nitrogen. Thus, when the insulating film **118** is formed, one or both of hydrogen and nitrogen are added to the conductive films **120a** and **120b** in contact with the insulating film **118**, so that the conductive films **120a** and **120b** have a high carrier density and can function as oxide conductive films.

(284) Furthermore, in the case where a silicon nitride film is formed by a PECVD method as the insulating film **118**, a deposition gas containing silicon, nitrogen, and ammonia are preferably used as a source gas. A small amount of ammonia compared with the amount of nitrogen is used, whereby ammonia is dissociated in the plasma and activated species are generated. The activated species cleave a bond between silicon and hydrogen that are contained in the deposition gas containing silicon and a triple bond of nitrogen. As a result, the bond between silicon and nitrogen is promoted, and a dense silicon nitride film having few bonds between silicon and hydrogen and few defects can be formed. On the other hand, when the amount of ammonia with respect to nitrogen is large, decomposition of the deposition gas containing silicon and nitrogen are not promoted, so that a sparse silicon nitride film in which bonds between silicon and hydrogen remain and defects are increased is formed. Therefore, in the source gas, the flow rate of nitrogen is preferably set to 5 times or more and 50 times or less or 10 times or more and 50 times or less that of ammonia.

(285) In this embodiment, with the use of a PECVD apparatus, a 100-nm-thick silicon nitride film is formed as the insulating film **118** using silane, nitrogen, and ammonia as a source gas. The flow rate of silane is 50 sccm, the flow rate of nitrogen is 5000 sccm, and the flow rate of ammonia is 100 sccm. The pressure in the treatment chamber is 100 Pa, the substrate temperature is 350° C., and high-frequency power of 1000 W is supplied to a parallel-plate electrode with a 27.12 MHz high-frequency power source. The PECVD apparatus is a parallel-plate PECVD apparatus in which the electrode area is 6000 cm.², and the power per unit area (power density) into which the supplied power is converted is 1.7×10^{-1} W/cm.².

(286) Through the above process, the transistor **100J** illustrated in FIGS. 9(A) and (B) can be manufactured.

(287) Furthermore, in the entire manufacturing process of the transistor **100J**, the substrate temperature is preferably lower than 400° C., more preferably lower than 375° C., and still more preferably higher than or equal to 180° C. and lower than or equal to 350° C. because the change in shape of the substrate (distortion or warp) can be significantly reduced even when a large-sized substrate is used. Note that as a step in which the substrate temperature is increased in the

manufacturing process of the transistor **100J**, typically, the following are given: the substrate temperature in the formation of the insulating films **106** and **107** (lower than 400° C., preferably higher than or equal to 250° C. and lower than or equal to 350° C.), the substrate temperature in the formation of the oxide semiconductor film **108** (higher than or equal to room temperature and lower than 340° C., preferably higher than or equal to 100° C. and lower than or equal to 200° C., more preferably higher than or equal to 100° C. and lower than 150° C.), the substrate temperature in the formation of the insulating films **116** and **118** (lower than 400° C., preferably lower than 375° C., more preferably higher than or equal to 180° C. and lower than or equal to 350° C.), the first heat treatment or the second heat treatment after the addition of oxygen (lower than 400° C., preferably lower than 375° C., more preferably higher than or equal to 180° C. and lower than or equal to 350° C.), and the like.

(288) <1-7. Manufacturing Method 2 of Transistor>

(289) Next, a method for manufacturing the transistor **100M** that is a semiconductor device of one embodiment of the present invention is described with reference to FIG. **18** to FIG. **20**. Note that FIG. **18** to FIG. **20** are cross-sectional views illustrating a method for manufacturing the semiconductor device; FIGS. **18(A)**, (C), and (E), FIGS. **19(A)** and (C), and FIG. **20(A)** are cross-sectional views in the channel length direction shown by X1-X2, and FIGS. **18(B)**, (D), and (F), FIGS. **19(B)** and (D), and FIG. **20(B)** are cross-sectional views in the channel width direction shown by Y1-Y2.

(290) Methods for forming the conductive film **104**, the insulating films **106** and **107**, the oxide semiconductor film **108**, the conductive film **112_1**, the conductive film **112a_2** (the regions **112a_2a** and **112a_2b**), the conductive film **112b_2** (the regions **112b_2a** and **112b_2b**), the conductive film **112a_3**, and the conductive film **112b_3** are the same as those of the method for forming the transistor **100J**; thus, FIG. **14** and FIG. **15** can be referred to.

(291) Note that in this embodiment, a laminated film in which a 50-nm-thick tungsten film as the conductive film **112_1** used in the conductive film **112**, a 200-nm-thick copper film as the conductive film **1122**, and a 50-nm-thick titanium film as the conductive film **1123** are stacked in this order is used.

(292) Next, the conductive film **112_1** is processed using the conductive films **112a_2**, **112b_2**, and **112_3** as masks, whereby the island-shaped conductive films **112a_1** and **112b_1** separated from each other are formed. When the process is performed, the conductive film **112a** that includes the conductive film **112a_1**, the conductive film **112a_2** (the regions **112a_2a** and **112a_2b**), and the conductive film **112a_3** and the conductive film **112b** that includes the conductive film **112b_1**, the conductive film **112b_2** (the regions **112b_2a** and **112b_2b**), and the conductive film **112b_3** are formed (see FIGS. **18(A)** and (B)).

(293) As a method for forming the conductive films **112a** and **112b**, a method similar to that for the transistor **100J** can be used.

(294) Next, the insulating films **114** and **116** and the insulating film **118** are formed over the oxide semiconductor film **108** and the conductive films **112a** and **112b** (see FIGS. **18(C)**, (D), (E), and (F)). As methods for forming the insulating films **114**, **116**, and **118**, methods similar to those for the transistor **100J** can be used.

(295) After the insulating film **118** is formed, heat treatment similar to the first heat treatment or the second heat treatment described above (hereinafter referred to as third heat treatment) may be performed. The heat treatment at a temperature of lower than 400° C., preferably lower than 375° C., more preferably higher than or equal to 180° C. and lower than or equal to 350° C. is thus performed after the addition of oxygen to the insulating film **116** when the insulating film **118** is formed, whereby oxygen or excess oxygen in the insulating film **116** can be transferred to the oxide semiconductor film **108** (particularly, the oxide semiconductor film **108b**) and compensate for oxygen vacancies in the oxide semiconductor film **108**.

(296) Here, oxygen transferred to the oxide semiconductor film **108** is described using FIG. **20**.

FIG. 20 is a model diagram illustrating oxygen transferred to the oxide semiconductor film **108** due to the substrate temperature at the time of forming the insulating film **118** (typically, lower than 375° C.) or the second heat treatment after the formation of the insulating film **118** (typically, lower than 375° C.). Note that in FIG. 20, oxygen (oxygen radicals, oxygen atoms, or oxygen molecules) shown in the oxide semiconductor film **108** is illustrated by arrows of broken lines.

(297) In the oxide semiconductor film **108** in FIG. 20, oxygen vacancies are compensated for by oxygen transferred from insulating films that are close to the oxide semiconductor film **108** (here, the insulating films **107** and **114**). Specifically, in the semiconductor device of one embodiment of the present invention, the insulating film **107** includes an excess oxygen region because an oxygen gas is used at the time of forming the oxide semiconductor film **108b** by sputtering and oxygen is added to the insulating film **107**. Furthermore, the insulating film **116** includes an excess oxygen region because an oxygen gas is used at the time of forming the oxide conductive film by sputtering and oxygen is added to the insulating film **116**. Thus, in the oxide semiconductor film **108** between the insulating films including the excess oxygen regions, oxygen vacancies can be favorably compensated for.

(298) Then, a mask is formed over the insulating film **118** through a lithography process, and the opening **152c** is formed in a desired region in the insulating films **114**, **116**, and **118**. Note that the opening **152c** is formed to reach the conductive film **112b** (see FIGS. 19(A) and (B)).

(299) Next, the conductive films **120a** and **120b** are formed over the insulating film **116** to cover the opening **152c** (see FIGS. 19(C) and (D)). For the conductive films **120a** and **120b**, a method similar to that for the transistor **100J** can be used.

(300) Through the above process, the transistor **100M** illustrated in FIG. 10(C) can be manufactured.

(301) The structure and method described above in this embodiment can be used in combination with the other structures and methods described in the other embodiments, as appropriate.

Embodiment 2

(302) In this embodiment, the composition of an oxide semiconductor, the structure of an oxide semiconductor, and the like that can be used in one embodiment of the present invention are described with reference to FIG. 21 to FIG. 28.

(303) <2-1. Composition of Oxide Semiconductor>

(304) First, the composition of an oxide semiconductor is described.

(305) An oxide semiconductor preferably contains at least indium or zinc. In particular, indium and zinc are preferably contained. In addition, aluminum, gallium, yttrium, tin, or the like is preferably contained. Furthermore, one or more kinds selected from boron, silicon, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, magnesium, and the like may be contained.

(306) Here, the case where an oxide semiconductor contains indium, an element M, and zinc is considered. Note that the element M is aluminum, gallium, yttrium, tin, or the like. Other elements that can be used as the element M are boron, silicon, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, magnesium, and the like. Note that a plurality of the above elements may be combined as the element M, in some cases.

(307) First, preferred ranges of the atomic ratio of indium, the element M, and zinc contained in an oxide semiconductor of one embodiment of the present invention are described with reference to FIG. 21(A), FIG. 21(B), and FIG. 21(C). Note that the proportion of oxygen atoms is not shown in FIG. 21. Furthermore, the terms of the atomic ratio of indium, the element M, and zinc contained in the oxide semiconductor are denoted by [In], [M], and [Zn], respectively.

(308) In FIG. 21(A), FIG. 21(B), and FIG. 21(C), broken lines indicate a line where the atomic ratio $[In]:[M]:[Zn]=(1+\alpha):(1-\alpha):1$, ($-1 \leq \alpha \leq 1$), a line where the atomic ratio $[In]:[M]:[Zn]=(1+\alpha):(1-\alpha):2$, a line where the atomic ratio $[In]:[M]:[Zn]=(1+\alpha):(1-\alpha):3$, a line where the atomic ratio $[In]:[M]:[Zn]=(1+\alpha):(1-\alpha):4$, and a line where the atomic ratio $[In]:[M]:[Zn]=(1+\alpha):(1-\alpha):5$.

(309) Furthermore, dashed-dotted lines indicate a line where the atomic ratio $[\text{In}]:[\text{M}]:[\text{Zn}]=1:1:\beta$ ($\beta \geq 0$), a line where the atomic ratio $[\text{In}]:[\text{M}]:[\text{Zn}]=1:2:\beta$, a line where the atomic ratio $[\text{In}]:[\text{M}]:[\text{Zn}]=1:3:\beta$, a line where the atomic ratio $[\text{In}]:[\text{M}]:[\text{Zn}]=1:4:\beta$, a line where the atomic ratio $[\text{In}]:[\text{M}]:[\text{Zn}]=2:1:\beta$, and a line where the atomic ratio $[\text{In}]:[\text{M}]:[\text{Zn}]=5:1:\beta$.

(310) Furthermore, a dashed double-dotted line indicates a line where the atomic ratio $[\text{In}]:[\text{M}]:[\text{Zn}]= (1+\gamma):2:(1-\gamma)$ ($-1 \leq \gamma \leq 1$). Furthermore, the oxide semiconductor shown in FIG. 21 with an atomic ratio of $[\text{In}]:[\text{M}]:[\text{Zn}]=0:2:1$ or a neighborhood thereof is likely to have a spinel crystal structure.

(311) FIG. 21(A) and FIG. 21(B) show examples of the preferred ranges of the atomic ratio of indium, the element M, and zinc contained in an oxide semiconductor of one embodiment of the present invention.

(312) FIG. 22 shows the crystal structure of InMZnO.sub.4 in which $[\text{In}]:[\text{M}]:[\text{Zn}]=1:1:1$ as an example. Furthermore, FIG. 22 shows the crystal structure of InMZnO.sub.4 observed from a direction parallel to a b-axis. Note that a metal element in a layer containing M, Zn, and oxygen (hereinafter, referred to as an (M,Zn) layer) in FIG. 22 represents the element M or zinc. In that case, the proportion of the element M is the same as that of zinc. The element M and zinc can be replaced with each other and are arranged randomly.

(313) InMZnO.sub.4 has a layered crystal structure (also referred to as a layered structure) and includes one layer that contains indium and oxygen (hereinafter, In layer) for every two (M,Zn) layers that contain the element M, zinc, and oxygen, as shown in FIG. 22.

(314) Furthermore, indium and the element M can be replaced with each other. Therefore, the element M in the (M,Zn) layer can be replaced with indium, and the layer can also be referred to as an (In,M,Zn) layer. In that case, a layered structure that contains two (In,M,Zn) layers with respect to one In layer is obtained.

(315) An oxide semiconductor whose atomic ratio $[\text{In}]:[\text{M}]:[\text{Zn}]=1:1:2$ has a layered structure that contains three (M,Zn) layers with respect to one In layer. That is, when $[\text{Zn}]$ is higher than $[\text{In}]$ and $[\text{M}]$, the proportion of the (M,Zn) layer to the In layer becomes higher when the oxide semiconductor is crystallized.

(316) Note that in the case where the number of (M,Zn) layers with respect to one In layer is not an integer in the oxide semiconductor, it might have a plurality of kinds of layered structures where the number of (M,Zn) layers with respect to one In layer is an integer. For example, in the case of $[\text{In}]:[\text{M}]:[\text{Zn}]=1:1:1.5$, layered structures in which a layered structure including two (M,Zn) layers with respect to one In layer and a layered structure including three (M,Zn) layers with respect to one In layer are mixed can be obtained in some cases.

(317) For example, in the case where the oxide semiconductor is deposited with a sputtering apparatus, a film having an atomic ratio deviated from the atomic ratio of a target is formed. In particular, $[\text{Zn}]$ in the film might be smaller than $[\text{Zn}]$ in the target depending on the substrate temperature in deposition.

(318) Furthermore, a plurality of phases (e.g., two phases or three phases) exist in the oxide semiconductor in some cases. For example, with an atomic ratio $[\text{In}]:[\text{M}]:[\text{Zn}]$ that is close to $0:2:1$, two phases of a spinel crystal structure and a layered crystal structure are likely to exist. In addition, with an atomic ratio $[\text{In}]:[\text{M}]:[\text{Zn}]$ that is close to $1:0:0$, two phases of a bixbyite crystal structure and a layered crystal structure are likely to exist. In the case where a plurality of phases exist in the oxide semiconductor, a grain boundary (also referred to as a grain boundary) might be formed between different crystal structures. In addition, when the indium content is increased, the carrier mobility (electron mobility) of the oxide semiconductor can be increased. This is because in an oxide semiconductor containing indium, the element M, and zinc, the s orbital of heavy metal mainly contributes to carrier transfer, and when the indium content is increased, overlaps of the s orbitals are increased; therefore, an oxide semiconductor having a high content of indium has higher carrier mobility than an oxide semiconductor having a low content of indium.

(319) In contrast, when the indium content and the zinc content in an oxide semiconductor become lower, carrier mobility becomes lower. Thus, with an atomic ratio of [In]:[M]:[Zn]=0:1:0 and an atomic ratio in the neighborhood thereof (e.g., a region C in FIG. 21(C)), insulation performance becomes better.

(320) Accordingly, an oxide semiconductor of one embodiment of the present invention preferably has an atomic ratio represented by a region A in FIG. 21(A) that is likely to have a layered structure with high carrier mobility and a few grain boundaries.

(321) Furthermore, a region B in FIG. 21(B) represents [In]:[M]:[Zn]=4:2:3 to 4.1 and the neighborhood thereof. The neighborhood includes an atomic ratio of [In]:[M]:[Zn]=5:3:4, for example. An oxide semiconductor with an atomic ratio represented by the region B is an excellent oxide semiconductor that has particularly high crystallinity and high carrier mobility.

(322) Note that conditions where an oxide semiconductor forms a layered structure are not uniquely determined by an atomic ratio. There is a difference in the degree of difficulty in forming a layered structure among atomic ratios. In contrast, even with the same atomic ratio, whether a layered structure is formed or not depends on a formation condition. Therefore, the illustrated regions represent an atomic ratio with which an oxide semiconductor has a layered structure, and boundaries of the region A to the region C are not clear.

(323) <2-2. Structure in which Oxide Semiconductor is Used for Transistor>

(324) Next, the case in which the above oxide semiconductor is used for a transistor is described.

(325) Note that when the above oxide semiconductor is used for a transistor, carrier scattering or the like at a grain boundary can be reduced; thus, the transistor can have high field-effect mobility. In addition, the transistor with high reliability can be obtained. Furthermore, an oxide semiconductor with low carrier density is preferably used for the transistor. For example, the carrier density of an oxide semiconductor is lower than $8 \times 10^{11} \text{ cm}^{-3}$, preferably lower than $1 \times 10^{11} \text{ cm}^{-3}$, more preferably lower than $1 \times 10^{10} \text{ cm}^{-3}$, and greater than or equal to $1 \times 10^{-9} \text{ cm}^{-3}$.

(326) Note that a highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor has few carrier generation sources and thus can have a low carrier density. Furthermore, a highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor has a low density of defect states and accordingly has a low density of trap states in some cases.

(327) Furthermore, charge trapped by the trap states in the oxide semiconductor takes a long time to disappear and may behave like fixed charge. Thus, the transistor whose channel region is formed in the oxide semiconductor having a high density of trap states has unstable electrical characteristics in some cases.

(328) Thus, in order to stabilize electrical characteristics of the transistor, a reduction in concentration of impurities in the oxide semiconductor is effective. Furthermore, in order to reduce the concentration of impurities in the oxide semiconductor, the concentration of impurities in a film adjacent thereto is preferably reduced. Impurities include hydrogen, nitrogen, alkali metal, alkaline earth metal, iron, nickel, silicon, and the like.

(329) Here, the influence of impurities in the oxide semiconductor is described.

(330) When silicon or carbon that is one of Group 14 elements is contained in the oxide semiconductor, defect states are formed in the oxide semiconductor. Thus, the concentration of silicon or carbon in the oxide semiconductor and the concentration of silicon or carbon in the vicinity of an interface with the oxide semiconductor (the concentration measured by secondary ion mass spectrometry (SIMS: Secondary Ion Mass Spectrometry)) is set to lower than or equal to $2 \times 10^{18} \text{ atoms/cm}^3$, preferably lower than or equal to $2 \times 10^{17} \text{ atoms/cm}^3$.

(331) Furthermore, when the oxide semiconductor contains alkali metal or alkaline earth metal, defect states are formed and carriers are generated, in some cases. Thus, a transistor including an oxide semiconductor that contains alkali metal or alkaline earth metal is likely to have normally-on

characteristics. Therefore, it is preferable to reduce the concentration of alkali metal or alkaline earth metal in the oxide semiconductor. Specifically, the concentration of alkali metal or alkaline earth metal in the oxide semiconductor obtained by SIMS is set to lower than or equal to 1×10^{18} atoms/cm³, and preferably lower than or equal to 2×10^{16} atoms/cm³. (332) Furthermore, when the oxide semiconductor contains nitrogen, it easily becomes n-type because electrons that are carriers are generated to increase the carrier density. As a result, a transistor whose semiconductor includes an oxide semiconductor that contains nitrogen is likely to have normally-on characteristics. For this reason, nitrogen in the oxide semiconductor is preferably reduced as much as possible; the nitrogen concentration in the oxide semiconductor measured by SIMS is set, for example, lower than 5×10^{19} atoms/cm³, preferably lower than or equal to 5×10^{18} atoms/cm³, more preferably lower than or equal to 1×10^{18} atoms/cm³, and still more preferably lower than or equal to 5×10^{17} atoms/cm³. (333) Furthermore, hydrogen contained in an oxide semiconductor reacts with oxygen bonded to a metal atom to be water, and thus an oxygen vacancy is formed in some cases. Entry of hydrogen into the oxygen vacancy generates an electron that is a carrier in some cases. Furthermore, in some cases, part of hydrogen is bonded to oxygen bonded to a metal atom to generate an electron that is a carrier. Thus, a transistor including an oxide semiconductor that contains hydrogen is likely to have normally-on characteristics. Accordingly, it is preferable that hydrogen in the oxide semiconductor be reduced as much as possible. Specifically, the hydrogen concentration in the oxide semiconductor measured by SIMS is set lower than 1×10^{20} atoms/cm³, preferably lower than 1×10^{19} atoms/cm³, more preferably lower than 5×10^{18} atoms/cm³, and still more preferably lower than 1×10^{18} atoms/cm³. (334) When an oxide semiconductor with sufficiently reduced impurities is used for a channel formation region in a transistor, stable electrical characteristics can be provided. (335) <2-3. Layered Structure of Oxide Semiconductor> (336) Next, the case where the oxide semiconductor has a two-layer structure or a three-layer structure is described. A band diagram of insulators that are in contact with a layered structure of an oxide semiconductor **51**, an oxide semiconductor **S2**, and an oxide semiconductor **S3** and a band diagram of insulators that are in contact with a layered structure of the oxide semiconductor **S2** and the oxide semiconductor **S3** are described with reference to FIG. **23**. (337) FIG. **23(A)** is an example of a band diagram of a layered structure including an insulator **I1**, the oxide semiconductor **51**, the oxide semiconductor **S2**, the oxide semiconductor **S3**, and an insulator **I2** in a film thickness direction. Furthermore, FIG. **23(B)** is an example of a band diagram of a layered structure including the insulator **I1**, the oxide semiconductor **S2**, the oxide semiconductor **S3**, and the insulator **I2** in a film thickness direction. Note that for easy understanding, the band diagrams show the energy level of the conduction band minimum (E_c) of the insulator **I1**, the oxide semiconductor **51**, the oxide semiconductor **S2**, the oxide semiconductor **S3**, and the insulator **I2**. (338) The energy level of the conduction band minimum of the oxide semiconductor **51** and the oxide semiconductor **S3** is closer to the vacuum level than that of the oxide semiconductor **S2**; typically, a difference in energy level between the conduction band minimum of the oxide semiconductor **S2** and the conduction band minimum of the oxide semiconductor **51** and the oxide semiconductor **S3** is preferably greater than or equal to 0.15 eV or greater than or equal to 0.5 eV, and less than or equal to 2 eV or less than or equal to 1 eV. That is, it is preferable that the electron affinity of the oxide semiconductor **S2** be higher than the electron affinity of the oxide semiconductors **S1** and **S3**, and the difference between the electron affinity of the oxide semiconductor **51** and the oxide semiconductor **S3** and the electron affinity of the oxide semiconductor **S2** be greater than or equal to 0.15 eV or greater than or equal to 0.5 eV, and less than or equal to 2 eV or less than or equal to 1 eV. (339) As shown in FIGS. **23(A)** and **23(B)**, the energy level of the conduction band minimum of

the oxide semiconductor 51, the oxide semiconductor S2, and the oxide semiconductor S3 gradually changes. In other words, the energy level of the conduction band minimum continuously changes or is continuously connected. In order to obtain such a band diagram, the density of defect states in a mixed layer formed at an interface between the oxide semiconductor 51 and the oxide semiconductor S2 or an interface between the oxide semiconductor S2 and the oxide semiconductor S3 is preferably decreased.

(340) Specifically, when the oxide semiconductor 51 and the oxide semiconductor S2 or the oxide semiconductor S2 and the oxide semiconductor S3 contain the same element (as a main component) in addition to oxygen, a mixed layer with a low density of defect states can be formed. For example, in the case where the oxide semiconductor S2 is an In—Ga—Zn oxide semiconductor, it is preferable to use an In—Ga—Zn oxide semiconductor, a Ga—Zn oxide semiconductor, gallium oxide, or the like as the oxide semiconductor 51 and the oxide semiconductor S3.

(341) At this time, the oxide semiconductor S2 becomes a main carrier path. Since the density of defect states at the interface between the oxide semiconductor 51 and the oxide semiconductor S2 and the interface between the oxide semiconductor S2 and the oxide semiconductor S3 can be decreased, the influence of interface scattering on carrier conduction is small, and high on-state current can be obtained.

(342) When an electron is trapped in a trap state, the trapped electron behaves like fixed charge; thus, the threshold voltage of the transistor is shifted in a positive direction. By providing the oxide semiconductor 51 and the oxide semiconductor S3, the trap state can be apart from the oxide semiconductor S2. This structure can prevent the positive shift of the threshold voltage of the transistor.

(343) A material whose conductivity is sufficiently lower than that of the oxide semiconductor S2 is used for the oxide semiconductor 51 and the oxide semiconductor S3. In that case, the oxide semiconductor S2, the interface between the oxide semiconductor S2 and the oxide semiconductor S1, and the interface between the oxide semiconductor S2 and the oxide semiconductor S3 mainly function as a channel region. For example, an oxide semiconductor with high insulation performance and the atomic ratio represented by the region C in FIG. 21(C) can be used as the oxide semiconductor S1 and the oxide semiconductor S3. Note that the region C in FIG. 21(C) represents the atomic ratio of [In]:[M]:[Zn]=0:1:0 or the neighborhood thereof.

(344) In the case where an oxide semiconductor with the atomic ratio represented by the region A is used as the oxide semiconductor S2, it is particularly preferable to use, as the oxide semiconductor S1 and the oxide semiconductor S3, an oxide semiconductor with [M]/[In] of greater than or equal to 1, preferably greater than or equal to 2. In addition, it is suitable to use, as the oxide semiconductor S3, an oxide semiconductor with sufficiently high insulation performance and [M]/([Zn]+[In]) of greater than or equal to 1.

(345) <2-4. Structure of Oxide Semiconductor>

(346) The structure of an oxide semiconductor is described below.

(347) An oxide semiconductor is classified into a single-crystal oxide semiconductor and a non-single-crystal oxide semiconductor. Non-single-crystal oxide semiconductors include a CAAC-OS (c-axis-aligned crystalline oxide semiconductor), a polycrystalline oxide semiconductor, an nc-OS (nanocrystalline oxide semiconductor), an amorphous-like oxide semiconductor (a-like OS: amorphous-like oxide semiconductor), and an amorphous oxide semiconductor.

(348) From another perspective, an oxide semiconductor is classified into an amorphous oxide semiconductor and a crystalline oxide semiconductor. Crystalline oxide semiconductors include a single-crystal oxide semiconductor, a CAAC-OS, a polycrystalline oxide semiconductor, and an nc-OS.

(349) An amorphous structure is generally thought to be isotropic and have no non-uniform structure, to be metastable and have no fixed atomic arrangement, to have a flexible bond angle,

and to have a short-range order but have no long-range order, for example.

(350) That is, a stable oxide semiconductor cannot be called a completely amorphous (completely amorphous) oxide semiconductor. Moreover, an oxide semiconductor that is not isotropic (e.g., with a periodic structure in a microscopic region) cannot be called a completely amorphous oxide semiconductor. In contrast, an a-like OS, which is not isotropic, has an unstable structure that contains a void (also referred to as a void). Because of its instability, an a-like OS is close to an amorphous oxide semiconductor in terms of physical properties.

(351) <<CAAC-OS>>

(352) First, a CAAC-OS is described.

(353) A CAAC-OS is one kind of oxide semiconductors having a plurality of c-axis aligned crystal parts (also referred to as pellets).

(354) The case where a CAAC-OS is analyzed by X-ray diffraction (XRD: X-Ray Diffraction) is described. For example, when the structure of a CAAC-OS including an InGaZnO.sub.4 crystal, which is classified into the space group R-3m, is analyzed by an out-of-plane method, a peak appears at a diffraction angle (2θ) of around 31° as shown in FIG. 24(A). This peak is derived from the (009) plane of the InGaZnO.sub.4 crystal, which indicates that crystals in the CAAC-OS have c-axis alignment and that the c-axes are aligned in the direction substantially perpendicular to a surface over which the CAAC-OS is formed (also referred to as a formation surface) or a top surface. Note that a peak sometimes appears at 2θ of around 36° in addition to the peak at 2θ of around 31° . The peak at 2θ of around 36° is attributed to a crystal structure classified into the space group Fd-3m. Thus, it is preferred that the CAAC-OS do not show the peak.

(355) On the other hand, in structural analysis by an in-plane method in which an X-ray is incident on the CAAC-OS in the direction parallel to the formation surface, a peak appears at 2θ of around 56° . This peak is attributed to the (110) plane of the InGaZnO.sub.4 crystal. Then, when analysis (ϕ scan) is performed with 2θ fixed at around 56° while the sample is rotated around a normal vector to the sample surface as an axis (ϕ axis), as shown in FIG. 24(B), a peak is not clearly observed. In contrast, in the case where single-crystal InGaZnO.sub.4 is subjected to ϕ scan with 2θ fixed at around 56° , as shown in FIG. 24(C), six peaks that are derived from crystal planes equivalent to the (110) plane are observed. Accordingly, the structural analysis using XRD shows that the orientations of the a-axes and b-axes are irregular in the CAAC-OS.

(356) Next, a CAAC-OS analyzed by electron diffraction is described. For example, when an electron beam with a probe diameter of 300 nm is incident on a CAAC-OS including an InGaZnO.sub.4 crystal in the direction parallel to the formation surface of the CAAC-OS, a diffraction pattern (also referred to as a selected-area electron diffraction pattern) shown in FIG. 24(D) may be shown. This diffraction pattern includes spots derived from the (009) plane of the InGaZnO.sub.4 crystal. Thus, the electron diffraction also indicates that pellets included in the CAAC-OS have c-axis alignment and that the c-axes are oriented in the direction substantially perpendicular to the formation surface or the top surface. Meanwhile, FIG. 24(E) shows a diffraction pattern obtained in such a manner that an electron beam with a probe diameter of 300 nm is incident on the same sample in the direction perpendicular to the sample surface. In FIG. 24(E), a ring-like diffraction pattern is observed. Thus, the electron diffraction using an electron beam with a probe diameter of 300 nm also indicates that the a-axes and b-axes of the pellets included in the CAAC-OS do not have orientation. Note that the first ring in FIG. 24(E) is considered to be derived from the (010) plane, the (100) plane, and the like of the InGaZnO.sub.4 crystal. Furthermore, the second ring in FIG. 24(E) is considered to be derived from the (110) plane and the like.

(357) Furthermore, in a combined analysis image (also referred to as a high-resolution TEM image) of a bright-field image and a diffraction pattern of a CAAC-OS, which is observed using a transmission electron microscope (TEM: Transmission Electron Microscope), a plurality of pellets can be observed. However, even in the high-resolution TEM image, a boundary between pellets,

that is, a crystal grain boundary (also referred to as a grain boundary) cannot be clearly observed in some cases. Thus, it can be said that in the CAAC-OS, a reduction in electron mobility due to the crystal grain boundary is less likely to occur.

(358) FIG. 25(A) shows a high-resolution TEM image of a cross section of the CAAC-OS that is observed from the direction substantially parallel to the sample surface. For observation of the high-resolution TEM image, a spherical aberration corrector (Spherical Aberration Corrector) function was used. The high-resolution TEM image obtained with a spherical aberration corrector function is particularly referred to as a Cs-corrected high-resolution TEM image. The Cs-corrected high-resolution TEM image can be observed with, for example, an atomic resolution analytical electron microscope JEM-ARM200F manufactured by JEOL Ltd.

(359) FIG. 25(A) shows pellets in which metal atoms are arranged in a layered manner. It is found that the size of a pellet is greater than or equal to 1 nm or greater than or equal to 3 nm. Thus, the pellet can also be referred to as a nanocrystal (nc: nanocrystal). Furthermore, the CAAC-OS can also be referred to as an oxide semiconductor including CANC (C-Axis Aligned nanocrystals). A pellet reflects unevenness of a formation surface or a top surface of the CAAC-OS film and is parallel to the formation surface or the top surface of the CAAC-OS.

(360) Furthermore, FIG. 25(B) and FIG. 25(C) show Cs-corrected high-resolution TEM images of a plane of the CAAC-OS observed from the direction substantially perpendicular to the sample surface. FIG. 25(D) and FIG. 25(E) are images obtained by image processing of FIG. 25(B) and FIG. 25(C). The method of image processing is described below. First, FIG. 25(B) is subjected to fast Fourier transform (FFT: Fast Fourier Transform) treatment to obtain an FFT image. Then, mask processing is performed such that a range of from 2.8 nm.^{sup.}−1 to 5.0 nm.^{sup.}−1 from the origin in the obtained FFT image remains. Next, the FFT image subjected to the mask processing is subjected to inverse fast Fourier transform (IFFT: Inverse Fast Fourier Transform) treatment to obtain a processed image. The image obtained in this manner is referred to as an FFT filtering image. The FFT filtering image is a Cs-corrected high-resolution TEM image from which a periodic component is extracted, and shows a lattice arrangement.

(361) In FIG. 25(D), a portion where the lattice arrangement is broken is shown by dashed lines. A region surrounded by dashed lines is to one pellet. The portion denoted by the dashed lines is a junction of pellets. The dashed lines draw a hexagon, which means that the pellet has a hexagonal shape. Note that the shape of the pellet is not always a regular hexagon but is a non-regular hexagon in many cases.

(362) In FIG. 25(E), a dotted line denotes a portion between a region with a regular lattice arrangement and another region with a regular lattice arrangement. A clear crystal grain boundary cannot be observed even in the vicinity of the dotted line. When a lattice point in the vicinity of the dotted line is regarded as a center and surrounding lattice points are joined, a distorted hexagon, a distorted pentagon, and/or a distorted heptagon can be formed, for example. That is, a lattice arrangement is distorted so that formation of a crystal grain boundary is inhibited. This is probably because the CAAC-OS can tolerate distortion owing to a low density of the atomic arrangement in an a-b plane direction, the interatomic bond distance changed by substitution of a metal element, and the like.

(363) As described above, the CAAC-OS has c-axis alignment, a plurality of pellets (nanocrystals) are connected in the a-b plane direction, and the crystal structure has distortion. For this reason, the CAAC-OS can also be referred to as a CAA crystal (c-axis-aligned a-b-plane-anchored crystal).

(364) The CAAC-OS is an oxide semiconductor with high crystallinity. Entry of impurities, formation of defects, or the like might decrease the crystallinity of an oxide semiconductor, which means that the CAAC-OS is an oxide semiconductor having few impurities and defects (e.g., oxygen vacancies).

(365) Note that an impurity is an element other than the main components of an oxide semiconductor, such as hydrogen, carbon, silicon, or a transition metal element. For example, an

element having stronger bonding force to oxygen than a metal element constituting part of an oxide semiconductor, such as silicon, extracts oxygen from the oxide semiconductor, which results in a disordered atomic arrangement and reduced crystallinity of the oxide semiconductor. Furthermore, a heavy metal such as iron or nickel, argon, carbon dioxide, or the like has a large atomic radius (or molecular radius), and thus disturbs the atomic arrangement and decreases the crystallinity of the oxide semiconductor.

(366) <<nc-OS>>

(367) Next, an nc-OS is described.

(368) The case where an nc-OS is analyzed by XRD is described. For example, when the structure of an nc-OS is analyzed by an out-of-plane method, a peak indicating orientation does not appear. That is, a crystal of an nc-OS does not have orientation.

(369) Furthermore, for example, when an electron beam with a probe diameter of 50 nm is incident on a 34-nm-thick region of a thinned nc-OS including an InGaZnO.sub.4 crystal in the direction parallel to the formation surface, a ring-like diffraction pattern (nanobeam electron diffraction pattern) shown in FIG. 26(A) is observed. Furthermore, FIG. 26(B) shows a diffraction pattern (nanobeam electron diffraction pattern) obtained when an electron beam with a probe diameter of 1 nm is incident on the same sample. In FIG. 26(B), a plurality of spots are observed in a ring-like region. Thus, ordering in an nc-OS is not observed when an electron beam with a probe diameter of 50 nm is incident on the nc-OS but ordering is observed when an electron beam with a probe diameter of 1 nm is incident on the nc-OS.

(370) Furthermore, when an electron beam with a probe diameter of 1 nm is incident on a region with a thickness less than 10 nm, an electron diffraction pattern in which spots are arranged in an approximately regular hexagonal shape as shown in FIG. 26(C) is observed in some cases. This means that an nc-OS has a well-ordered region, that is, a crystal, in the thickness range of less than nm. Note that an electron diffraction pattern having regularity is not observed in some regions because crystals are aligned in various directions.

(371) FIG. 26(D) shows a Cs-corrected high-resolution TEM image of a cross section of an nc-OS observed from the direction substantially parallel to the formation surface. In the high-resolution TEM image, the nc-OS includes a region where a crystal part is observed, such as the part indicated by auxiliary lines, and a region where a crystal part is not clearly observed. In most cases, the size of a crystal part included in the nc-OS is greater than or equal to 1 nm and less than or equal to 10 nm, in particular, greater than or equal to 1 nm and less than or equal to 3 nm. Note that an oxide semiconductor including a crystal part whose size is greater than 10 nm and less than or equal to 100 nm may be referred to as a microcrystalline oxide semiconductor (micro crystalline oxide semiconductor). In a high-resolution TEM image of the nc-OS, for example, a crystal grain boundary is not clearly observed in some cases. Note that there is a possibility that the origin of the nanocrystal is the same as that of a pellet in a CAAC-OS. Thus, a crystal part of the nc-OS may be referred to as a pellet in the following description.

(372) As described above, in the nc-OS, a microscopic region (for example, a region with a size greater than or equal to 1 nm and less than or equal to 10 nm, in particular, a region with a size greater than or equal to 1 nm and less than or equal to 3 nm) has a periodic atomic arrangement. Furthermore, there is no regularity of crystal orientation between different pellets in the nc-OS. Thus, the orientation of the whole film is not observed. Accordingly, in some cases, the nc-OS cannot be distinguished from an a-like OS or an amorphous oxide semiconductor, depending on an analysis method.

(373) Note that since there is no regularity of crystal orientation between the pellets (nanocrystals), the nc-OS can also be referred to as an oxide semiconductor including RANC (Random Aligned nanocrystals) or an oxide semiconductor including NANC (Non-Aligned nanocrystals).

(374) The nc-OS is an oxide semiconductor that has higher regularity than an amorphous oxide semiconductor. Thus, the nc-OS has a lower density of defect states than the a-like OS and the

amorphous oxide semiconductor. Note that there is no regularity of crystal orientation between different pellets in the nc-OS. Therefore, the nc-OS has a higher density of defect states than the CAAC-OS.

(375) <<a-like OS>>

(376) An a-like OS has a structure between that of an nc-OS and that of an amorphous oxide semiconductor.

(377) FIG. 27 shows high-resolution cross-sectional TEM images of an a-like OS. Here, FIG. 27(A) is the high-resolution cross-sectional TEM image of the a-like OS that is taken at the start of the electron irradiation. FIG. 27(B) is the high-resolution cross-sectional TEM image of the a-like OS that is taken after the irradiation with electrons (e) at $4.3 \times 10^{18} \text{ e}^-/\text{nm}^2$. FIG. 27(A) and FIG. 27(B) show that striped bright regions extending vertically are observed in the a-like OS from the start of the electron irradiation. It can also be found that the shape of the bright region changes after the electron irradiation. Note that the bright region is presumably a void or a low-density region.

(378) The a-like OS has an unstable structure because it contains a void. To verify that an a-like OS has an unstable structure compared with a CAAC-OS and an nc-OS, a change in structure caused by electron irradiation is described below.

(379) An a-like OS, an nc-OS, and a CAAC-OS are prepared as samples. Each of the samples is an In—Ga—Zn oxide.

(380) First, a high-resolution cross-sectional TEM image of each sample is obtained. The high-resolution cross-sectional TEM images show that all the samples have crystal parts.

(381) Note that it is known that a unit cell of an $\text{InGaZnO}_{3.4}$ crystal has a structure in which nine layers including three In—O layers and six Ga—Zn—O layers are stacked in the c-axis direction in a layered manner. The distance between the adjacent layers is equivalent to the lattice spacing on the (009) plane (also referred to as d value), and the value is calculated to be 0.29 nm from crystal structural analysis. Accordingly, a portion where the spacing between lattice fringes is greater than or equal to 0.28 nm and less than or equal to 0.30 nm is regarded as a crystal part of $\text{InGaZnO}_{3.4}$ in the following description. Note that each of the lattice fringes corresponds to the a-b plane of the $\text{InGaZnO}_{3.4}$ crystal.

(382) FIG. 28 is an example of investigating the average size of crystal parts (at 22 points to 30 points) in each sample. Note that the length of the lattice fringe is regarded as the crystal part size. FIG. 28 indicates that the crystal part size in the a-like OS increases in accordance with the cumulative electron dose in obtaining TEM images, for example. As shown in FIG. 28, a crystal part with a size of approximately 1.2 nm (also referred to as an initial nucleus) at the start of TEM observation grows to a size of approximately 1.9 nm at a cumulative electron (e) dose of $4.2 \times 10^{18} \text{ e}^-/\text{nm}^2$. In contrast, the crystal part sizes in the nc-OS and the CAAC-OS show few changes from the start of electron irradiation to a cumulative electron dose of $4.2 \times 10^{18} \text{ e}^-/\text{nm}^2$. As shown in FIG. 28, the crystal part sizes in the nc-OS and the CAAC-OS are approximately 1.3 nm and approximately 1.8 nm, respectively, regardless of the cumulative electron dose. Note that for the electron beam irradiation and TEM observation, a Hitachi transmission electron microscope H-9000NAR was used. The conditions of the electron beam irradiation were as follows: the accelerating voltage was 300 kV; the current density was $6.7 \times 10^{-5} \text{ e}^-/(\text{nm}^2 \cdot \text{s})$; and the diameter of an irradiation region was 230 nm.

(383) In this manner, growth of the crystal part in the a-like OS may be induced by electron irradiation. In contrast, in the nc-OS and the CAAC-OS, growth of the crystal part is hardly induced by electron irradiation. This implies that the a-like OS has an unstable structure compared with the nc-OS and the CAAC-OS.

(384) Furthermore, the a-like OS has a lower density than the nc-OS and the CAAC-OS because it contains a void. Specifically, the density of the a-like OS is higher than or equal to 78.6% and lower than 92.3% of the density of the single-crystal oxide semiconductor having the same

composition. Furthermore, the density of the nc-OS and the density of the CAAC-OS are each higher than or equal to 92.3% and lower than 100% of the density of the single-crystal oxide semiconductor having the same composition. It is difficult to deposit an oxide semiconductor having a density lower than 78% of the density of the single-crystal oxide semiconductor.

(385) For example, in an oxide semiconductor in which In:Ga:Zn is 1:1:1 [atomic ratio], the density of single-crystal InGaZnO₄ with a rhombohedral crystal structure is 6.357 g/cm³.

(386) Accordingly, in the oxide semiconductor in which In:Ga:Zn is 1:1:1 [atomic ratio], the density of the a-like OS is higher than or equal to 5.0 g/cm³ and lower than 5.9 g/cm³, for example. Furthermore, in the oxide semiconductor in which In:Ga:Zn is 1:1:1 [atomic ratio], the density of the nc-OS and that of the CAAC-OS are each higher than or equal to 5.9 g/cm³ and lower than 6.3 g/cm³, for example.

(387) Note that in the case where single crystals with the same composition do not exist, single crystals with different compositions are combined at an appropriate ratio, whereby a density corresponding to that of a single crystal with the desired composition can be estimated. The density corresponding to that of the single crystal with the desired composition is estimated using a weighted average with respect to the combination ratio of the single crystals with different compositions. Note that it is preferable to combine as few kinds of single crystals as possible to estimate the density.

(388) As described above, oxide semiconductors have various structures and various properties. Note that an oxide semiconductor may be a laminated film including two or more kinds of an amorphous oxide semiconductor, an a-like OS, an nc-OS, and a CAAC-OS, for example.

(389) <2-5. Carrier Density of Oxide Semiconductor>

(390) Next, the carrier density of an oxide semiconductor is described below.

(391) Factors affecting the carrier density of an oxide semiconductor include oxygen vacancies (V_O) in the oxide semiconductor, impurities in the oxide semiconductor, and the like.

(392) As the number of oxygen vacancies in the oxide semiconductor increases, the density of defect states increases when hydrogen is bonded to the oxygen vacancies (this state is also referred to as V_{OH}). Alternatively, the density of defect states also increases with an increase in the number of impurities in the oxide semiconductor. Hence, the carrier density of an oxide semiconductor can be controlled by controlling the density of defect states in the oxide semiconductor.

(393) Here, a transistor using the oxide semiconductor in a channel region is considered.

(394) The carrier density of the oxide semiconductor is preferably reduced in the case where suppression of the negative shift of the threshold voltage of the transistor or reduction in the off-state current of the transistor is intended. In the case where the carrier density of the oxide semiconductor is reduced, the impurity concentration in the oxide semiconductor is reduced so that the density of defect states can be reduced. In this specification and the like, a state with a low impurity concentration and a low density of defect states is referred to as a highly purified intrinsic or substantially highly purified intrinsic state. The carrier density of a highly purified intrinsic oxide semiconductor is lower than $8 \times 10^{15} \text{ cm}^{-3}$, preferably lower than $1 \times 10^{11} \text{ cm}^{-3}$, more preferably lower than $1 \times 10^{10} \text{ cm}^{-3}$ and is higher than or equal to $1 \times 10^{-9} \text{ cm}^{-3}$.

(395) By contrast, the carrier density of the oxide semiconductor is preferably increased in the case where improvement in the on-state current of the transistor or improvement in the field-effect mobility of the transistor is intended. In the case where the carrier density of the oxide semiconductor is increased, the impurity concentration or the density of defect states in the oxide semiconductor is slightly increased. Alternatively, the bandgap of the oxide semiconductor is narrowed. For example, an oxide semiconductor that has a slightly high impurity concentration or a slightly high density of defect states in the range where an on/off ratio is obtained in the I_d - V_g characteristics of the transistor can be regarded as substantially intrinsic. Furthermore, an oxide

semiconductor that has a high electron affinity and thus has a narrow bandgap to increase the density of thermally excited electrons (carriers) can be regarded as substantially intrinsic. Note that when an oxide semiconductor with higher electron affinity is used, the transistor has lower threshold voltage.

(396) The oxide semiconductor with an increased carrier density has somewhat n-type conductivity; thus, it can be referred to as a “Slightly-n” oxide semiconductor. The carrier density of a substantially intrinsic oxide semiconductor is preferably higher than or equal to $1 \times 10^{15} \text{ cm}^{-3}$ and lower than $1 \times 10^{18} \text{ cm}^{-3}$, more preferably higher than or equal to $1 \times 10^{17} \text{ cm}^{-3}$ and lower than or equal to $1 \times 10^{17} \text{ cm}^{-3}$, still more preferably higher than or equal to $1 \times 10^{19} \text{ cm}^{-3}$ and lower than or equal to $5 \times 10^{16} \text{ cm}^{-3}$, yet more preferably higher than or equal to $1 \times 10^{10} \text{ cm}^{-3}$ and lower than or equal to $1 \times 10^{16} \text{ cm}^{-3}$, and yet still preferably higher than or equal to $1 \times 10^{11} \text{ cm}^{-3}$ and lower than or equal to $1 \times 10^{15} \text{ cm}^{-3}$.

(397) Note that the structure described in this embodiment can be used in combination with the structures described in the other embodiments or the other examples, as appropriate.

Embodiment 3

(398) In this embodiment, an example of a display device that includes the transistor described in the above embodiments is described below using FIG. 29 to FIG. 35.

(399) FIG. 29 is a top view illustrating an example of a display device. A display device 700 in FIG. 29 includes a pixel portion 702 provided over a first substrate 701, a source driver circuit portion 704 and a gate driver circuit portion 706 that are provided over the first substrate 701, a sealant 712 provided to surround the pixel portion 702, the source driver circuit portion 704, and the gate driver circuit portion 706, and a second substrate 705 provided to face the first substrate 701. Note that the first substrate 701 and the second substrate 705 are sealed with the sealant 712. That is, the pixel portion 702, the source driver circuit portion 704, and the gate driver circuit portion 706 are sealed with the first substrate 701, the sealant 712, and the second substrate 705. Note that although not illustrated in FIG. 29, a display element is provided between the first substrate 701 and the second substrate 705.

(400) Furthermore, in the display device 700, an FPC terminal portion 708 (FPC: Flexible printed circuit) that is electrically connected to the pixel portion 702, the source driver circuit portion 704, the gate driver circuit portion 706, and the gate driver circuit portion 706 is provided in a region different from the region that is over the first substrate 701 and surrounded by the sealant 712. Furthermore, an FPC 716 is connected to the FPC terminal portion 708, and a variety of signals and the like are supplied from the FPC 716 to the pixel portion 702, the source driver circuit portion 704, and the gate driver circuit portion 706. Furthermore, a signal line 710 is connected to the pixel portion 702, the source driver circuit portion 704, the gate driver circuit portion 706, and the FPC terminal portion 708. Through the signal line 710, a variety of signals and the like are supplied from the FPC 716 to the pixel portion 702, the source driver circuit portion 704, the gate driver circuit portion 706, and the FPC terminal portion 708.

(401) Furthermore, a plurality of gate driver circuit portions 706 may be provided in the display device 700. Furthermore, an example of the display device 700 in which the source driver circuit portion 704 and the gate driver circuit portion 706 are formed over the first substrate 701 where the pixel portion 702 is also formed is described; however, the structure is not limited thereto. For example, only the gate driver circuit portion 706 may be formed over the first substrate 701, or only the source driver circuit portion 704 may be formed over the first substrate 701. In this case, a substrate over which a source driver circuit, a gate driver circuit, or the like is formed (e.g., a driver circuit board formed using a single-crystal semiconductor film or a polycrystalline semiconductor film) may be formed on the first substrate 701. Note that there is no particular limitation on the method for connecting the separately formed driver circuit board, and a COG (Chip On Glass) method, a wire bonding method, or the like can be used.

(402) Furthermore, the pixel portion **702**, the source driver circuit portion **704**, and the gate driver circuit portion **706** included in the display device **700** include a plurality of transistors, and the transistor that is a semiconductor device of one embodiment of the present invention can be used.

(403) Furthermore, the display device **700** can include a variety of elements. Examples of the elements include electroluminescent (EL) element (e.g., an EL element containing an organic material and an inorganic material, an organic EL element, an inorganic EL element, or an LED), a light-emitting transistor element (a transistor that emits light depending on current), an electron emitter, a liquid crystal element, an electronic ink element, an electrophoretic element, an electrowetting element, a plasma display (PDP), an MEMS (micro electro mechanical systems) display (e.g., a grating light valve (GLV), a digital micromirror device (DMD), a digital micro shutter (DMS) element, or an interferometric modulation (IMOD) element), a piezoelectric ceramic display, and the like.

(404) Furthermore, an example of a display device using an EL element is an EL display. Examples of a display device using an electron emitter include a field emission display (FED) and an SED-type flat panel display (SED: Surface-conduction Electron-emitter Display). Examples of a display device using a liquid crystal element include a liquid crystal display (a transmissive liquid crystal display, a transfective liquid crystal display, a reflective liquid crystal display, a direct-view liquid crystal display, or a projection liquid crystal display). Examples of a display device including an electronic ink element or an electrophoretic element include electronic paper. Note that in the case where a transfective liquid crystal display or a reflective liquid crystal display is obtained, some or all of pixel electrodes function as reflective electrodes. For example, some or all of pixel electrodes contain aluminum, silver, or the like. Furthermore, in this case, a memory circuit such as an SRAM can be provided under the reflective electrodes. Thus, the power consumption can be further reduced.

(405) Note that as a display system of the display device **700**, a progressive system, an interlace system, or the like can be employed. Furthermore, color elements controlled in pixels at the time of color display are not limited to three colors of RGB (R, G, and B represent red, green, and blue, respectively). For example, four pixels of an R pixel, a G pixel, a B pixel, and a W (white) pixel may be used. Alternatively, a color element may be composed of two colors of R, G, and B as in PenTile layout, and the two colors may differ among color elements. Alternatively, one or more of yellow, cyan, magenta, and the like may be added to RGB. Note that the size of a display region may differ between dots of color elements. Note that the disclosed invention is not limited to a color display device and can also be used for a monochrome display device.

(406) Furthermore, a coloring layer (also referred to as a color filter) may be used so that a display device performs full-color display using white light emission (W) for a backlight (e.g., an organic EL element, an inorganic EL element, an LED, or a fluorescent lamp). For the coloring layer, for example, red (R), green (G), blue (B), yellow (Y), and the like can be used in appropriate combination. With the use of the coloring layer, high color reproducibility can be obtained compared with the case without the coloring layer. Here, by providing a region with a coloring layer and a region without a coloring layer, white light in the region without the coloring layer may be directly utilized for display. By partly providing the region without the coloring layer, a decrease in the luminance of a bright image due to the coloring layer can be inhibited, and approximately 20% to 30% of power consumption can be reduced in some cases. Note that in the case where full-color display is performed using a self-luminous element such as an organic EL element or an inorganic EL element, elements may emit light of their respective colors R, G, B, Y, and W. By using a self-luminous element, power consumption may be further reduced compared with the case of using a coloring layer.

(407) Furthermore, as a coloring system, the following may be used: the above-described system (color filter system) in which part of white light emission is converted into red, green, and blue through color filters; a system (three-color system) in which red light emission, green light

emission, and blue light emission are used; and a system (color conversion system or quantum dot system) in which part of blue light emission is converted into red or green.

(408) In this embodiment, structures using a liquid crystal element and an EL element as display elements are described using FIG. 30 and FIG. 32. Note that FIG. 30 is a cross-sectional view taken along dashed-dotted line Q-R in FIG. 29 and is the structure using a liquid crystal element as a display element. Furthermore, FIG. 32 is a cross-sectional view taken along dashed-dotted line Q-R in FIG. 29 and is the structure using an EL element as a display element.

(409) Common portions between FIG. 30 and FIG. 32 are described first, and then, different portions are described below.

(410) <3-1. Description of Common Portions in Display Devices>

(411) The display device 700 in FIG. 30 and FIG. 32 includes a lead wiring portion 711, the pixel portion 702, the source driver circuit portion 704, and the FPC terminal portion 708. Furthermore, the lead wiring portion 711 includes the signal line 710. Furthermore, the pixel portion 702 includes a transistor 750 and a capacitor 790. Furthermore, the source driver circuit portion 704 includes a transistor 752.

(412) The transistor 750 and the transistor 752 each have a structure similar to that of the transistor 100 described above. Note that the transistor 750 and the transistor 752 may have the structures of the other transistors described in the above embodiments.

(413) The transistor used in this embodiment includes an oxide semiconductor film that is highly purified and in which formation of an oxygen vacancy is suppressed. The transistor can have low off-state current. Accordingly, an electrical signal such as an image signal can be held for a long time, and a long writing interval can be set in an on state. Accordingly, the frequency of refresh operation can be reduced, which suppresses power consumption.

(414) In addition, the transistor used in this embodiment can have relatively high field-effect mobility and thus is capable of high-speed operation. For example, when such a transistor capable of high-speed operation is used in a liquid crystal display device, a switching transistor in a pixel portion and a driver transistor used in a driver circuit portion can be formed over the same substrate. That is, no additional semiconductor device formed using a silicon wafer or the like is needed as a driver circuit; therefore, the number of components of the semiconductor device can be reduced. In addition, by using the transistor capable of high-speed operation in the pixel portion, a high-quality image can be provided.

(415) The capacitor 790 includes a lower electrode formed through a step of processing the same conductive film as a conductive film functioning as a first gate electrode of the transistor 750 and an upper electrode formed through a step of processing the same conductive film as a conductive film functioning as a source electrode and a drain electrode of the transistor 750. Furthermore, between the lower electrode and the upper electrode, an insulating film formed through a step of forming the same insulating film as an insulating film functioning as a first gate insulating film of the transistor 750 is provided. That is, the capacitor 790 has a layered structure in which an insulating film functioning as a dielectric film is positioned between the pair of electrodes.

(416) Furthermore, in FIG. 30 and FIG. 32, a planarization insulating film 770 is provided over the transistor 750, the transistor 752, and the capacitor 790.

(417) For the planarization insulating film 770, a heat-resistant organic material such as a polyimide resin, an acrylic resin, a polyimide amide resin, a benzocyclobutene resin, a polyamide resin, or an epoxy resin can be used. Note that the planarization insulating film 770 may be formed by stacking a plurality of insulating films formed using these materials. Furthermore, a structure without the planarization insulating film 770 may be employed.

(418) Furthermore, although FIG. 30 and FIG. 32 illustrate an example in which the transistor 750 included in the pixel portion 702 and the transistor 752 included in the source driver circuit portion 704 have the same structure, there is no limitation thereto. For example, different transistors may be used for the pixel portion 702 and the source driver circuit portion 704. Specifically, a structure

in which a staggered transistor is used in the pixel portion **702** and the inverted staggered transistor described in Embodiment 1 is used in the source driver circuit portion **704**, a structure in which the inverted staggered transistor described in Embodiment 1 is used in the pixel portion **702** and a staggered transistor is used in the source driver circuit portion **704**, and the like are given. Note that the source driver circuit portion **704** can be replaced by a gate driver circuit portion.

(419) Furthermore, the signal line **710** is formed through the same process as the conductive films functioning as source electrodes and drain electrodes of the transistors **750** and **752**. In the case where a material containing a copper element is used for the signal line **710**, for example, signal delay or the like due to wiring resistance is less likely to occur, which enables display on a large screen.

(420) Furthermore, the FPC terminal portion **708** includes a connecting electrode **760**, an anisotropic conductive film **780**, and the FPC **716**. Note that the connecting electrode **760** is formed through the same process as the conductive films functioning as the source electrodes and the drain electrodes of the transistors **750** and **752**. Furthermore, the connecting electrode **760** is electrically connected to a terminal included in the FPC **716** through the anisotropic conductive film **780**.

(421) Furthermore, for example, a glass substrate can be used as the first substrate **701** and the second substrate **705**. Furthermore, as each of the first substrate **701** and the second substrate **705**, a flexible substrate may be used. Examples of the flexible substrate include a plastic substrate.

(422) Furthermore, a structure body **778** is provided between the first substrate **701** and the second substrate **705**. The structure body **778** is a columnar spacer obtained by selective etching of an insulating film and is provided to control the distance (cell gap) between the first substrate **701** and the second substrate **705**. Note that a spherical spacer may be used as the structure body **778**.

(423) Furthermore, a light-blocking film **738** functioning as a black matrix, a coloring film **736** functioning as a color filter, and an insulating film **734** in contact with the light-blocking film **738** and the coloring film **736** are provided on the second substrate **705** side.

(424) <3-2. Structure Example of Display Device Using Liquid Crystal Element>

(425) The display device **700** in FIG. **30** includes a liquid crystal element **775**. The liquid crystal element **775** includes a conductive film **772**, a conductive film **774**, and a liquid crystal layer **776**. The conductive film **774** is provided on the second substrate **705** side and has a function of a counter electrode. The display device **700** in FIG. **30** can display an image in such a manner that transmission or non-transmission of light is controlled by the alignment state in the liquid crystal layer **776** that is changed depending on the voltage applied to the conductive film **772** and the conductive film **774**.

(426) Furthermore, the conductive film **772** is electrically connected to the conductive film functioning as the source electrode and the drain electrode of the transistor **750**. The conductive film **772** is formed over the planarization insulating film **770** and functions as a pixel electrode, that is, one electrode of the display element. Furthermore, the conductive film **772** has a function of a reflective electrode. The display device **700** in FIG. **30** is what is called a reflective color liquid crystal display device that performs display by utilizing external light that is reflected by the conductive film **772** and then passes through the coloring film **736**.

(427) A conductive film that transmits visible light or a conductive film that reflects visible light can be used as the conductive film **772**. For example, a material containing one kind selected from indium (In), zinc (Zn), and tin (Sn) is preferably used for the conductive film that transmits visible light. For example, a material containing aluminum or silver is preferably used for the conductive film that reflects visible light. In this embodiment, a conductive film that reflects visible light is used as the conductive film **772**.

(428) Note that although FIG. **30** illustrates an example in which the conductive film **772** is connected to the conductive film functioning as the drain electrode of the transistor **750**, there is no limitation thereto. For example, as illustrated in FIG. **31**, a structure in which the conductive film

772 is electrically connected to the conductive film functioning as the drain electrode of the transistor 750 through a conductive film 777 functioning as a connecting electrode may be employed. Note that the conductive film 777 is formed through a step of processing the same conductive film as a conductive film functioning as a second gate electrode of the transistor 750 and thus can be formed without an increase in the number of manufacturing steps.

(429) Furthermore, the display device 700 in FIG. 30 is a reflective color liquid crystal display device but is not limited thereto; for example, the display device 700 may be a transmissive color liquid crystal display device using a conductive film that transmits visible light as the conductive film 772. Alternatively, the display device 700 may be what is called a transfective color liquid crystal display device in which a reflective color liquid crystal display device and a transmissive color liquid crystal display device are combined.

(430) Here, FIG. 33 illustrates an example of a transmissive color liquid crystal display device. FIG. 33 is a cross-sectional view taken along dashed-dotted line Q-R in FIG. 29 and illustrates a structure in which a liquid crystal element is used as the display element. Furthermore, the display device 700 illustrated in FIG. 33 is an example of a structure in which a horizontal electric field mode (e.g., an FFS mode) is used as a driving mode of the liquid crystal element. In the case of the structure illustrated in FIG. 33, an insulating film 773 is provided over the conductive film 772 functioning as a pixel electrode, and the conductive film 774 is provided over the insulating film 773. In such a case, the conductive film 774 has a function of a common electrode (also referred to as a common electrode), and an electric field generated between the conductive film 772 and the conductive film 774 through the insulating film 773 can control the alignment state in the liquid crystal layer 776.

(431) Furthermore, although not illustrated in FIG. 30 and FIG. 33, one or both of the conductive film 772 and the conductive film 774 may be provided with an alignment film on a side in contact with the liquid crystal layer 776. Furthermore, although not illustrated in FIG. 30 and FIG. 33, an optical member (optical substrate) or the like, such as a polarizing member, a retardation member, or an anti-reflection member, may be provided as appropriate. For example, circular polarization by a polarizing substrate and a retardation substrate may be used. In addition, a backlight, a sidelight, or the like may be used as a light source.

(432) In the case where a liquid crystal element is used as the display element, thermotropic liquid crystal, low-molecular liquid crystal, high-molecular liquid crystal, polymer dispersed liquid crystal, ferroelectric liquid crystal, anti-ferroelectric liquid crystal, or the like can be used. These liquid crystal materials exhibit a cholesteric phase, a smectic phase, a cubic phase, a chiral nematic phase, an isotropic phase, or the like depending on conditions.

(433) Furthermore, in the case where a horizontal electric field mode is employed, liquid crystal exhibiting a blue phase for which an alignment film is unnecessary may be used. The blue phase is one of liquid crystal phases, which is generated just before a cholesteric phase changes into an isotropic phase when the temperature of cholesteric liquid crystal is increased. Since the blue phase appears only in a narrow temperature range, a liquid crystal composition in which a chiral material is mixed to account for several weight % or more is used for the liquid crystal layer in order to improve the temperature range. The liquid crystal composition containing liquid crystal exhibiting a blue phase and a chiral material has a short response time and optical isotropy, which makes the alignment process unneeded. Furthermore, an alignment film does not need to be provided, and thus, rubbing treatment is not necessary; accordingly, electrostatic discharge damage caused by the rubbing treatment can be prevented, and defects and damage of a liquid crystal display device in the manufacturing process can be reduced. Moreover, the liquid crystal material that exhibits a blue phase has small viewing angle dependence.

(434) Furthermore, in the case where a liquid crystal element is used as the display element, a TN (Twisted Nematic) mode, an IPS (In-Plane-Switching) mode, an FFS (Fringe Field Switching) mode, an ASM (Axially Symmetric aligned Micro-cell) mode, an OCB (Optical Compensated

Birefringence) mode, an FLC (Ferroelectric Liquid Crystal) mode, an AFLC (AntiFerroelectric Liquid Crystal) mode, or the like can be used.

(435) Furthermore, a normally black liquid crystal display device such as a transmissive liquid crystal display device employing a vertical alignment (VA) mode may be used. There are some examples of a vertical alignment mode; for example, an MVA (Multi-Domain Vertical Alignment) mode, a PVA (Patterned Vertical Alignment) mode, or an ASV mode can be employed.

(436) <3-3. Display Device Using Light-Emitting Element>

(437) The display device **700** illustrated in FIG. **32** includes a light-emitting element **782**. The light-emitting element **782** includes the conductive film **772**, an EL layer **786**, and a conductive film **788**. The display device **700** illustrated in FIG. **32** can display an image when the EL layer **786** of the light-emitting element **782** emits light. Note that the EL layer **786** contains an organic compound or an inorganic compound such as a quantum dot.

(438) Materials that can be used for an organic compound include a fluorescent material, a phosphorescent material, and the like. Furthermore, materials that can be used for a quantum dot include a colloidal quantum dot material, an alloyed quantum dot material, a core-shell quantum dot material, a core quantum dot material, and the like. Furthermore, a material including element groups of Group 12 and Group 16, Group 13 and Group 15, or Group 14 and Group 16 may be used. Alternatively, a quantum dot material containing an element such as cadmium (Cd), selenium (Se), zinc (Zn), sulfur (S), phosphorus (P), indium (In), tellurium (Te), lead (Pb), gallium (Ga), arsenic (As), or aluminum (Al) may be used.

(439) Furthermore, in the display device **700** in FIG. **32**, an insulating film **730** is provided over the planarization insulating film **770** and the conductive film **772**. The insulating film **730** covers part of the conductive film **772**. Note that the light-emitting element **782** has a top-emission structure. Thus, the conductive film **788** has a light-transmitting property and transmits light emitted from the EL layer **786**. Note that although the top-emission structure is described as an example in this embodiment, the structure is not limited thereto. For example, a bottom-emission structure in which light is emitted to the conductive film **772** side or a dual-emission structure in which light is emitted to both the conductive film **772** and the conductive film **788** may also be employed.

(440) Furthermore, the coloring film **736** is provided to overlap with the light-emitting element **782**, and the light-blocking film **738** is provided in the lead wiring portion **711** and the source driver circuit portion **704** to overlap with the insulating film **730**. Furthermore, the coloring film **736** and the light-blocking film **738** are covered with the insulating film **734**. Furthermore, a space between the light-emitting element **782** and the insulating film **734** is filled with a sealing film **732**. Note that the structure of the display device **700** is not limited to the example in FIG. **32**, in which the coloring film **736** is provided. For example, a structure without the coloring film **736** may be employed in the case where the EL layer **786** is formed by separate coloring.

(441) <3-4. Structure Example of Display Device Provided with Input/Output Device>

(442) Furthermore, an input/output device may be provided in the display device **700** illustrated in FIG. **32** and FIG. **33**. Examples of the input/output device include a touch panel.

(443) FIG. **34** and FIG. **35** illustrate structures in which the touch panel **791** is provided in the display device **700** in FIG. **32** and FIG. **33**.

(444) FIG. **34** is a cross-sectional view of the structure in which the touch panel **791** is provided in the display device **700** illustrated in FIG. **32**, and FIG. **35** is a cross-sectional view of the structure in which the touch panel **791** is provided in the display device **700** illustrated in FIG. **33**.

(445) First, the touch panel **791** illustrated in FIG. **34** and FIG. **35** is described below.

(446) The touch panel **791** illustrated in FIG. **34** and FIG. **35** is what is called an in-cell touch panel provided between the substrate **705** and the coloring film **736**. The touch panel **791** is formed on the substrate **705** side before the light-blocking film **738** and the coloring film **736** are formed.

(447) Note that the touch panel **791** includes the light-blocking film **738**, an insulating film **792**, an electrode **793**, an electrode **794**, an insulating film **795**, an electrode **796**, and an insulating film

797. A change in the mutual capacitance between the electrode 793 and the electrode 794 can be sensed when an object to be sensed, such as a finger or a stylus, approaches, for example.

(448) Furthermore, a portion where the electrode 793 intersects with the electrode 794 is clearly illustrated above the transistor 750 illustrated in FIG. 34 and FIG. 35. The electrode 796 is electrically connected to the two electrodes 793 between which the electrode 794 is positioned, through openings provided in the insulating film 795. Note that a structure in which a region where the electrode 796 is provided is provided in the pixel portion 702 is illustrated in FIG. 34 and FIG. 35 as an example; however, there is no limitation thereto; for example, the region may be formed in the source driver circuit portion 704.

(449) The electrode 793 and the electrode 794 are provided in a region overlapping with the light-blocking film 738. Furthermore, as illustrated in FIG. 34, it is preferable that the electrode 793 be provided not to overlap with the light-emitting element 782. Furthermore, as illustrated in FIG. 35, it is preferable that the electrode 793 be provided not to overlap with the liquid crystal element 775. In other words, the electrode 793 has an opening in a region overlapping with the light-emitting element 782 and the liquid crystal element 775. That is, the electrode 793 has a mesh shape. With such a structure, a structure in which the electrode 793 does not block light emitted from the light-emitting element 782 can be obtained. Alternatively, a structure in which the electrode 793 does not block light transmitted through the liquid crystal element 775 can be obtained. Thus, since a reduction in luminance caused by providing the touch panel 791 is extremely small, a display device with high visibility and reduced power consumption can be obtained. Note that the electrode 794 can have a similar structure.

(450) Furthermore, since the electrode 793 and the electrode 794 do not overlap with the light-emitting element 782, a metal material having low visible light transmittance can be used for the electrode 793 and the electrode 794. Alternatively, since the electrode 793 and the electrode 794 do not overlap with the liquid crystal element 775, a metal material having low visible light transmittance can be used for the electrode 793 and the electrode 794.

(451) Thus, the resistance of the electrode 793 and the electrode 794 can be reduced compared with an electrode using an oxide material having high visible light transmittance, whereby the sensitivity of the sensor of the touch panel can be increased.

(452) For example, a conductive nanowire may be used for the electrodes 793, 794, and 796. The nanowire has a mean diameter of greater than or equal to 1 nm and less than or equal to 100 nm, preferably greater than or equal to 5 nm and less than or equal to 50 nm, more preferably greater than or equal to 5 nm and less than or equal to 25 nm. Furthermore, as the nanowire, a carbon nanotube or a metal nanowire such as an Ag nanowire, a Cu nanowire, or an Al nanowire is used.

(453) For example, in the case where an Ag nanowire is used for any one of or all of electrodes 793, 794, and 796, the visible light transmittance can be greater than or equal to 89% and the sheet resistance can be greater than or equal to 40 Ω /square and less than or equal to 100 Ω /square.

(454) Furthermore, although the structure of the in-cell touch panel is illustrated in FIG. 34 and FIG. 35, there is no limitation thereto. For example, what is called an on-cell touch panel that is formed over the display device 700, or what is called an out-cell touch panel that is attached to the display device 700 may be used. In this manner, the display device of one embodiment of the present invention can be used in combination with various types of touch panels.

(455) Note that the structures described in this embodiment can be used in combination with the structures described in the other embodiments, as appropriate.

Embodiment 4

(456) In this embodiment, a display device including a semiconductor device of one embodiment of the present invention is described using FIG. 36.

(457) <4. Circuit Configuration of Display Device>

(458) A display device illustrated in FIG. 36(A) includes a region including pixels of display elements (hereinafter, referred to as a pixel portion 502), a circuit portion that is provided outside

the pixel portion **502** and includes a circuit for driving the pixels (hereinafter, referred to as a driver circuit portion **504**), circuits having a function of protecting elements (hereinafter, referred to as protection circuits **506**), and a terminal portion **507**. Note that the protection circuits **506** are not necessarily provided.

(459) Part or the whole of the driver circuit portion **504** is desirably formed over a substrate over which the pixel portion **502** is formed. Thus, the number of components and the number of terminals can be reduced. When part or the whole of the driver circuit portion **504** is not formed over the substrate over which the pixel portion **502** is formed, the part or the whole of the driver circuit portion **504** can be mounted by COG or TAB (Tape Automated Bonding).

(460) The pixel portion **502** includes a plurality of circuits for driving display elements arranged in X rows (X is a natural number of 2 or more) and Y columns (Y is a natural number of 2 or more) (hereinafter, referred to as pixel circuits **501**), and the driver circuit portion **504** includes driver circuits such as a circuit for outputting a signal (scan signal) to select a pixel (hereinafter, referred to as a gate driver **504a**) and a circuit for supplying a signal (data signal) to drive a display element in a pixel (hereinafter, referred to as a source driver **504b**).

(461) The gate driver **504a** includes a shift register or the like. The gate driver **504a** receives a signal for driving the shift register through the terminal portion **507** and outputs a signal. For example, the gate driver **504a** receives a start pulse signal, a clock signal, or the like and outputs a pulse signal. The gate driver **504a** has a function of controlling the potentials of wirings supplied with scan signals (hereinafter, referred to as scan lines GL_1 to GL_X). Note that a plurality of gate drivers **504a** may be provided so that the scan lines GL_1 to GL_X are separately controlled by the plurality of gate drivers **504a**. Alternatively, the gate driver **504a** has a function of supplying an initialization signal. Note that without being limited thereto, the gate driver **504a** can supply another signal.

(462) The source driver **504b** includes a shift register or the like. The source driver **504b** receives a signal (image signal) from which a data signal is generated, as well as a signal for driving the shift register, through the terminal portion **507**. The source driver **504b** has a function of generating a data signal to be written to the pixel circuit **501** from the image signal. In addition, the source driver **504b** has a function of controlling output of a data signal in response to a pulse signal produced by input of a start pulse, a clock signal, or the like. Furthermore, the source driver **504b** has a function of controlling the potentials of wirings supplied with data signals (hereinafter, referred to as data lines DL_1 to DL_Y). Alternatively, the source driver **504b** has a function of supplying an initialization signal. Without being limited thereto, the source driver **504b** can supply another signal.

(463) The source driver **504b** is formed using a plurality of analog switches, for example. The source driver **504b** can output, as data signals, signals obtained by time-dividing the image signal by sequentially turning on the plurality of analog switches. Furthermore, the source driver **504b** may be formed using a shift register or the like.

(464) A pulse signal and a data signal are input to each of the plurality of pixel circuits **501** through one of the plurality of scan lines GL supplied with scan signals and one of the plurality of data lines DL supplied with data signals, respectively. Furthermore, Writing and holding of the data of the data signal to and in each of the plurality of pixel circuits **501** are controlled by the gate driver **504a**. For example, to the pixel circuit **501** in the m-th row and the n-th column, a pulse signal is input from the gate driver **504a** through the scan line GL_m (m is a natural number of X or less), and a data signal is input from the source driver **504b** through the data line DL_n (n is a natural number of Y or less) in accordance with the potential of the scan line GL_m.

(465) The protection circuit **506** in FIG. 36(A) is connected to, for example, the scan line GL that is a wiring between the gate driver **504a** and the pixel circuit **501**. Alternatively, the protection circuit **506** is connected to the data line DL that is a wiring between the source driver **504b** and the pixel circuit **501**. Alternatively, the protection circuit **506** can be connected to a wiring between the gate

driver **504a** and the terminal portion **507**. Alternatively, the protection circuit **506** can be connected to a wiring between the source driver **504b** and the terminal portion **507**. Note that the terminal portion **507** refers to a portion having terminals for inputting power, control signals, and image signals from external circuits to the display device.

(466) The protection circuit **506** is a circuit that electrically connects a wiring, which is connected thereto, to another wiring when a potential out of a certain range is supplied to the wiring.

(467) As illustrated in FIG. **36(A)**, the protection circuits **506** provided for the pixel portion **502** and the driver circuit portion **504** can improve the resistance of the display device to overcurrent generated by ESD (Electro Static Discharge: Electrostatic discharge) or the like. Note that the configuration of the protection circuits **506** is not limited thereto; for example, a configuration in which the protection circuit **506** is connected to the gate driver **504a**, or a configuration in which the protection circuit **506** is connected to the source driver **504b** can be employed. Alternatively, a configuration in which the protection circuit **506** is connected to the terminal portion **507** can be employed.

(468) Furthermore, FIG. **36(A)** illustrates an example in which the driver circuit portion **504** is formed using the gate driver **504a** and the source driver **504b**; however, the configuration is not limited thereto. For example, a configuration in which only the gate driver **504a** is formed and a separately prepared substrate over which a source driver circuit is formed (e.g., a driver circuit board formed using a single-crystal semiconductor film or a polycrystalline semiconductor film) is mounted may be employed.

(469) Furthermore, the plurality of pixel circuits **501** in FIG. **36(A)** can have the configuration illustrated in FIG. **36(B)**, for example.

(470) The pixel circuit **501** in FIG. **36(B)** includes a liquid crystal element **570**, a transistor **550**, and a capacitor **560**. As the transistor **550**, the transistor described in the above embodiment can be used.

(471) The potential of one of a pair of electrodes of the liquid crystal element **570** is set as appropriate in accordance with the specifications of the pixel circuit **501**. The alignment state of the liquid crystal element **570** depends on data written thereto. Note that a common potential (a common potential) may be supplied to the one of the pair of electrodes of the liquid crystal element **570** included in each of the plurality of pixel circuits **501**. Furthermore, the potential supplied to the one of the pair of electrodes of the liquid crystal element **570** in the pixel circuit **501** may differ between rows.

(472) As a driving method of the display device including the liquid crystal element **570**, the following can be used: a TN mode, an STN mode, a VA mode, an ASM (Axially Symmetric Aligned Micro-cell) mode, an OCB (Optically Compensated Birefringence) mode, an FLC (Ferroelectric Liquid Crystal) mode, an AFLC (AntiFerroelectric Liquid Crystal) mode, an MVA mode, a PVA (Patterned Vertical Alignment) mode, an IPS mode, an FFS mode, and a TBA (Transverse Bend Alignment) mode. Furthermore, other driving methods of the display device include an ECB (Electrically Controlled Birefringence) mode, a PDLC (Polymer Dispersed Liquid Crystal) mode, a PNLC (Polymer Network Liquid Crystal) mode, and a guest-host mode. Note that without being limited thereto, various liquid crystal elements and driving methods can be used.

(473) In the pixel circuit **501** in the m-th row and the n-th column, one of a source electrode and a drain electrode of the transistor **550** is electrically connected to the data line DL_n, and the other is electrically connected to the other of the pair of electrodes of the liquid crystal element **570**.

Furthermore, a gate electrode of the transistor **550** is electrically connected to the scan line GL_m. The transistor **550** has a function of controlling whether data of a data signal is written.

(474) One of a pair of electrodes of the capacitor **560** is electrically connected to a wiring through which a potential is supplied (hereinafter, a potential supply line VL), and the other is electrically connected to the other of the pair of electrodes of the liquid crystal element **570**. Note that the potential of the potential supply line VL is appropriately set in accordance with the specifications

of the pixel circuit **501**. The capacitor **560** has a function of a storage capacitor for storing written data.

(475) For example, in the display device including the pixel circuits **501** in FIG. **36(B)**, the gate driver **504a** in FIG. **36(A)** sequentially selects the pixel circuits **501** row by row to turn on the transistors **550**, and data of data signals is written.

(476) When the transistor **550** is turned off, the pixel circuit **501** to which the data has been written is brought into a holding state. This operation is sequentially performed row by row; thus, an image can be displayed.

(477) Alternatively, the plurality of pixel circuits **501** in FIG. **36(A)** can have the configuration illustrated in FIG. **36(C)**, for example.

(478) Furthermore, the pixel circuit **501** in FIG. **36(C)** includes transistors **552** and **554**, a capacitor **562**, and a light-emitting element **572**. The transistor described in the above embodiment can be used as any one or both of the transistor **552** and the transistor **554**.

(479) One of a source electrode and a drain electrode of the transistor **552** is electrically connected to a wiring through which a data signal is supplied (hereinafter, referred to as a signal line DL_n). Furthermore, a gate electrode of the transistor **552** is electrically connected to a wiring through which a gate signal is supplied (hereinafter, referred to as the scan line GL_m).

(480) The transistor **552** has a function of controlling whether data of a data signal is written.

(481) One of a pair of electrodes of the capacitor **562** is electrically connected to a wiring through which a potential is supplied (hereinafter, referred to as a potential supply line VL_a), and the other is electrically connected to the other of the source electrode and the drain electrode of the transistor **552**.

(482) The capacitor **562** functions as a storage capacitor for storing written data.

(483) One of a source electrode and a drain electrode of the transistor **554** is electrically connected to the potential supply line VL_a. Furthermore, a gate electrode of the transistor **554** is electrically connected to the other of the source electrode and the drain electrode of the transistor **552**. One of an anode and a cathode of the light-emitting element **572** is electrically connected to a potential supply line VL_b, and the other is electrically connected to the other of the source electrode and the drain electrode of the transistor **554**.

(484) As the light-emitting element **572**, an organic electroluminescent element (also referred to as an organic EL element) can be used, for example. Note that the light-emitting element **572** is not limited thereto and an inorganic EL element comprising an inorganic material may be used.

(485) Note that a high power supply potential VDD is supplied to one of the potential supply line VL_a and the potential supply line VL_b, and a low power supply potential VSS is supplied to the other.

(486) In the display device including the pixel circuits **501** in FIG. **36(C)**, the gate driver **504a** in FIG. **36(A)** sequentially selects the pixel circuits **501** row by row to turn on the transistors **552**, and data of data signals is written.

(487) When the transistor **552** is turned off, the pixel circuit **501** to which the data has been written is brought into a holding state. Furthermore, the amount of current flowing between the source electrode and the drain electrode of the transistor **554** is controlled in accordance with the potential of the written data signal, and the light-emitting element **572** emits light with a luminance corresponding to the amount of flowing current. This operation is sequentially performed row by row; thus, an image can be displayed.

(488) Note that the structures described in this embodiment can be used in combination with the structures described in the other embodiments, as appropriate.

Embodiment 5

(489) In this embodiment, circuit configuration examples in which the transistors described in the above embodiments can be used are described with reference to FIG. **37** to FIG. **40**.

(490) Note that in the following description in this embodiment, the transistor including an oxide

semiconductor described in the above embodiment is referred to as an OS transistor.

(491) <5. Configuration Example of Inverter Circuit>

(492) FIG. 37(A) is a circuit diagram of an inverter that can be used for a shift register, a buffer, or the like included in the driver circuit. An inverter **800** outputs a signal whose logic is inverted from that of a signal of an input terminal IN to an output terminal OUT. The inverter **800** includes a plurality of OS transistors. A signal S.sub.BG is a signal that can switch electrical characteristics of the OS transistors.

(493) FIG. 37(B) is an example of the inverter **800**. The inverter **800** includes an OS transistor **810** and an OS transistor **820**. The inverter **800** can be formed using only n-channel transistors and thus can be formed at lower cost than an inverter (a CMOS inverter) formed using a CMOS (Complementary Metal Oxide Semiconductor).

(494) Note that the inverter **800** including the OS transistors can be provided over a CMOS including Si transistors. Since the inverter **800** can be provided to overlap with the CMOS circuit, no additional area is required for the inverter **800**, and thus, an increase in the circuit area can be suppressed.

(495) The OS transistors **810** and **820** include a first gate functioning as a front gate, a second gate functioning as a back gate, a first terminal functioning as one of a source and a drain, and a second terminal functioning as the other of the source and the drain.

(496) The first gate of the OS transistor **810** is connected to the second terminal of the OS transistor **810**. The second gate of the OS transistor **810** is connected to a wiring that supplies the signal S.sub.BG. The first terminal of the OS transistor **810** is connected to a wiring that supplies a voltage VDD. The second terminal of the OS transistor **810** is connected to the output terminal OUT.

(497) The first gate of the OS transistor **820** is connected to the input terminal IN. The second gate of the OS transistor **820** is connected to the input terminal IN. The first terminal of the OS transistor **820** is connected to the output terminal OUT. The second terminal of the OS transistor **820** is connected to a wiring that supplies a voltage VS S.

(498) FIG. 37(C) is a timing chart for explaining the operation of the inverter **800**. The timing chart in FIG. 37(C) shows changes in a signal waveform of the input terminal IN, a signal waveform of the output terminal OUT, a signal waveform of the signal S.sub.BG, and the threshold voltage of the OS transistor **810**.

(499) The signal S.sub.BG is supplied to the second gate of the OS transistor **810**, so that the threshold voltage of the OS transistor **810** can be controlled.

(500) The signal S.sub.BG includes a voltage V.sub.BG_A for shifting the threshold voltage in the negative direction and a voltage V.sub.BG_B for shifting the threshold voltage in the positive direction. The threshold voltage of the OS transistor **810** can be shifted in the negative direction to be a threshold voltage V.sub.TH_A when the voltage V.sub.BG_A is applied to the second gate. Furthermore, the threshold voltage of the OS transistor **810** can be shifted in the positive direction to be a threshold voltage V.sub.TH_B when the voltage V.sub.BG_B is applied to the second gate. To visualize the above description, FIG. 38(A) shows an I.sub.d-V.sub.g curve, which is one of the electrical characteristics of a transistor.

(501) With a high voltage such as the voltage V.sub.BG_A as the voltage of the second gate, the electrical characteristics of the OS transistor **810** can be shifted to match a curve shown by a dashed line **840** in FIG. 38(A). Furthermore, with a low voltage such as the voltage V.sub.BG_B as the voltage of the second gate, the electrical characteristics of the OS transistor **810** can be shifted to match a curve shown by a solid line **841** in FIG. 38(A). As shown in FIG. 38(A), switching the signal S.sub.BG between the voltage V.sub.BG_A and the voltage V.sub.BG_B enables the threshold voltage of the OS transistor **810** to be shifted in the positive direction or the negative direction.

(502) The shift of the threshold voltage in the positive direction toward the threshold voltage

V.sub.TH_B can make current less likely to flow in the OS transistor **810**. FIG. **38(B)** visualizes this state.

(503) As illustrated in FIG. **38(B)**, a current I.sub.B that flows in the OS transistor **810** can be extremely low. Thus, when a signal supplied to the input terminal IN is at a high level and the OS transistor **820** is in an on state (ON), the voltage of the output terminal OUT can be sharply decreased.

(504) Since a state in which current is less likely to flow in the OS transistor **810** as illustrated in FIG. **38(B)** can be obtained, a change in a signal waveform **831** of the output terminal in the timing chart in FIG. **37(C)** can be made steep. Shoot-through current flowing between the wiring that supplies the voltage VDD and the wiring that supplies the voltage VS S can be low, leading to low-power operation.

(505) Furthermore, the shift of the threshold voltage in the negative direction toward the threshold voltage V.sub.TH_A can make current flow easily in the OS transistor **810**. FIG. **38(C)** visualizes this state. As illustrated in FIG. **38(C)**, a current I.sub.A flowing at this time can be higher than at least the current Is. Thus, when a signal supplied to the input terminal IN is at a low level and the OS transistor **820** is in an off state (OFF), the voltage of the output terminal OUT can be sharply increased. Since a state in which current is likely to flow in the OS transistor **810** as illustrated in FIG. **38(C)** can be obtained, a change in a signal waveform **832** of the output terminal in the timing chart in FIG. **37(C)** can be made steep.

(506) Note that the threshold voltage of the OS transistor **810** is preferably controlled by the signal S.sub.BG before the state of the OS transistor **820** is switched, i.e., before time T1 or T2. For example, as shown in FIG. **37(C)**, it is preferable that the threshold voltage of the OS transistor **810** be switched from the threshold voltage V.sub.TH_A to the threshold voltage V.sub.TH_B before time T1 at which the level of the signal supplied to the input terminal IN is switched to a high level. Moreover, as shown in FIG. **37(C)**, it is preferable that the threshold voltage of the OS transistor **810** be switched from the threshold voltage V.sub.TH_B to the threshold voltage V.sub.TH_A before time T2 at which the level of the signal supplied to the input terminal IN is switched to a low level.

(507) Note that although the timing chart in FIG. **37(C)** shows the configuration in which the signal S.sub.BG is switched in accordance with the signal supplied to the input terminal IN, a different configuration may be employed. For example, a configuration in which voltage for controlling the threshold voltage is held by the second gate of the OS transistor **810** in a floating state may be employed. FIG. **39(A)** illustrates a circuit configuration example that can achieve the configuration.

(508) In FIG. **39(A)**, an OS transistor **850** is additionally included in the circuit configuration illustrated in FIG. **37(B)**. A first terminal of the OS transistor **850** is connected to the second gate of the OS transistor **810**. Furthermore, a second terminal of the OS transistor **850** is connected to a wiring that supplies the voltage V.sub.BG_B (or the voltage V.sub.BG_A). A first gate of the OS transistor **850** is connected to a wiring that supplies a signal S.sub.F. A second gate of the OS transistor **850** is connected to the wiring that supplies the voltage V.sub.BG_B (or the voltage V.sub.BG_A).

(509) The operation in FIG. **39(A)** is described with reference to a timing chart in FIG. **39(B)**.

(510) The voltage for controlling the threshold voltage of the OS transistor **810** is supplied to the second gate of the OS transistor **810** before time T3 at which the level of the signal supplied to the input terminal IN is switched to a high level. The signal S.sub.F is set to a high level and the OS transistor **850** is turned on, so that the voltage V.sub.BG_B for controlling the threshold voltage is supplied to a node N.sub.BG.

(511) The OS transistor **850** is turned off after the voltage of the node N.sub.BG becomes V.sub.BG_B. Since the off-state current of the OS transistor **850** is extremely low, the threshold voltage V.sub.BG_B that has been held by the node N.sub.BG can be held while the OS transistor **850** remains off. Thus, the number of times the voltage V.sub.BG_B is supplied to the second gate

of the OS transistor **850** can be reduced and accordingly, the power consumption for rewriting the voltage $V_{\text{sub.BG_B}}$ can be reduced.

(512) Note that although in the circuit configurations in FIG. 37(B) and FIG. 39(A), the voltage is supplied to the second gate of the OS transistor **810** by control from the outside, a different configuration may be employed. For example, a configuration in which voltage for controlling the threshold voltage is generated on the basis of the signal supplied to the input terminal IN and is supplied to the second gate of the OS transistor **810** may be employed. FIG. 40(A) illustrates a circuit configuration example that can achieve the configuration.

(513) In FIG. 40(A), a CMOS inverter **860** is provided between the input terminal IN and the second gate of the OS transistor **810** in the circuit configuration illustrated in FIG. 37(B). An input terminal of the CMOS inverter **860** is connected to the input terminal IN. An output terminal of the CMOS inverter **860** is connected to the second gate of the OS transistor **810**.

(514) The operation in FIG. 40(A) is described using a timing chart in FIG. 40(B). The timing chart in FIG. 40(B) shows changes in a signal waveform of the input terminal IN, a signal waveform of the output terminal OUT, an output waveform IN_B of the CMOS inverter **860**, and the threshold voltage of the OS transistor **810**.

(515) The output waveform IN_B that is a signal whose logic is inverted from that of the signal supplied to the input terminal IN can be a signal that controls the threshold voltage of the OS transistor **810**. Thus, the threshold voltage of the OS transistor **810** can be controlled as described in FIG. 38(A) to FIG. 38(C). For example, the signal supplied to the input terminal IN is at a high level and the OS transistor **820** is turned on at time T4 in FIG. 40(B). At this time, the output waveform IN_B is at a low level. Accordingly, current can be made less likely to flow in the OS transistor **810**; thus, an increase in the voltage of the output terminal OUT can be sharply decreased.

(516) Moreover, the signal supplied to the input terminal IN is at a low level and the OS transistor **820** is turned off at time T5 in FIG. 40(B). At this time, the output waveform IN_B is at a high level. Accordingly, current can easily flow in the OS transistor **810**; thus, the voltage of the output terminal OUT can be sharply increased.

(517) As described above, in the configuration of this embodiment, the voltage of the back gate in the inverter including the OS transistor is switched in accordance with the logic of the signal of the input terminal IN. In such a configuration, the threshold voltage of the OS transistor can be controlled. The control of the threshold voltage of the OS transistor by the signal supplied to the input terminal IN can cause a steep change in the voltage of the output terminal OUT. Moreover, shoot-through current between the wirings that supply power supply voltages can be reduced. Thus, power consumption can be reduced.

(518) Note that the structures described in this embodiment can be used in combination with the structures described in the other embodiments, as appropriate.

Embodiment 6

(519) In this embodiment, examples of a semiconductor device in which the transistor including an oxide semiconductor (OS transistor) described in the above embodiments is used in a plurality of circuits are described using FIG. 41 to FIG. 44.

(520) <6. Circuit Configuration Example of Semiconductor Device>

(521) FIG. 41(A) is a block diagram of a semiconductor device **900**. The semiconductor device **900** includes a power supply circuit **901**, a circuit **902**, a voltage generation circuit **903**, a circuit **904**, a voltage generation circuit **905**, and a circuit **906**. The power supply circuit **901** is a circuit that generates a voltage $V_{\text{sub.ORG}}$ used as a reference.

(522) The voltage $V_{\text{sub.ORG}}$ is not necessarily one voltage and can be a plurality of voltages. The voltage $V_{\text{sub.ORG}}$ can be generated on the basis of a voltage $V_{\text{sub.0}}$ supplied from the outside of the semiconductor device **900**. The semiconductor device **900** can generate the voltage $V_{\text{sub.ORG}}$ on the basis of one power supply voltage supplied from the outside. Thus, the semiconductor

device **900** can operate without supply of a plurality of power supply voltages from the outside. (523) The circuits **902**, **904**, and **906** operate with different power supply voltages. For example, the power supply voltage of the circuit **902** is a voltage applied on the basis of the voltage $V_{sub.ORG}$ and the voltage $V_{sub.SS}$ ($V_{sub.ORG} > V_{sub.SS}$). Furthermore, for example, the power supply voltage of the circuit **904** is a voltage applied on the basis of a voltage $V_{sub.POG}$ and the voltage $V_{sub.SS}$ ($V_{sub.POG} > V_{sub.ORG}$). Furthermore, for example, the power supply voltages of the circuit **906** are voltages applied on the basis of the voltage $V_{sub.ORG}$ and a voltage $V_{sub.NEG}$ ($V_{sub.ORG} > V_{sub.SS} > V_{sub.NEG}$). Note that when the voltage $V_{sub.SS}$ is equal to a ground potential (GND), the kinds of voltages generated in the power supply circuit **901** can be reduced.

(524) The voltage generation circuit **903** is a circuit that generates the voltage $V_{sub.POG}$. The voltage generation circuit **903** can generate the voltage $V_{sub.POG}$ on the basis of the voltage $V_{sub.ORG}$ supplied from the power supply circuit **901**. Thus, the semiconductor device **900** including the circuit **904** can operate on the basis of one power supply voltage supplied from the outside.

(525) The voltage generation circuit **905** is a circuit that generates the voltage $V_{sub.NEG}$. The voltage generation circuit **905** can generate the voltage $V_{sub.NEG}$ on the basis of the voltage $V_{sub.ORG}$ supplied from the power supply circuit **901**. Thus, the semiconductor device **900** including the circuit **906** can operate on the basis of one power supply voltage supplied from the outside.

(526) FIG. **41(B)** is an example of the circuit **904** that operates with the voltage $V_{sub.POG}$ and FIG. **41(C)** is an example of a waveform of a signal for operating the circuit **904**.

(527) In FIG. **41(B)**, a transistor **911** is illustrated. A signal supplied to a gate of the transistor **911** is generated on the basis of, for example, the voltage $V_{sub.POG}$ and the voltage $V_{sub.SS}$. The signal is the voltage $V_{sub.POG}$ to turn on the transistor **911** and is the voltage $V_{sub.SS}$ to turn off the transistor **911**. As illustrated in FIG. **41(C)**, the voltage $V_{sub.POG}$ is higher than the voltage $V_{sub.ORG}$. Thus, an operation for bringing a source (S) and a drain (D) of the transistor **911** into a conduction state can be performed more reliably. As a result, the circuit **904** can be a circuit with reduced frequency of malfunction.

(528) FIG. **41(D)** illustrates an example of the circuit **906** that operates with the voltage $V_{sub.NEG}$ and FIG. **41(E)** illustrates an example of a waveform of a signal for operating the circuit **906**. FIG. **41(D)** illustrates a transistor **912** having a back gate. A signal supplied to a gate of the transistor **912** is generated on the basis of, for example, the voltage $V_{sub.ORG}$ and the voltage $V_{sub.SS}$. The signal is the voltage $V_{sub.ORG}$ to turn on the transistor **911** and is the voltage $V_{sub.SS}$ to turn off the transistor **911**. Furthermore, a voltage applied to the back gate of the transistor **912** is generated on the basis of the voltage $V_{sub.NEG}$. As illustrated in FIG. **41(E)**, the voltage $V_{sub.NEG}$ is lower than the voltage $V_{sub.SS}$ (GND). Thus, the threshold voltage of the transistor **912** can be controlled to shift in the positive direction. Thus, the transistor **912** can be more reliably turned off and a current flowing between a source (S) and a drain (D) can be reduced. As a result, the frequency of malfunction of the circuit **906** can be reduced and power consumption thereof can be reduced.

(529) Note that the voltage $V_{sub.NEG}$ may be directly supplied to the back gate of the transistor **912**. Alternatively, a signal supplied to the gate of the transistor **912** may be generated on the basis of the voltage $V_{sub.ORG}$ and the voltage $V_{sub.NEG}$ and the signal may be supplied to the back gate of the transistor **912**.

(530) Furthermore, FIGS. **42(A)** and (B) illustrate a modification example of FIGS. **41(D)** and (E).

(531) In a circuit diagram illustrated in FIG. **42(A)**, a transistor **922** whose conduction state can be controlled by a control circuit **921** is shown between the voltage generation circuit **905** and the circuit **906**. The transistor **922** is an n-channel OS transistor. The control signal $S_{sub.BG}$ output from the control circuit **921** is a signal for controlling the conduction state of the transistor **922**.

Furthermore, transistors **912A** and **912B** included in the circuit **906** are OS transistors like the transistor **922**.

(532) A timing chart in FIG. **42(B)** shows changes in the potential of the control signal S.sub.BG and the potential of a node N.sub.BG that indicates the states of potentials of back gates of the transistors **912A** and **912B**. When the control signal S.sub.BG is at a high level, the transistor **922** is turned on and the voltage of the node N.sub.BG becomes V.sub.NEG. Then, when the control signal S.sub.BG is at a low level, the node N.sub.BG is brought into an electrically floating state. Since the transistor **922** is an OS transistor, its off-state current is small. Accordingly, even when the node N.sub.BG is in an electrically floating state, the voltage V.sub.NEG that has been supplied can be held.

(533) Furthermore, FIG. **43(A)** illustrates an example of a circuit configuration that can be used for the above-described voltage generation circuit **903**. The voltage generation circuit **903** illustrated in FIG. **43(A)** is a five-stage charge pump including diodes D1 to D5, capacitors C1 to C5, and an inverter INV. A clock signal CLK is supplied to the capacitors C1 to C5 directly or through the inverter INV. When the power supply voltage of the inverter INV is a voltage applied on the basis of the voltage V.sub.ORG and the voltage V.sub.SS, the voltage V.sub.POG, which has been increased to a positive voltage having a positively quintupled value of the voltage V.sub.ORG by the clock signal CLK, can be obtained. Note that the forward voltage of the diodes D1 to D5 is 0 V. Furthermore, the number of stages of the charge pump can be changed to obtain a desired voltage V.sub.POG.

(534) Furthermore, FIG. **43(B)** illustrates an example of a circuit configuration that can be used for the above-described voltage generation circuit **905**. The voltage generation circuit **905** illustrated in FIG. **43(B)** is a four-stage charge pump including the diodes D1 to D5, the capacitors C1 to C5, and the inverter INV. The clock signal CLK is supplied to the capacitors C1 to C5 directly or through the inverter INV. When the power supply voltage of the inverter INV is a voltage applied on the basis of the voltage V.sub.ORG and the voltage V.sub.SS, the voltage V.sub.NEG, which has been reduced from the ground voltage, i.e., the voltage V.sub.SS to a negative voltage having a negatively quadrupled value of the voltage V.sub.ORG by the clock signal CLK, can be obtained. Note that the forward voltage of the diodes D1 to D5 is 0 V. Furthermore, the number of stages of the charge pump can be changed to obtain a desired voltage V.sub.NEG.

(535) Note that the circuit configuration of the voltage generation circuit **903** is not limited to the configuration in the circuit diagram illustrated in FIG. **43(A)**. For example, modification examples of the voltage generation circuit **903** are illustrated in FIG. **44(A)** to FIG. **44(C)**. Note that modification examples of the voltage generation circuit **903** can be obtained by changing voltages supplied to wirings or arrangement of elements in voltage generation circuits **903A** to **903C** illustrated in FIG. **44(A)** to FIG. **44(C)**.

(536) The voltage generation circuit **903A** illustrated in FIG. **44(A)** includes transistors M1 to M10, capacitors C11 to C14, and an inverter INV'. The clock signal CLK is supplied to gates of the transistors M1 to M10 directly or through the inverter INV1. By the clock signal CLK, the voltage V.sub.POG, which has been increased to a positive voltage having a positively quadrupled value of the voltage V.sub.ORG, can be obtained. Note that the number of stages can be changed to obtain a desired voltage V.sub.POG. In the voltage generation circuit **903A** in FIG. **44(A)**, off-state current of the transistors M1 to M10 can be small when the transistors are OS transistors, and leakage of charge held in the capacitors C11 to C14 can be suppressed. Accordingly, the voltage V.sub.ORG can be efficiently increased to the voltage V.sub.POG.

(537) Furthermore, the voltage generation circuit **903B** illustrated in FIG. **44(B)** includes transistors M11 to M14, capacitors C15 and C16, and an inverter INV2. The clock signal CLK is supplied to gates of the transistors M11 to M14 directly or through the inverter INV2. By the clock signal CLK, the voltage V.sub.POG, which has been increased to a positive voltage having a positively doubled value of the voltage V.sub.ORG, can be obtained. In the voltage generation

circuit **903B** in FIG. **44(B)**, off-state current of the transistors **M11** to **M14** can be small when the transistors are OS transistors, and leakage of charge held in the capacitors **C15** and **C16** can be suppressed. Accordingly, the voltage $V_{sub.ORG}$ can be efficiently increased to the voltage $V_{sub.POG}$. The voltage generation circuit **903C** illustrated in FIG. **44(C)** includes an inductor **Ind1**, a transistor **M15**, a diode **D6**, and a capacitor **C17**. The conduction state of the transistor **M15** is controlled by a control signal **EN**. Owing to the control signal **EN**, the voltage $V_{sub.POG}$ increased from the voltage $V_{sub.ORG}$ can be obtained. Since the voltage generation circuit **903C** in FIG. **44(C)** increases the voltage using the inductor **Ind1**, the voltage can be increased with high changing efficiency.

(538) As described above, in the structures of this embodiment, a voltage required for circuits included in a semiconductor device can be internally generated. Thus, in the semiconductor device, the number of power supply voltages supplied from the outside can be reduced. Note that the structures and the like described in this embodiment can be used in combination with the structures described in the other embodiments, as appropriate.

Embodiment 7

(539) In this embodiment, a display module and electronic devices which include a semiconductor device of one embodiment of the present invention are described using FIG. **45** to FIG. **48**.

(540) <7-1. Display Module>

(541) A display module **7000** illustrated in FIG. **45** includes a touch panel **7004** connected to an FPC **7003**, a display panel **7006** connected to an FPC **7005**, a backlight **7007**, a frame **7009**, a printed board **7010**, and a battery **7011**, between an upper cover **7001** and a lower cover **7002**.

(542) The semiconductor device of one embodiment of the present invention can be used for the display panel **7006**, for example.

(543) The shapes and sizes of the upper cover **7001** and the lower cover **7002** can be changed as appropriate in accordance with the sizes of the touch panel **7004** and the display panel **7006**.

(544) As the touch panel **7004**, a resistive or capacitive touch panel can be used with it overlapping with the display panel **7006**. Furthermore, a counter substrate (sealing substrate) of the display panel **7006** can have a touch panel function. Furthermore, a photosensor may be provided in each pixel of the display panel **7006** to obtain an optical touch panel.

(545) The backlight **7007** includes a light source **7008**. Note that the structure in which the light source **7008** is provided over the backlight **7007** is illustrated as an example in FIG. **45**; however, there is no limitation thereto. For example, a structure in which the light source **7008** is provided at an end portion of the backlight **7007** and a light diffusion plate is further used may be employed. Note that the backlight **7007** need not be provided in the case where a self-luminous light-emitting element such as an organic EL element is used or in the case of a reflective panel or the like.

(546) The frame **7009** has a function of protecting the display panel **7006** and a function of an electromagnetic shield for blocking electromagnetic waves generated by the operation of the printed board **7010**. The frame **7009** may also have a function of a radiator plate.

(547) The printed board **7010** includes a power supply circuit and a signal processing circuit for outputting a video signal and a clock signal. A power source for supplying power to the power supply circuit may be an external commercial power source or a power source using the battery **7011** provided separately. The battery **7011** can be omitted in the case where a commercial power source is used.

(548) Furthermore, the display module **7000** may be additionally provided with a member such as a polarizing plate, a retardation plate, or a prism sheet.

(549) <7-2. Electronic Device 1>

(550) Next, FIG. **46(A)** to FIG. **46(E)** illustrate examples of electronic devices.

(551) FIG. **46(A)** is an external view of a camera **8000** to which a finder **8100** is attached.

(552) The camera **8000** includes a housing **8001**, a display portion **8002**, operation buttons **8003**, a shutter button **8004**, and the like. Furthermore, an detachable lens **8006** is attached to the camera

8000.

(553) Although the lens **8006** of the camera **8000** here is detachable from the housing **8001** for replacement, the lens **8006** may be included in the housing.

(554) Images can be taken with the camera **8000** at the press of the shutter button **8004**. In addition, the display portion **8002** has a function of a touch panel; thus, images can be taken at the touch of the display portion **8002**. The housing **8001** of the camera **8000** includes a mount including an electrode, so that the finder **8100**, a stroboscope, or the like can be connected to the housing.

(555) The finder **8100** includes a housing **8101**, a display portion **8102**, a button **8103**, and the like.

(556) The housing **8101** includes a mount for engagement with the mount of the camera **8000** so that the finder **8100** can be attached to the camera **8000**. Furthermore, the mount includes an electrode, and an image or the like received from the camera **8000** through the electrode can be displayed on the display portion **8102**.

(557) The button **8103** has a function of a power button. The on/off of display of the display portion **8102** can be switched with the button **8103**.

(558) A display device of one embodiment of the present invention can be used in the display portion **8002** of the camera **8000** and the display portion **8102** of the finder **8100**.

(559) Note that although the camera **8000** and the finder **8100** are separate and detachable electronic devices in FIG. **46(A)**, the housing **8001** of the camera **8000** may include a finder having a display device.

(560) FIG. **46(B)** is an external view of a head-mounted display **8200**.

(561) The head-mounted display **8200** includes a mounting portion **8201**, a lens **8202**, a main body **8203**, a display portion **8204**, a cable **8205**, and the like. Furthermore, the mounting portion **8201** includes a battery **8206**.

(562) Power is supplied from the battery **8206** to the main body **8203** through the cable **8205**. The main body **8203** includes a wireless receiver or the like and can display received video data, such as image data, on the display portion **8204**. Furthermore, the movement of the eyeball and the eyelid of a user is captured by a camera provided in the main body **8203** and then coordinates of the points the user looks at are calculated using the captured data to utilize the eyepoint of the user as an input means.

(563) Furthermore, a plurality of electrodes to be in contact with the user may be provided in the mounting portion **8201**. The main body **8203** may have a function of sensing current flowing through the electrodes with the movement of the user's eyeball to determine the points the user looks at. Furthermore, the main body may have a function of sensing current flowing through the electrodes to monitor the user's pulse. Furthermore, the mounting portion **8201** may include sensors, such as a temperature sensor, a pressure sensor, and an acceleration sensor and a function of displaying the user's biological information on the display portion **8204** may be provided. Furthermore, the movement of the user's head or the like may be sensed to change an image displayed on the display portion **8204** in synchronization with the movement.

(564) The display device of one embodiment of the present invention can be used in the display portion **8204**.

(565) FIGS. **46(C)**, **(D)**, and **(E)** are external views of a head-mounted display **8300**. The head-mounted display **8300** includes a housing **8301**, a display portion **8302**, fixing bands **8304**, and a pair of lenses **8305**.

(566) A user can see display on the display portion **8302** through the lenses **8305**. Note that it is suitable that the display portion **8302** be curved. When the display portion **8302** is provided with it curved, a user can feel high realistic sensation.

(567) Note that the display device of one embodiment of the present invention can be used in the display portion **8302**. The display device including the semiconductor device of one embodiment of the present invention has an extremely high resolution; thus, even when an image is magnified using the lenses **8305** as illustrated in FIG. **46(E)**, the user does not perceive pixels, and thus a

more realistic image can be displayed.

(568) <7-3. Electronic Device 2>

(569) Next, FIG. 47(A) to FIG. 47(G) illustrate examples of electronic devices that are different from the electronic devices illustrated in FIG. 46(A) to FIG. 46(E).

(570) The electronic devices illustrated in FIG. 47(A) to FIG. 47(G) include a housing **9000**, a display portion **9001**, a speaker **9003**, an operation key **9005** (including a power switch or an operation switch), a connection terminal **9006**, a sensor **9007** (a sensor having a function of measuring force, displacement, position, speed, acceleration, angular velocity, rotational frequency, distance, light, liquid, magnetism, temperature, chemical substance, sound, time, hardness, electric field, current, voltage, electric power, radiation, flow rate, humidity, gradient, oscillation, odor, or infrared ray), a microphone **9008**, and the like.

(571) The electronic devices in FIG. 47(A) to FIG. 47(G) have a variety of functions. For example, the electronic devices can have a function of displaying a variety of information (e.g., a still image, a moving image, and a text image) on the display portion, a touch panel function, a function of displaying a calendar, date, time, and the like, a function of controlling processing with a variety of software (programs), a wireless communication function, a function of being connected to a variety of computer networks with a wireless communication function, a function of transmitting and receiving a variety of data with a wireless communication function, a function of reading out a program or data stored in a recording medium and displaying it on the display portion, and the like. Note that functions of the electronic devices in FIG. 47(A) to FIG. 47(G) are not limited thereto, and the electronic devices can have a variety of functions. Furthermore, although not illustrated in FIG. 47(A) to FIG. 47(G), the electronic devices may have a plurality of display portions.

Furthermore, the electronic devices may be provided with a camera and the like to have a function of taking a still image, a function of taking a moving image, a function of storing the taken image in a memory medium (an external memory medium or a memory medium incorporated in the camera), a function of displaying the taken image on the display portion, or the like.

(572) The details of the electronic devices in FIG. 47(A) to FIG. 47(G) are described below.

(573) FIG. 47(A) is a perspective view illustrating a television device **9100**. The television device **9100** can include the display portion **9001** having a large screen size of, for example, 50 inches or more, or 100 inches or more.

(574) FIG. 47(B) is a perspective view illustrating a portable information terminal **9101**. The portable information terminal **9101** has, for example, one or more functions selected from a telephone set, a notebook, and an information browsing system. Specifically, the portable information terminal can be used as a smartphone. Note that the portable information terminal **9101** may include a speaker **9003**, a connection terminal **9006**, a sensor **9007**, or the like. Furthermore, the portable information terminal **9101** can display text and image information on its plurality of surfaces. For example, three operation buttons **9050** (also referred to as operation icons or simply as icons) can be displayed on one surface of the display portion **9001**. Furthermore, information **9051** indicated by dashed rectangles can be displayed on another surface of the display portion **9001**. Note that examples of the information **9051** include display indicating reception of an e-mail, SNS (social networking service), a telephone call, or the like, the title of an e-mail, SNS, or the like, the sender of an e-mail, SNS, or the like, date, time, remaining battery, and reception strength of an antenna. Alternatively, in place of the information **9051**, the operation buttons **9050** or the like may be displayed on the position where the information **9051** is displayed.

(575) FIG. 47(C) is a perspective view illustrating a portable information terminal **9102**. The portable information terminal **9102** has a function of displaying information on three or more surfaces of the display portion **9001**. Here, an example in which information **9052**, information **9053**, and information **9054** are displayed on different surfaces is illustrated. For example, a user of the portable information terminal **9102** can see the display (here, the information **9053**) on the portable information terminal **9102** put in a breast pocket of the clothes. Specifically, a caller's

phone number, name, or the like of an incoming call is displayed in a position that can be seen from above the portable information terminal **9102**. The user can see the display without taking out the portable information terminal **9102** from the pocket and decide whether to answer the call.

(576) FIG. **47(D)** is a perspective view illustrating a watch-type portable information terminal **9200**. The portable information terminal **9200** is capable of executing a variety of applications such as a mobile phone, e-mailing, viewing and editing texts, music reproduction, Internet communication, and a computer game. Furthermore, the display surface of the display portion **9001** is curved, and display can be performed on the curved display surface. Furthermore, the portable information terminal **9200** can employ near field communication conformable to a communication standard. For example, hands-free calling can be achieved by mutual communication with a headset capable of wireless communication. Moreover, the portable information terminal **9200** includes the connection terminal **9006** and can perform direct data communication with another information terminal via a connector. Furthermore, charging through the connection terminal **9006** is possible. Note that the charging operation may be performed by wireless power feeding without using the connection terminal **9006**.

(577) FIGS. **47(E)**, **(F)**, and **(G)** are perspective views illustrating a foldable portable information terminal **9201**. Furthermore, FIG. **47(E)** is the perspective view of the portable information terminal **9201** that is opened, FIG. **47(F)** is the perspective view of the portable information terminal **9201** that is shifted from one of the opened state and the folded state to the other, and FIG. **47(G)** is the perspective view of the portable information terminal **9201** that is folded. The portable information terminal **9201** is highly portable when folded, and is highly browsable when opened owing to a seamless large display region. The display portion **9001** of the portable information terminal **9201** is supported by three housings **9000** joined by hinges **9055**. By being folded at the hinges **9055** between the two housings **9000**, the portable information terminal **9201** can be reversibly changed in shape from the opened state to the folded state. For example, the portable information terminal **9201** can be bent with a radius of curvature greater than or equal to 1 mm and less than or equal to 150 mm.

(578) FIGS. **48(A)** and **(B)** are perspective views of a display device including a plurality of display panels. Note that FIG. **48(A)** is the perspective view of the plurality of display panels that are wound, and FIG. **48(B)** is the perspective view of the plurality of display panels that are unwound.

(579) A display device **9500** illustrated in FIGS. **48(A)** and **(B)** includes a plurality of display panels **9501**, an axis portion **9511**, and a bearing **9512**. Furthermore, the plurality of display panels **9501** include a display region **9502** and a light-transmitting region **9503**.

(580) Furthermore, the plurality of display panels **9501** are flexible. Furthermore, two adjacent display panels **9501** are provided to partly overlap with each other. For example, the light-transmitting regions **9503** of the two adjacent display panels **9501** can overlap with each other. A display device having a large screen can be obtained with the plurality of display panels **9501**. Furthermore, the display device is highly versatile because the display panels **9501** can be wound depending on its use.

(581) Furthermore, although the display regions **9502** of the adjacent display panels **9501** are separated from each other in FIGS. **48(A)** and **(B)**, without limitation thereto, the display regions **9502** of the adjacent display panels **9501** may overlap with each other without any space so that a continuous display region **9502** is obtained, for example.

(582) The electronic devices described in this embodiment are characterized by having a display portion for displaying some sort of information. Note that the semiconductor device of one embodiment of the present invention can also be used for an electronic device that does not have a display portion.

(583) Note that the structures described in this embodiment can be used in combination with the structures described in the other embodiments, as appropriate.

Embodiment 8

(584) In this embodiment, a semiconductor device (memory device) that can retain stored contents even when not powered and that has an unlimited number of times of writing, and a CPU including the semiconductor device are described using FIG. 49 to FIG. 51. The CPU described in this embodiment can be used for the electronic device described in the above embodiment, for example.

(585) <8-1. Memory Device>

(586) An example of a semiconductor device (memory device) which can retain stored contents even when not powered and which has an unlimited number of times of writing is shown in FIG.

49. Note that FIG. 49(B) is a circuit diagram of FIG. 49(A).

(587) The semiconductor device illustrated in FIGS. 49(A) and (B) includes a transistor **3200** using a first semiconductor material, a transistor **3300** using a second semiconductor material, and a capacitor **3400**.

(588) The first and second semiconductor materials preferably have different energy gaps. For example, the first semiconductor material can be a semiconductor material other than an oxide semiconductor (silicon (including strained silicon), germanium, silicon germanium, silicon carbide, gallium arsenide, aluminum gallium arsenide, indium phosphide, gallium nitride, an organic semiconductor, and the like), and the second semiconductor material can be an oxide semiconductor. A transistor using a material other than an oxide semiconductor, such as single crystal silicon, can operate at high speed easily. On the other hand, a transistor using an oxide semiconductor has a low off-state current.

(589) The transistor **3300** is a transistor in which a channel is formed in a semiconductor layer including an oxide semiconductor. Since the off-state current of the transistor **3300** is small, the use of the transistor **3300** enables stored contents to be retained for a long period. In other words, power consumption can be sufficiently reduced because a semiconductor memory device in which refresh operation is unnecessary or the frequency of refresh operation is extremely low can be provided.

(590) In FIG. 49(B), a first wiring **3101** is electrically connected to a source electrode of the transistor **3200**, and a second wiring **3102** is electrically connected to a drain electrode of the transistor **3200**. A third wiring **3103** is electrically connected to one of a source electrode and a drain electrode of the transistor **3300**, and a fourth wiring **3104** is electrically connected to a gate electrode of the transistor **3300**. A gate electrode of the transistor **3200** and the other of the source electrode and the drain electrode of the transistor **3300** are electrically connected to one electrode of the capacitor **3400**, and a fifth wiring **3105** is electrically connected to the other electrode of the capacitor **3400**.

(591) A characteristic of the semiconductor device in FIG. 49(A) that the potential of the gate electrode of the transistor **3200** can be retained is taken advantage of, whereby writing, retaining, and reading of data can be performed as follows.

(592) Writing and retaining of data are described. First, the potential of the fourth wiring **3104** is set to a potential at which the transistor **3300** is turned on, so that the transistor **3300** is turned on. Accordingly, the potential of the third wiring **3103** is supplied to the gate electrode of the transistor **3200** and the capacitor **3400**. That is, a predetermined charge is supplied to the gate of the transistor **3200** (writing). Here, one of charges providing two different potential levels (hereinafter referred to as a Low-level charge and a High-level charge) is supplied. After that, the potential of the fourth wiring **3104** is set to a potential at which the transistor **3300** is turned off, so that the transistor **3300** is turned off. Thus, the charge supplied to the gate of the transistor **3200** is held (retaining).

(593) Since the off-state current of the transistor **3300** is extremely small, the charge of the gate of the transistor **3200** is retained for a long time.

(594) Next, reading of data is described. An appropriate potential (a reading potential) is supplied to the fifth wiring **3105** while a predetermined potential (a constant potential) is supplied to the first wiring **3101**, whereby the potential of the second wiring **3102** varies depending on the amount of

charge retained in the gate of the transistor **3200**. This is because in the case where the transistor **3200** is an n-channel type, an apparent threshold voltage $V_{\text{sub.th_H}}$ when the High-level charge is given to the gate electrode of the transistor **3200** is lower than an apparent threshold voltage $V_{\text{sub.th_L}}$ when the Low-level charge is given to the gate electrode of the transistor **3200**. Here, an apparent threshold voltage refers to the potential of the fifth wiring **3105** which is needed to turn “on” the transistor **3200**. Thus, the potential of the fifth wiring **3105** is set to a potential V_0 which is between $V_{\text{sub.th_H}}$ and $V_{\text{sub.th_L}}$, whereby charge supplied to the gate of the transistor **3200** can be determined. For example, in the case where the High-level charge is supplied in writing and the potential of the fifth wiring **3105** is $V_0 (>V_{\text{sub.th_H}})$, the transistor **3200** is turned “on”. In the case where the Low-level charge is supplied, even when the potential of the fifth wiring **3105** is $V_0 (<V_{\text{sub.th_L}})$, the transistor **3200** remains “off”. Thus, the retained data can be read by determining the potential of the second wiring **3102**.

(595) The semiconductor device illustrated in FIG. **49(C)** is different from that in FIG. **49(A)** in that the transistor **3200** is not provided. Also in this case, writing and retaining operation of data can be performed in a manner similar to that described above.

(596) Next, reading of data of the semiconductor device illustrated in FIG. **49(C)** is described. When the transistor **3300** is turned on, the third wiring **3103** which is in a floating state and the capacitor **3400** are electrically connected to each other, and the charge is redistributed between the third wiring **3103** and the capacitor **3400**. As a result, the potential of the third wiring **3103** is changed. The amount of change in the potential of the third wiring **3103** varies depending on the potential of the one electrode of the capacitor **3400** (or the charge accumulated in the capacitor **3400**).

(597) For example, the potential of the third wiring **3103** after the charge redistribution is $(CB \times VB_0 + C \times V)/(CB + C)$, where V is the potential of the one electrode of the capacitor **3400**, C is the capacitance of the capacitor **3400**, CB is the capacitance component of the third wiring **3103**, and VB_0 is the potential of the third wiring **3103** before the charge redistribution. Thus, it can be found that, assuming that the memory cell can be in two states in which the potential of the one electrode of the capacitor **3400** is V_1 and V_0 ($V_1 > V_0$), the potential of a bit line BL in the case of retaining the potential V_1 ($= (CB \times VB_0 + C \times V_1)/(CB + C)$) is higher than the potential of the bit line BL in the case of retaining the potential V_0 ($= (CB \times VB_0 + C \times V_0)/(CB + C)$).

(598) Then, by comparing the potential of the third wiring **3103** with a predetermined potential, data can be read.

(599) In this case, a transistor using the first semiconductor material may be used for a driver circuit for driving a memory cell, and a transistor using the second semiconductor material as the transistor **3300** may be stacked over the driver circuit.

(600) When using a transistor which uses an oxide semiconductor in a channel formation region and has an extremely small off-state current, the semiconductor device described in this embodiment can retain stored contents for an extremely long time. In other words, refresh operation becomes unnecessary or the frequency of the refresh operation can be extremely low, which leads to a sufficient reduction in power consumption. Moreover, stored contents can be retained for a long time even when power is not supplied (note that a potential is preferably fixed).

(601) Furthermore, in the semiconductor device described in this embodiment, high voltage is not needed for writing data and there is no problem of deterioration of elements. Unlike in a conventional nonvolatile memory, for example, it is not necessary to inject and extract electrons into and from a floating gate; thus, a problem such as deterioration of a gate insulating film is not caused at all. That is, the semiconductor device described in this embodiment does not have a limit on the number of times data can be rewritten, which is a problem of a conventional nonvolatile memory, and the reliability thereof is drastically improved. Furthermore, data is written depending on the on/off state of the transistor, whereby high-speed operation can be easily achieved.

(602) Note that the above memory device can also be applied to an LSI such as a DSP (Digital

Signal Processor), a custom LSI, or a PLD (Programmable Logic Device), and RF-ID (Radio Frequency Identification), in addition to a CPU (Central Processing Unit), for example.

(603) <8-2. CPU>

(604) A CPU including the above memory device is described below.

(605) FIG. 50 is a block diagram illustrating a configuration example of the CPU including the above memory device.

(606) The CPU illustrated in FIG. 50 includes, over a substrate **2190**, an ALU **2191** (ALU: Arithmetic logic unit, arithmetic circuit), an ALU controller **2192**, an instruction decoder **2193**, an interrupt controller **2194**, a timing controller **2195**, a register **2196**, a register controller **2197**, a bus interface **2198** (Bus I/F), a rewritable ROM **2199**, and a ROM interface **2189** (ROM I/F). A semiconductor substrate, an SOI substrate, a glass substrate, or the like is used as the substrate **2190**. The ROM **2199** and the ROM interface **2189** may be provided over a separate chip. Needless to say, the CPU in FIG. 50 is just an example in which the configuration is simplified, and an actual CPU may have a variety of configurations depending on the application. For example, the CPU may have the following configuration: a configuration including the CPU illustrated in FIG. 50 or an arithmetic circuit is considered as one core; a plurality of the cores are included; and the cores operate in parallel. The number of bits that the CPU can process in an internal arithmetic circuit or in a data bus can be, for example, 8, 16, 32, or 64.

(607) An instruction that is input to the CPU through the bus interface **2198** is input to the instruction decoder **2193** and decoded therein, and then, input to the ALU controller **2192**, the interrupt controller **2194**, the register controller **2197**, and the timing controller **2195**.

(608) The ALU controller **2192**, the interrupt controller **2194**, the register controller **2197**, and the timing controller **2195** conduct various controls on the basis of the decoded instruction. Specifically, the ALU controller **2192** generates signals for controlling the operation of the ALU **2191**. While the CPU is executing a program, the interrupt controller **2194** processes an interrupt request from an external input/output device or a peripheral circuit, depending on its priority or a mask state. The register controller **2197** generates an address of the register **2196**, and reads/writes data from/to the register **2196** depending on the state of the CPU.

(609) The timing controller **2195** generates signals for controlling operation timings of the ALU **2191**, the ALU controller **2192**, the instruction decoder **2193**, the interrupt controller **2194**, and the register controller **2197**. For example, the timing controller **2195** includes an internal clock generation portion for generating an internal clock signal on the basis of a reference clock signal, and supplies the internal clock signal to the above circuits.

(610) In the CPU illustrated in FIG. 50, a memory device is provided in the register **2196**.

(611) In the CPU illustrated in FIG. 50, the register controller **2197** selects retaining operation in the register **2196** in accordance with an instruction from the ALU **2191**. That is, the register controller **2197** selects whether data is retained by a flip-flop or by a capacitor in the memory device included in the register **2196**. When data retaining by the flip-flop is selected, a power supply voltage is supplied to the memory device in the register **2196**. When data retaining by the capacitor is selected, the data is rewritten in the capacitor, and supply of the power supply voltage to the memory device in the register **2196** can be stopped.

(612) FIG. 51 is an example of a circuit diagram of a memory element that can be used for the register **2196**. A memory element **2200** includes a circuit **2201** in which stored data is volatile when power supply is stopped, a circuit **2202** in which stored data is nonvolatile even when power supply is stopped, a switch **2203**, a switch **2204**, a logic element **2206**, a capacitor **2207**, and a circuit **2220** having a selecting function. The circuit **2202** includes a capacitor **2208**, a transistor **2209**, and a transistor **2210**. Note that the memory element **2200** may further include another element such as a diode, a resistor, or an inductor, as needed.

(613) Here, the above-described memory device can be used as the circuit **2202**. When supply of a power supply voltage to the memory element **2200** is stopped, a ground potential (0 V) or a

potential at which the transistor **2209** in the circuit **2202** is turned off continues to be input to a gate of the transistor **2209**. For example, the gate of the transistor **2209** is grounded through a load such as a resistor.

(614) Shown is an example in which the switch **2203** is formed using a transistor **2213** having one conductivity type (e.g., an n-channel type) and the switch **2204** is formed using a transistor **2214** having a conductivity type opposite to the one conductivity type (e.g., a p-channel type). Here, a first terminal of the switch **2203** corresponds to one of a source and a drain of the transistor **2213**, a second terminal of the switch **2203** corresponds to the other of the source and the drain of the transistor **2213**, and conduction or non-conduction between the first terminal and the second terminal of the switch **2203** (i.e., the on/off state of the transistor **2213**) is selected by a control signal RD input to a gate of the transistor **2213**. A first terminal of the switch **2204** corresponds to one of a source and a drain of the transistor **2214**, a second terminal of the switch **2204** corresponds to the other of the source and the drain of the transistor **2214**, and conduction or non-conduction between the first terminal and the second terminal of the switch **2204** (i.e., the on/off state of the transistor **2214**) is selected by the control signal RD input to a gate of the transistor **2214**.

(615) One of a source and a drain of the transistor **2209** is electrically connected to one of a pair of electrodes of the capacitor **2208** and a gate of the transistor **2210**. Here, the connection portion is referred to as a node M2. One of a source and a drain of the transistor **2210** is electrically connected to a wiring that can supply a low power supply potential (e.g., a GND line), and the other is electrically connected to the first terminal of the switch **2203** (the one of the source and the drain of the transistor **2213**). The second terminal of the switch **2203** (the other of the source and the drain of the transistor **2213**) is electrically connected to the first terminal of the switch **2204** (the one of the source and the drain of the transistor **2214**). The second terminal of the switch **2204** (the other of the source and the drain of the transistor **2214**) is electrically connected to a wiring that can supply a power supply potential VDD. The second terminal of the switch **2203** (the other of the source and the drain of the transistor **2213**), the first terminal of the switch **2204** (the one of the source and the drain of the transistor **2214**), an input terminal of the logic element **2206**, and one of a pair of electrodes of the capacitor **2207** are electrically connected to each other. Here, the connection portion is referred to as a node M1. A constant potential can be input to the other of the pair of electrodes of the capacitor **2207**. For example, a low power supply potential (e.g., GND) or a high power supply potential (e.g., VDD) can be input to the other of the pair of electrodes of the capacitor **2207**. The other of the pair of electrodes of the capacitor **2207** is electrically connected to the wiring that can supply a low power supply potential (e.g., a GND line). A constant potential can be input to the other of the pair of electrodes of the capacitor **2208**. For example, a low power supply potential (e.g., GND) or a high power supply potential (e.g., VDD) can be input to the other of the pair of electrodes of the capacitor **2208**. The other of the pair of electrodes of the capacitor **2208** is electrically connected to the wiring that can supply a low power supply potential (e.g., a GND line).

(616) Note that the capacitor **2207** and the capacitor **2208** can be omitted when the parasitic capacitance of the transistor or the wiring, or the like is actively utilized.

(617) A control signal WE is input to a first gate (a first gate electrode) of the transistor **2209**. As for the switches **2203** and **2204**, a conduction state or a non-conduction state between the first terminal and the second terminal is selected by the control signal RD that is different from the control signal WE. When the first terminal and the second terminal of one of the switches are in the conduction state, the first terminal and the second terminal of the other of the switches are in the non-conduction state.

(618) A signal corresponding to data retained in the circuit **2201** is input to the other of the source and the drain of the transistor **2209**. FIG. 51 illustrates an example in which a signal output from the circuit **2201** is input to the other of the source and the drain of the transistor **2209**. A signal output from the second terminal of the switch **2203** (the other of the source and the drain of the

transistor **2213**) becomes an inverted signal obtained in such a manner that the logic value of the signal is inverted by the logic element **2206**, and the inverted signal is input to the circuit **2201** through the circuit **2220**.

(619) Note that FIG. **51** shows the example where a signal output from the second terminal of the switch **2203** (the other of the source and the drain of the transistor **2213**) is input to the circuit **2201** through the logic element **2206** and the circuit **2220**; however, there is no limitation thereto. The signal output from the second terminal of the switch **2203** (the other of the source and the drain of the transistor **2213**) may be input to the circuit **2201** without its logic value being inverted. For example, in the case where the circuit **2201** includes a node in which a signal obtained by inversion of the logic value of a signal input from the input terminal is retained, the signal output from the second terminal of the switch **2203** (the other of the source and the drain of the transistor **2213**) can be input to the node.

(620) In FIG. **51**, the transistors used in the memory element **2200** except for the transistor **2209** can be a transistor in which a channel is formed in a layer formed using a semiconductor other than an oxide semiconductor or in the substrate **2190**. For example, the transistor can be a transistor whose channel is formed in a silicon layer or a silicon substrate. A transistor in which a channel is formed in an oxide semiconductor film can be used for all the transistors used in the memory element **2200**. Alternatively, in the memory element **2200**, a transistor in which a channel is formed in an oxide semiconductor film can be included besides the transistor **2209**, and a transistor in which a channel is formed in a layer formed using a semiconductor other than an oxide semiconductor or the substrate **2190** can be used for the rest of the transistors.

(621) As the circuit **2201** in FIG. **51**, for example, a flip-flop can be used. As the logic element **2206**, for example, an inverter, a clocked inverter, or the like can be used.

(622) In a period during which the memory element **2200** is not supplied with the power supply voltage, the semiconductor device described in this embodiment can retain data stored in the circuit **2201** with the use of the capacitor **2208** that is provided in the circuit **2202**.

(623) The off-state current of a transistor in which a channel is formed in an oxide semiconductor film is extremely small. For example, the off-state current of a transistor in which a channel is formed in an oxide semiconductor film is significantly smaller than that of a transistor in which a channel is formed in silicon having crystallinity. Thus, when the transistor in which a channel is formed in an oxide semiconductor film is used as the transistor **2209**, a signal is retained in the capacitor **2208** for a long time also in a period during which the power supply voltage is not supplied to the memory element **2200**. The memory element **2200** can accordingly retain the stored contents (data) also in a period during which the supply of the power supply voltage is stopped.

(624) Since the memory element is characterized by performing pre-charge operation by providing the switches **2203** and **2204**, the time required for the circuit **2201** to retain original data again after the supply of the power supply voltage is restarted can be shortened.

(625) In the circuit **2202**, a signal retained by the capacitor **2208** is input to the gate of the transistor **2210**. Thus, after supply of the power supply voltage to the memory element **2200** is restarted, the signal retained by the capacitor **2208** can be converted into the one corresponding to the state (the on state or the off state) of the transistor **2210** to be read from the circuit **2202**. Consequently, an original signal can be accurately read even when a potential corresponding to the signal retained by the capacitor **2208** changes to some degree.

(626) By using the above-described memory element **2200** in a memory device such as a register or a cache memory included in a processor, data in the memory device can be prevented from being lost owing to the stop of the supply of the power supply voltage. Furthermore, shortly after the supply of the power supply voltage is restarted, the memory device can be returned to the state before the power supply is stopped. Thus, the power supply can be stopped even for a short time in the whole processor or one or a plurality of logic circuits included in the processor, resulting in lower power consumption.

(627) Note that the example where the memory element **2200** is used in a CPU is described in this embodiment; however, the memory element **2200** can also be applied to an LSI such as a DSP (Digital Signal Processor), a custom LSI, or a PLD (Programmable Logic Device), and RF-ID (628) (Radio Frequency Identification).
(629) At least part of this embodiment can be implemented in combination with the other embodiment in this specification, as appropriate.

Embodiment 9

(630) In this embodiment, an imaging device that can use the semiconductor device of one embodiment of the present invention is described using FIG. 52 and FIG. 53.

(631) FIGS. 52(A) to 52(C) illustrate circuit configuration examples of an imaging device.

(632) <9. Imaging Device>

(633) An imaging device **610** including the circuit illustrated in FIG. 52(A) includes a photoelectric conversion element **601**, a transistor **602**, a transistor **604**, and a capacitor **606**. One of a source and a drain of the transistor **602** is electrically connected to the photoelectric conversion element **601**, and the other of the source and the drain of the transistor **602** is electrically connected to a gate of the transistor **604** through a node **607** (a charge accumulation portion).

(634) An OS transistor is preferably used as the transistor **602**. Since the off-state current of the OS transistor can be extremely low, the capacitor **606** can be small. Alternatively, the capacitor **606** can be omitted as illustrated in FIG. 52(B). Furthermore, when an OS transistor is used as the transistor **602**, the potential of the node **607** is less likely to be changed. Thus, an imaging device which is less likely to be affected by noise can be provided.

(635) For example, the transistor shown as an example in the above embodiment, or the like can be used as the transistor **602**. Note that an OS transistor may be used as the transistor **604**.

(636) A diode element in which a pn junction or a pin junction is formed in a silicon substrate can be used as the photoelectric conversion element **601**. Alternatively, a pin diode element using an amorphous silicon film, a microcrystalline silicon film, or the like may be used. Alternatively, a diode-connected transistor may be used. Further alternatively, a variable resistor or the like utilizing a photoelectric effect may be formed using silicon, germanium, selenium, or the like.

(637) The photoelectric conversion element may be formed using a material capable of generating charges by absorbing radiation. Examples of the material capable of generating charges by absorbing radiation include lead iodide, mercury iodide, gallium arsenide, CdTe, and CdZn.

(638) The imaging device **610** including the circuit in FIG. 52(C) is illustrated in which a photodiode is used as the photoelectric conversion element **601**. The imaging device **610** illustrated in FIG. 52(C) includes the photoelectric conversion element **601**, the transistor **602**, a transistor **603**, the transistor **604**, a transistor **605**, and the capacitor **606**. One of the source and the drain of the transistor **602** is electrically connected to a cathode of the photoelectric conversion element **601**, and the other is electrically connected to the node **607**. An anode of the photoelectric conversion element **601** is electrically connected to a wiring **611**. One of a source and a drain of the transistor **603** is electrically connected to the node **607**, and the other is electrically connected to a wiring **608**. The gate of the transistor **604** is electrically connected to the node **607**, one of a source and a drain of the transistor **604** is electrically connected to a wiring **609**, and the other is electrically connected to one of a source and a drain of the transistor **605**. The other of the source and the drain of the transistor **605** is electrically connected to the wiring **608**. One electrode of the capacitor **606** is electrically connected to the node **607**, and the other electrode is electrically connected to the wiring **611**.

(639) The transistor **602** can function as a transfer transistor. A gate of the transistor **602** is supplied with a transfer signal TX. The transistor **603** can function as a reset transistor. A gate of the transistor **603** is supplied with a reset signal RST. The transistor **604** can function as an amplifier transistor. The transistor **605** can function as a selection transistor. A gate of the transistor **605** is supplied with a selection signal SEL. Moreover, VDD is supplied to the wiring **608** and VSS is

supplied to the wiring **611**.

(640) Next, operations of the imaging device **610** including the circuit illustrated in FIG. 52(C) are described. First, the transistor **603** is turned on so that VDD is supplied to the node **607** (reset operation). Then, the transistor **603** is turned off so that VDD is retained at the node **607**. Next, the transistor **602** is turned on so that the potential of the node **607** is changed in accordance with the amount of light received by the photoelectric conversion element **601** (accumulation operation). After that, the transistor **602** is turned off so that the potential of the node **607** is retained. Next, the transistor **605** is turned on so that a potential corresponding to the potential of the node **607** is output from the wiring **609** (selection operation). Measuring the potential of the wiring **609** can determine the amount of light received by the photoelectric conversion element **601**.

(641) An OS transistor is preferably used for the transistors **602** and **603**. Since the off-state current of the OS transistor can be extremely low as described above, the capacitor **606** can be small. Alternatively, the capacitor **606** can be omitted. Furthermore, when an OS transistors are used as the transistors **602** and **603**, the potential of the node **607** is less likely to be changed. Thus, an imaging device which is less likely to be affected by noise can be provided. A high-resolution imaging device can be obtained when the imaging devices **610** including any of the circuits illustrated in FIGS. 52(A) to 52(C) are arranged in a matrix.

(642) For example, when the imaging devices **610** are arranged in a 1920×1080 matrix, an imaging device can be obtained which can take an image with what is called full high definition (also referred to as “2K resolution”, “2K1K”, “2K”, and the like). For example, when the imaging devices **610** are arranged in a 4096×2160 matrix, an imaging device can be obtained which can take an image with what is called ultra-high definition (also referred to as “4K resolution”, “4K2K”, “4K”, and the like). For example, when the imaging devices **610** are arranged in a 8192×4320 matrix, an imaging device can be obtained which can take an image with what is called super high definition (also referred to as “8K resolution”, “8K4K”, “8K”, and the like). Using a larger number of display elements, an imaging device can be obtained which can take an image with 16K or 32K resolution.

(643) FIGS. 53(A) and (B) illustrate a structure example of the imaging device **610** using the above-described transistors. FIGS. 53(A) and (B) are cross-sectional views of the imaging device **610**.

(644) In the imaging device **610** illustrated in FIG. 53(A), an n-type semiconductor is used for a substrate **641**. A p-type semiconductor **1221** of the photoelectric conversion element **601** is provided in the substrate **641**. A portion of the substrate **641** functions as an n-type semiconductor **1223** of the photoelectric conversion element **601**.

(645) The transistor **604** is provided on the substrate **641**. The transistor **604** can function as an n-channel transistor. A well **1220** of a p-type semiconductor is provided in a portion of the substrate **641**. The well **1220** can be provided by a method similar to that for forming the p-type semiconductor **1221**. The well **1220** and the p-type semiconductor **1221** can be formed at the same time.

(646) In the imaging device **610** illustrated in FIG. 53(B), the transistor **604** and the transistor **605** are provided on the substrate **641**. The transistor **604** can function as an n-channel transistor. The transistor **605** can function as a p-channel transistor.

(647) In the imaging device **610** illustrated in FIG. 53(B), the photoelectric conversion element **601** is provided over the substrate **641**.

(648) The photoelectric conversion element **601** illustrated in FIG. 53(B) includes a photoelectric conversion layer **681** between an electrode **686** formed with a metal material or the like and a light-transmitting conductive layer **682**. FIG. 53(B) illustrates the mode using a selenium-based material for the photoelectric conversion layer **681**. The photoelectric conversion element **601** using a selenium-based material has a characteristic of high external quantum efficiency with respect to visible light. With the photoelectric conversion element, a highly sensitive sensor in which the

amplification of electrons with respect to the amount of incident light is large owing to an avalanche phenomenon can be obtained. Furthermore, the selenium-based material has a high light-absorption coefficient, which leads to an advantage that the photoelectric conversion layer **681** can be formed thin.

(649) Amorphous selenium or crystalline selenium can be used as the selenium-based material. Crystalline selenium can be obtained by, for example, depositing amorphous selenium and then performing heat treatment. When the crystal grain size of crystalline selenium is smaller than a pixel pitch, variation in characteristics between pixels can be reduced. Moreover, crystalline selenium has characteristics of higher spectral sensitivity and light-absorption coefficient for visible light than amorphous selenium.

(650) Although the photoelectric conversion layer **681** is illustrated as a single layer, gallium oxide, cerium oxide, or the like as a hole injection blocking layer may be provided on the light reception side of the selenium-based material, and nickel oxide, antimony sulfide, or the like as an electron injection blocking layer may be provided on the electrode **686** side.

(651) Furthermore, the photoelectric conversion layer **681** may be a layer including a compound of copper, indium, and selenium (CIS). Alternatively, the photoelectric conversion layer **681** may be a layer including a compound of copper, indium, gallium, and selenium (CIGS). With CIS and CIGS, a photoelectric conversion element that can utilize an avalanche phenomenon as in the case of using a single layer of selenium can be formed.

(652) Furthermore, CIS and CIGS are p-type semiconductors, and cadmium sulfide, zinc sulfide, or the like, which is an n-type semiconductor, may be provided in contact with the p-type semiconductor in order to form a junction.

(653) It is preferable to apply a relatively high voltage (e.g., 10 V or higher) to the photoelectric conversion element in order to cause the avalanche phenomenon. Since the OS transistor has higher drain withstand voltage than a Si transistor, the application of a relatively high voltage to the photoelectric conversion element is easy. Thus, by combination of the OS transistor having high drain withstand voltage and a photoelectric conversion element using the selenium-based material for the photoelectric conversion layer, a highly sensitive and highly reliable imaging device can be obtained.

(654) For the light-transmitting conductive layer **682**, the following can be used: indium tin oxide; indium tin oxide containing silicon; indium oxide containing zinc; zinc oxide; zinc oxide containing gallium; zinc oxide containing aluminum; tin oxide; tin oxide containing fluorine; tin oxide containing antimony; graphene; or the like. The light-transmitting conductive layer **682** is not limited to a single layer, and may be a stacked layer of different films.

(655) Alternatively, a pin diode element or the like using an amorphous silicon film, a microcrystalline silicon film, or the like may be used as the photoelectric conversion element **601**. In the photodiode, an n-type semiconductor layer, an i-type semiconductor layer, and a p-type semiconductor layer are stacked in this order. Amorphous silicon is preferably used for the i-type semiconductor layer. For the p-type semiconductor layer and the n-type semiconductor layer, amorphous silicon, microcrystalline silicon, or the like that includes a dopant imparting the corresponding conductivity type can be used. A photodiode in which a photoelectric conversion layer is formed using amorphous silicon has high sensitivity in a visible light wavelength region, and therefore can easily sense weak visible light.

(656) Note that a pn or pin diode element is preferably provided such that the p-type semiconductor layer serves as a light-receiving surface. When the p-type semiconductor layer serves as a light-receiving surface, the output current of the photoelectric conversion element **601** can be increased.

(657) The photoelectric conversion element **601** formed using the selenium-based material, amorphous silicon, or the like can be formed through general semiconductor manufacturing processes such as a deposition process, a lithography process, and an etching process

(658) This embodiment can be implemented in combination with the structures described in the

other embodiments and the like, as appropriate.

Example 1

(659) In this example, samples including conductive films that can be used for the semiconductor device of one embodiment of the present invention were fabricated and the cross-sectional shapes of the samples were observed with a scanning transmission electron microscope (Scanning Transmission Electron Microscope: STEM). In addition, XPS (X-ray photoelectron spectroscopy) analysis of the samples was performed.

(660) <Sample Fabrication>

(661) For Sample A1, 35-nm-thick tantalum nitride and 200-nm-thick copper were deposited in this order on a glass substrate with a size of 720 mm×600 mm with a sputtering apparatus.

(662) For Sample A2, 35-nm-thick tantalum nitride and 200-nm-thick copper were deposited in this order on a glass substrate with a size of 720 mm×600 mm with a sputtering apparatus. Then, the copper surface was silicified. The conditions of the silicification treatment are as follows. Plasma was discharged in an atmosphere containing an ammonia gas with a PECVD apparatus so that an oxide film formed on the copper surface was removed first. Then, a silane gas was introduced into a PECVD apparatus, and the copper surface was exposed to the silane gas to form copper silicide. Note that the substrate temperature during removal of the oxide film on the copper surface was 350° C. The substrate temperature during formation of copper silicide was 220° C., and a silane gas at a flow rate of 300 sccm and a nitrogen gas at a flow rate of 500 sccm were used.

(663) <Cross-Sectional Observation>

(664) The cross-sectional shapes of the fabricated Samples A1 and A2 were observed with a STEM. FIG. 54(A) is a cross-sectional STEM photograph of Sample A1, and FIG. 54(B) is a cross-sectional STEM photograph of Sample A2.

(665) The cross-sectional STEM photographs of FIGS. 54(A) and (B) show that a film different from copper (copper silicide layer) was formed on the outermost surface of a film in Sample A2.

(666) <Xps Analysis>

(667) Then, XPS analysis of the surface of Sample A2 was performed. Table 1 shows the composition found by XPS. Note that the detection depth of the surface in XPS is approximately 5 nm.

(668) TABLE-US-00001 TABLE 1 Cu O C Si N Composition (atomic %) 24.1 37.6 18.2 11.3 8.8

(669) Copper and silicon were detected in the surface of Sample A2, and the ratio of silicon to copper (Si/Cu) was 0.47.

(670) Next, FIG. 55 shows results obtained by performing XPS analysis in the depth direction of Sample A2. FIG. 55 shows a profile of copper (Cu), silicon (Si), tantalum (Ta), oxygen (O), and nitrogen (N) in the depth direction from the film surface of Sample A2 to a glass substrate.

(671) As shown in FIG. 55, copper (Cu) and silicon (Si) were detected in the vicinity of the film surface.

(672) Next, FIGS. 56(A) and (B) show the spectrum of Cu2p.sub.3/2 and the spectrum of Si2p obtained by XPS analysis in the depth direction of Sample A2, respectively. Note that the horizontal axis represents binding energy (Binding Energy).

(673) In FIG. 56(A), peaks in the range from 931 eV to 934 eV are attributed to a Cu—Si group. In FIG. 56(B), peaks in the range from 98 eV to 100 eV are attributed to a metal-Si group.

(674) Based on the above, it was confirmed that copper silicide (copper silicide) having a bond of Cu and Si was formed on the film surface in Sample A2.

(675) The structures described above in this example can be used in combination with the other embodiment or example, as appropriate.

Example 2

(676) In this example, transistors of one embodiment of the present invention were formed and the electrical characteristics were evaluated.

(677) Samples B1 and B2 corresponding to the transistor 100L illustrated in FIGS. 10(A) and (B)

were fabricated as transistors for electrical characteristics evaluation. Note that Sample B1 is a transistor of one embodiment of the present invention, and Sample B2 is a transistor for comparison.

(678) The transistor **100L** includes the conductive film **104** functioning as a gate electrode over the substrate **102**, the insulating films **106** and **107** functioning as a first gate insulating film over the substrate **102** and the conductive film **104**, the oxide semiconductor film **108** over the insulating film **107**, the conductive film **112a** electrically connected to the oxide semiconductor film **108** and functioning as a source electrode, the conductive film **112b** electrically connected to the oxide semiconductor film **108** and functioning as a drain electrode, the insulating films **114** and **116** functioning as a second gate insulating film over the oxide semiconductor film **108** and the conductive films **112a** and **112b**, the insulating film **118** functioning as a protective insulating film over the insulating film **116**, and **120b** functioning as a second gate electrode over the insulating film **118**. The oxide semiconductor film **108** includes the oxide semiconductor films **108b** and **108c**.

(679) The conductive film **112a** includes the conductive films **112a_1**, **112a_2**, and **112a_3**, and the conductive film **112b** includes the conductive film **112b_1**, **112b_2**, and **112b_3**. The conductive film **112a_2** includes the region **112a_2b** in contact with the insulating film **114** at the end portion, and the conductive film **112b_2** includes the region **112b_2b** in contact with the insulating film **114** at the end portion.

(680) <Method for Forming Transistor>

(681) Next, methods for forming transistors corresponding to Samples B1 and B2 are described.

(682) Note that Embodiment 1 can be referred to for the methods for forming the transistors.

(683) <<Fabrication of Sample B1>>

(684) The conductive film **104** was formed over the substrate **102**. As the substrate **102**, a glass substrate was used. For the conductive film **104**, a 10-nm-thick titanium film, a 100-nm-thick copper film, and a 50-nm-thick tantalum nitride film were formed in this order with a sputtering apparatus.

(685) Next, the insulating films **106** and **107** were formed over the substrate **102** and the conductive film **104**. As the insulating film **106**, a 400-nm-thick silicon nitride film was formed with a PECVD apparatus. As the insulating film **107**, a 15-nm-thick silicon oxynitride film was formed with a PECVD apparatus.

(686) Then, the oxide semiconductor film **108** was formed over the insulating film **107**.

(687) As the oxide semiconductor film **108**, an In—Ga—Zn oxide (also referred to as IGZO) was used. A 10-nm-thick IGZO film was formed as the oxide semiconductor film **108b** of the oxide semiconductor film **108** with a sputtering apparatus. Note that the IGZO film was formed under the conditions where the substrate temperature was 170° C., the argon gas and the oxygen gas were introduced into a chamber such that the flow rate ratio of the oxygen gas to the argon gas was 30%, the pressure was 0.2 Pa, and an AC power of 1500 W was supplied to a metal oxide sputtering target (In:Ga:Zn=4:2:4.1 [atomic ratio]). Then, a 20-nm-thick IGZO film was formed as the oxide semiconductor film **108c** with a sputtering apparatus. Note that the IGZO film was formed under the conditions where the substrate temperature was 170° C., the argon gas and the oxygen gas were introduced into a chamber such that the flow rate ratio of the oxygen gas to the argon gas was 50%, the pressure was 0.2 Pa, and an AC power of 500 W was supplied to a metal oxide sputtering target (In:Ga:Zn=1:1:1.2 [atomic ratio]). After formation of the oxide semiconductor film **108**, heat treatment was performed at 350° C. for 1 hour.

(688) Next, a conductive film to be the conductive films **112a** and **112b** was formed over the insulating film **107** and the oxide semiconductor film **108**. For the conductive film, a 50-nm-thick tungsten film, a 200-nm-thick copper film, and a 5-nm-thick tungsten film were successively formed in a vacuum with a sputtering apparatus. Then, a resist mask was formed over the

conductive film, and a desired region of the 5-nm-thick tungsten film and the 200-nm-thick copper film was etched. After the removal of the resist mask, silicide treatment of the exposed copper surface was performed. Then, the copper surface was silicified. The conditions of the silicification treatment are as follows. Plasma was discharged in an atmosphere containing an ammonia gas with a PECVD apparatus so that an oxide film formed on the copper surface was removed first. Then, a silane gas was introduced into a PECVD apparatus, and the copper surface was exposed to the silane gas to form copper silicide. Note that the substrate temperature during removal of the oxide film on the copper surface was 350° C. The substrate temperature during formation of copper silicide was 220° C., and a silane gas at a flow rate of 300 sccm and a nitrogen gas at a flow rate of 500 sccm were used. Then, a resist mask was formed over the conductive film subjected to silicide treatment and a desired region of the 50-nm-thick tungsten film was etched, so that the conductive films **112a** and **112b** were formed. Note that the resist mask was removed after the formation of the conductive films **112a** and **112b**.

(689) Next, an aqueous phosphoric acid solution (an aqueous solution obtained by diluting an aqueous solution with a phosphoric acid concentration of 85% with pure water by 100 times) was applied from above the insulating film **107**, the oxide semiconductor film **108**, and the conductive films **112a** and **112b**, so that part of the surface of the oxide semiconductor film **108** which is not covered with the conductive films **112a** and **112b** was removed.

(690) After that, the insulating films **114** and **116** were formed over the insulating film **107**, the oxide semiconductor film **108**, and the conductive films **112a** and **112b**. As the insulating film **114**, a 40-nm-thick silicon oxynitride film was formed with a PECVD apparatus. As the insulating film **116**, a 400-nm-thick silicon oxynitride film was formed with a PECVD apparatus. Note that the insulating films **114** and **116** were formed successively in a vacuum with a PECVD apparatus.

(691) The insulating film **114** was formed under the conditions where the substrate temperature was 220° C., a silane gas at a flow rate of 50 sccm and a dinitrogen monoxide gas at a flow rate of 2000 sccm were introduced into a chamber, the pressure was 20 Pa, and an RF power of 100 W was supplied between parallel-plate electrodes provided in a PECVD apparatus. The insulating film **116** was formed under the conditions where the substrate temperature was 220° C., a silane gas at a flow rate of 160 sccm and a dinitrogen monoxide gas at a flow rate of 4000 sccm were introduced into a chamber, the pressure was 200 Pa, and an RF power of 1500 W was supplied between parallel-plate electrodes provided in a PECVD apparatus.

(692) Next, heat treatment was performed at 300° C. under an atmosphere containing nitrogen for 1 hour.

(693) Then, the insulating film **118** was formed over the insulating film **116**. As the insulating film **118**, a 100-nm-thick silicon nitride film was formed with a PECVD apparatus. The insulating film **118** was formed under the conditions where the substrate temperature was 220° C., a silane gas at a flow rate of 50 sccm, a nitrogen gas at a flow rate of 5000 sccm, and an ammonia gas at a flow rate of 100 sccm were introduced into a chamber, the pressure was 100 Pa, and an RF power of 1000 W was supplied between parallel-plate electrodes provided in a PECVD apparatus.

(694) Then, as a conductive film, a 100-nm-thick ITSO film was formed over the insulating film **118** with a sputtering apparatus. The formation conditions of the ITSO film were as follows: the substrate temperature was room temperature, an argon gas at a flow rate of 72 sccm and an oxygen gas at a flow rate of 5 sccm were introduced into a chamber, and the pressure was 0.15 Pa. Note that the composition of the metal oxide target used for the ITSO film was

In.sub.2O.sub.3:SnO.sub.2:SiO.sub.2=85:10:5 [weight %].

(695) In the above process, the transistor corresponding to Sample B1 was formed.

(696) <<Fabrication of Sample B2>>

(697) The conductive film **104** was formed over the substrate **102**. As the substrate **102**, a glass substrate was used. As the conductive film **104**, a 100-nm-thick tungsten film was formed with a sputtering apparatus.

(698) Next, the insulating films **106** and **107** were formed over the substrate **102** and the conductive film **104**. As the insulating film **106**, a 400-nm-thick silicon nitride film was formed with a PECVD apparatus. As the insulating film **107**, a 50-nm-thick silicon oxynitride film was formed with a PECVD apparatus.

(699) Then, the oxide semiconductor film **108** was formed over the insulating film **107**.

(700) A 20-nm-thick IGZO film was formed as the oxide semiconductor film **108b** of the oxide semiconductor film **108** with a sputtering apparatus. Note that the IGZO film was formed under the conditions where the substrate temperature was 170° C., the argon gas and the oxygen gas were introduced into a chamber such that the flow rate ratio of the oxygen gas to the argon gas $\square\square\square\square\square\square\square\square$ oxygen gas was 30%, the pressure was 0.2 Pa, and an AC power of 1500 W was supplied to a metal oxide sputtering target (In:Ga:Zn=4:2:4.1 [atomic ratio]). Then, a 30-nm-thick IGZO film was formed as the oxide semiconductor film **108c** with a sputtering apparatus. Note that the IGZO film was formed under the conditions where the substrate temperature was 170° C., the argon gas and the oxygen gas were introduced into a chamber such that the flow rate ratio of the oxygen gas to the argon gas $\square\square\square\square\square\square\square\square$ oxygen gas was 50%, the pressure was 0.2 Pa, and an AC power of 500 W was supplied to a metal oxide sputtering target (In:Ga:Zn=1:1:1.2 [atomic ratio]). After formation of the oxide semiconductor film **108**, heat treatment was performed at 350° C. for 1 hour.

(701) Next, a conductive film was formed over the insulating film **107** and the oxide semiconductor film **108**, a resist mask was formed over the conductive film, and a desired region was etched to form the conductive films **112a** and **112b**. For the conductive films **112a** and **112b**, a 50-nm-thick tungsten film and a 200-nm-thick copper film were successively formed in a vacuum with a sputtering apparatus. Note that the resist mask was removed after the formation of the conductive films **112a** and **112b**.

(702) Then, the insulating film **114** and the insulating film **116** were formed over the insulating film **107**, the oxide semiconductor film **108**, and the conductive films **112a** and **112b**. As the insulating film **114**, a 40-nm-thick silicon oxynitride film was formed with a PECVD apparatus. As the insulating film **116**, a 400-nm-thick silicon oxynitride film was formed with a PECVD apparatus. Note that the insulating films **114** and **116** were formed successively in a vacuum with a PECVD apparatus.

(703) The insulating film **114** was formed under the conditions where the substrate temperature was 220° C., a silane gas at a flow rate of 50 sccm and a dinitrogen monoxide gas at a flow rate of 2000 sccm were introduced into a chamber, the pressure was 20 Pa, and an RF power of 100 W was supplied between parallel-plate electrodes provided in a PECVD apparatus. The insulating film **116** was formed under the conditions where the substrate temperature was 220° C., a silane gas at a flow rate of 160 sccm and a dinitrogen monoxide gas at a flow rate of 4000 sccm were introduced into a chamber, the pressure was 200 Pa, and an RF power of 1500 W was supplied between parallel-plate electrodes provided in a PECVD apparatus. Next, heat treatment was performed at 300° C. under an atmosphere containing nitrogen for 1 hour.

(704) Then, the insulating film **118** was formed over the insulating film **116**. As the insulating film **118**, a 100-nm-thick silicon nitride film was formed with a PECVD apparatus. The insulating film **118** was formed under the conditions where the substrate temperature was 220° C., a silane gas at a flow rate of 50 sccm, a nitrogen gas at a flow rate of 5000 sccm, and an ammonia gas at a flow rate of 100 sccm were introduced into a chamber, the pressure was 100 Pa, and an RF power of 1000 W was supplied between parallel-plate electrodes provided in a PECVD apparatus.

(705) Then, as a conductive film, a 100-nm-thick ITSO film was formed over the insulating film **118** with a sputtering apparatus. The formation conditions of the ITSO film were as follows: the substrate temperature was room temperature, an argon gas at a flow rate of 72 sccm and an oxygen gas at a flow rate of 5 sccm were introduced into a chamber, and the pressure was 0.15 Pa. Note that the composition of the metal oxide target used for the ITSO film was

In.sub.2O.sub.3:SnO.sub.2:SiO.sub.2=85:10:5 [weight %].

(706) In the above process, the transistor corresponding to Sample B2 was formed.

(707) Note that the transistors with the following three kinds of sizes were formed: a channel length of 2 μm and a channel width of 50 μm ; a channel length of 3 μm and a channel width of $\square$ 50 μm ; and a channel length of 6 μm and a channel width of 50 μm . In the following, Sample B1 with a channel length of 2 μm is referred to as Sample B1-1, Sample B1 with a channel length of 3 μm is referred to as Sample B1-2, and Sample B1 with a channel length of 6 μm is referred to as Sample B1-3. The same applies to Sample B2.

(708) <Evaluation of Electrical Characteristics of Transistors>

(709) The electrical characteristics of the fabricated transistors corresponding to Samples B1 and B2 were evaluated. The electrical characteristics of the transistors were drain current (I_d)-gate voltage (V_g) characteristics and field-effect mobility (μ_{FE}) obtained from the I_d - V_g characteristics. FIG. 57 and FIG. 58 show the electrical characteristics of the transistors of Samples B1 and B2. FIG. 57(A) and FIG. 58(A) show the characteristics of the transistor with a channel length 2 μm and a channel width of 50 μm , FIG. 57(B) and FIG. 58(B) show the characteristics of the transistor with a channel length of 3 μm and a channel width of 50 μm , and FIG. 57(C) to FIG. 58(C) show the characteristics of the transistor with a channel length of 6 μm and a channel width of 50 μm .

(710) In FIG. 57 and FIG. 58, the voltage (V_d) between the source electrode and the drain electrode was set to 0.1 V and 20 V, and V_g was applied from -15 V to 20 V in 0.25 V steps. In FIG. 57 and FIG. 58, the first vertical axis represents the drain current (I_d), the second vertical axis represents the field-effect mobility (μ_{FE}) when $V_d=20$ V, and the horizontal axis represents the gate voltage (V_g). Data of 10 transistors are superimposed on each other. All the transistors exhibit little variation and normally-off characteristics.

(711) As shown in FIG. 57 and FIG. 58, the following results were obtained: the threshold voltages of Sample B2 were different between when V_d was 0.1 V and when V_d was 20 V; and the transistor of Sample B1 had higher field-effect mobility (μ_{FE}) than that of Sample B2. This indicated that when the end portions of a source electrode and a drain electrode that contain copper of a transistor are subjected to silicide treatment, the transistor having excellent electrical characteristics can be provided.

(712) The structure described above in this example can be used in combination with the other embodiment or example, as appropriate.

Example 3

(713) In this example, samples including conductive films that can be used for the semiconductor device of one embodiment of the present invention were fabricated and the cross-sectional shapes of the samples were observed with a scanning transmission electron microscope (Scanning Transmission Electron Microscope: STEM). In addition, XPS (X-ray photoelectron spectroscopy) analysis of the samples was performed.

(714) <Sample Fabrication>

(715) For Sample C1, a 100-nm-thick silicon oxynitride (SiON) film was formed over a glass substrate with a size of 720 mm $\times$ 600 mm with a PECVD apparatus. Then, a conductive film was formed over the silicon oxynitride film, a resist mask was formed over the conductive film, and a desired region was etched. For the conductive film, a 5-nm-thick tungsten (W) film, a 200-nm-thick copper (Cu) film, and a 5-nm-thick tungsten (W) film were successively formed in a vacuum with a sputtering apparatus. Note that the resist mask was removed after the formation of the conductive film.

(716) For Sample C2, a 100-nm-thick silicon oxynitride (SiON) film was formed over a glass substrate with a size of 720 mm $\times$ 600 mm with a PECVD apparatus. Next, a 50-nm-thick IGZO film was formed as an oxide semiconductor film over the silicon oxynitride film with a sputtering apparatus. Then, a conductive film was formed over the oxide semiconductor film, a resist mask was formed over the conductive film, and a desired region was etched. For the conductive film, a 5-

nm-thick tungsten film, a 200-nm-thick copper film, and a 5-nm-thick tungsten film were successively formed in a vacuum with a sputtering apparatus. The resist mask was removed after the formation of the conductive film. Then, the copper surface was silicified. The conditions of the silicification treatment are as follows. Plasma was discharged in an atmosphere containing an ammonia gas with a PECVD apparatus so that an oxide film formed on the copper surface was removed first. Then, a silane gas was introduced into a PECVD apparatus, and the copper surface was exposed to the silane gas to form copper silicide. Note that the substrate temperature during removal of the oxide film on the copper surface was 350° C. The substrate temperature during formation of copper silicide was 350° C., and a silane gas at a flow rate of 10 sccm and a nitrogen gas at a flow rate of 1000 sccm were used.

(717) <Cross-Sectional Observation and EDX Analysis Results>

(718) The cross-sectional shapes of the fabricated Samples C1 and C2 were observed with a STEM. FIG. 59(A) is a cross-sectional STEM photograph of Sample C1, and FIG. 59(B) is a cross-sectional STEM photograph of Sample C2.

(719) The cross-sectional STEM photograph of FIG. 59(B) shows that a layer different from copper (copper silicide layer) was formed at the end portion of the conductive film of Sample C2.

(720) Elemental analysis was performed using EDX (Energy Dispersive X-ray Spectroscopy, energy dispersive X-ray analysis method) of the end portion of the conductive film of Sample C2. The EDX analysis results are shown in FIG. 60.

(721) As shown in FIG. 60, copper (Cu) and silicon (Si) were detected from the end portion of the conductive film of Sample C2. That is, it was confirmed that copper silicide containing Cu and Si was formed at the end portion of the conductive film of Sample C2.

(722) The structures described above in this example can be used in combination with the other embodiment or example, as appropriate.

REFERENCE NUMERALS

(723) I1 insulator I2 insulator S1 oxide semiconductor S2 oxide semiconductor S3 oxide semiconductor 100 transistor 100A transistor 100B transistor 100C transistor 100D transistor 100E transistor 100F transistor 100G transistor 100H transistor 100J transistor 100K transistor 100L transistor 100M transistor 100N transistor 100P transistor 100Q transistor 102 substrate 104 conductive film 106 insulating film 107 insulating film 108 oxide semiconductor film 108a oxide semiconductor film 108b oxide semiconductor film 108c oxide semiconductor film 112 conductive film 1121 conductive film 1122 conductive film 1123 conductive film 112a conductive film 112a_1 conductive film 112a_2 conductive film 112a_2a region 112a_2b region 112a_3 conductive film 112b conductive film 112b_1 conductive film 112b_2 conductive film 112b_2a region 112b_2b region 112b_3 conductive film 112c conductive film 112c_1 conductive film 112c_2 conductive film 112c_2a region 112c_2b region 112c_3 conductive film 114 insulating film 116 insulating film 118 insulating film 120a conductive film 120b conductive film 141a mask 141b mask 142a mask 142b mask 151 opening 151a opening 151b opening 152a opening 152b opening 152c opening 152d opening 191 target 192 plasma 193 target 194 plasma 195 plasma 501 pixel circuit 502 pixel portion 504 driver circuit portion 504a gate driver 504b source driver 506 protection circuit 507 terminal portion 550 transistor 552 transistor 554 transistor 560 capacitor 562 capacitor 570 liquid crystal element 572 light-emitting element 601 photoelectric conversion element 602 transistor 603 transistor 604 transistor 605 transistor 606 capacitor 607 node 608 wiring 609 wiring 610 imaging device 611 wiring 641 substrate 681 photoelectric conversion layer 682 light-transmitting conductive layer 686 electrode 700 display device 701 substrate 702 pixel portion 704 source driver circuit portion 705 substrate 706 gate driver circuit portion 708 FPC terminal portion 710 signal line 711 wiring portion 712 sealant 716 FPC 730 insulating film 732 sealing film 734 insulating film 736 coloring film 738 light blocking film 750 transistor 752 transistor 760 connecting electrode 770 planarization insulating film 772 conductive film 773 insulating film 774 conductive film 775 liquid crystal element 776 liquid crystal layer 777 conductive film 778

structure body **780** anisotropic conductive film **782** light-emitting element **786** EL layer **788** conductive film **790** capacitor **791** touch panel **792** insulating film **793** electrode **794** electrode **795** insulating film **796** electrode **797** insulating film **800** inverter **810** OStransistor **820** OStransistor **831** signal waveform **832** signal waveform **840** dashed line **841** solid line **850** OS transistor **860** CMOS inverter **900** semiconductor device **901** power supply circuit **902** circuit **903** voltage generation circuit **903A** voltage generation circuit **903B** voltage generation circuit **903C** voltage generation circuit **904** circuit **905** voltage generation circuit **906** circuit **911** transistor **912** transistor **912A** transistor **912B** transistor **921** control circuit **922** transistor **1220** well **1221** p-type semiconductor **1223** n-type semiconductor **2189** ROM interface **2190** substrate **2191** ALU **2192** ALU controller **2193** instruction decoder **2194** interrupt controller **2195** timing controller **2196** register **2197** register controller **2198** bus interface **2199** ROM **2200** memory element **2201** circuit **2202** circuit **2203** switch **2204** switch **2206** logic element **2207** capacitor **2208** capacitor **2209** transistor **2210** transistor **2213** transistor **2214** transistor **2220** circuit **3101** wiring **3102** wiring **3103** wiring **3104** wiring **3105** wiring **3200** transistor **3300** transistor **3400** capacitor **7000** display module **7001** upper cover **7002** lower cover **7003** FPC **7004** touch panel **7005** FPC **7006** display panel **7007** backlight **7008** light source **7009** frame **7010** printed board **7011** battery **8000** camera **8001** housing **8002** display portion **8003** operation button **8004** shutter button **8006** lens **8100** finder **8101** housing **8102** display portion **8103** button **8200** head-mounted display **8201** mounting portion **8202** lens **8203** main body **8204** display portion **8205** cable **8206** battery **8300** head-mounted display **8301** housing **8302** display portion **8304** fixing instrument **8305** lens **9000** housing **9001** display portion **9003** speaker **9005** operation key **9006** connection terminal **9007** sensor **9008** microphone **9050** operation button **9051** information **9052** information **9053** information **9054** information **9055** hinge **9100** television device **9101** portable information terminal **9102** portable information terminal **9200** portable information terminal **9201** portable information terminal **9500** display device **9501** display panel **9502** display region **9503** region **9511** axis portion **9512** bearing

Claims

1. A semiconductor device comprising: a substrate; a first conductive film over the substrate; a first insulating film over the first conductive film; a second insulating film over the first insulating film; an oxide semiconductor film and a second conductive film over the second insulating film; a third insulating film over the oxide semiconductor film and the second conductive film; a fourth insulating film over the third insulating film; and a third conductive film over the fourth insulating film, wherein the first conductive film is configured to function as a first gate electrode of a transistor, wherein the third conductive film is configured to function as a second gate electrode of the transistor, wherein the oxide semiconductor film comprises a channel formation region of the transistor, wherein the second conductive film comprises a fourth conductive film, a fifth conductive film, and a sixth conductive film, wherein the fourth conductive film is in contact with the first conductive film in a first opening provided in the first insulating film and the second insulating film, wherein the third conductive film is in contact with the sixth conductive film in a second opening provided in the third insulating film and the fourth insulating film, and wherein an end portion of the fifth conductive film comprises copper and silicon.
2. A semiconductor device comprising: a substrate; a first conductive film over the substrate; a first insulating film over the first conductive film; an oxide semiconductor film and a second conductive film over the first insulating film; a second insulating film over the oxide semiconductor film and the second conductive film; and a third conductive film over the second insulating film, wherein the oxide semiconductor film comprises a channel formation region of a transistor, wherein the first conductive film is electrically connected to the third conductive film via the second conductive film, and wherein the second conductive film is formed by processing a conductive film

functioning as one of a source electrode and a drain electrode of the transistor.

3. A semiconductor device comprising: a transistor, the transistor comprising: a gate electrode; a gate insulating film over the gate electrode; an oxide semiconductor film overlapping with the gate electrode with the gate insulating film therebetween; a source electrode comprising a region in contact with a top surface of the oxide semiconductor film; a drain electrode comprising a region in contact with the top surface of the oxide semiconductor film; and a first insulating film over the oxide semiconductor film, the source electrode, and the drain electrode, wherein the semiconductor device further comprises: a pixel electrode over the first insulating film; and a common electrode overlapping with the pixel electrode with a second insulating film therebetween, wherein the oxide semiconductor film comprises a first oxide semiconductor film and a second oxide semiconductor film over the first oxide semiconductor film, wherein the oxide semiconductor film comprises a first region overlapping with the source electrode, wherein the oxide semiconductor film comprises a second region overlapping with the drain electrode, wherein, in the first region of the oxide semiconductor film, a thickness of the second oxide semiconductor film is greater than a thickness of the first oxide semiconductor film, wherein, in the second region of the oxide semiconductor film, a thickness of the second oxide semiconductor film is greater than a thickness of the first oxide semiconductor film, wherein each of the source electrode and the drain electrode comprises a first conductive film and a second conductive film in contact with a top surface of the first conductive film, wherein, in a channel length direction of the transistor, an end portion of the first conductive film comprises a region which protrudes farther than an end portion of the second conductive film, wherein the first insulating film comprises a region in contact with the top surface of the first conductive film and a region in contact with a side surface of the first conductive film, wherein the pixel electrode is electrically connected to one of the source electrode and the drain electrode, wherein the common electrode comprises a region overlapping with the oxide semiconductor film with the first insulating film therebetween, wherein the first conductive film comprises titanium, wherein the second conductive film comprises copper, and wherein the end portion of the second conductive film comprises a compound including copper.

4. The semiconductor device according to claim 3, wherein each of the first oxide semiconductor film and the second oxide semiconductor film comprises In, Zn, and M, and wherein M is represented by Al, Ga, Y, or Sn.
